
Non-Obviously Manipulable Mechanism Design

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ABSTRACT

This thesis studies the design of mechanisms without obvious manipulations. The goal of mechanism design is to produce algorithms for strategic agents in which it is in each agent’s best interests to act truthfully. Typically this assumes that agents are perfectly rational, as in strategyproofness where agents are able to compare outcomes of the mechanism for each possible joint action profile of the other agents. When relaxing this assumption, bounded rationality instead supposes that agents are somehow cognitively limited. As a result this influences the constraints required for a mechanism to be incentive compatible. In Li’s strengthening of strategyproofness (S. Li, 2017), a mechanism is said to be *obviously strategyproof (OSP)* if truthtelling is “obvious” over dishonesty, and agents are assumed to only tell the truth when this is the case. Roughly this means that for each agent the worst outcome from acting truthfully is no worse than the best outcome from acting dishonestly when interacting with the mechanism. This thesis concentrates on a relaxation of strategyproofness known as “non-obvious manipulability” (Trojan and Morrill, 2020), which takes a more optimistic approach and assumes agents will only lie if it is obvious. A manipulation is said to be obvious if either the best (respectively, worst) outcome from dishonesty is strictly greater than the best (respectively, worst) outcome from truthtelling, where the extremes are taken over all possible actions of the other players. If a mechanism has no obvious manipulations then it is *not obviously manipulable (NOM)*. With NOM, we may safely ignore some manipulations under the assumption that they will not be recognised by the cognitively limited agents, as long as they are not obvious. This grants us some flexibility as the mechanism designer, and in this thesis we apply this new solution concept to the study of mechanism design using payments to align the incentives of the agents. We firstly investigate the conditions under which we can define NOM mechanisms in general domains. Starting with direct mechanisms, we characterise the class of social choice functions that may be implemented as NOM mechanisms, and restricting attention

to the popular setting of single-parameter domains we obtain a more precise characterisation on both the class of implementable functions as well as their associated payment rules. We then extend our new techniques to study which functions may be implemented as indirect NOM mechanisms, along the way showing a partial “revelation principle” for two settings in which direct mechanisms are just as powerful as the wider class of indirect ones. Applying our machinery to the settings of bilateral trade and general auction environments for both goods and chores, we show that while there remain fundamental incompatibilities of NOM with other desired properties such as efficiency, individual rationality, and budget balance, NOM provides much flexibility over other solution concepts for imperfectly rational agents and can produce mechanisms with desirable performance along a range of metrics.

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DECLARATION

I hereby declare that this thesis represents my own work which has been done after registration for the degree of PhD at King’s College London, and has not been previously included in a thesis or dissertation submitted to this or any other institution for a degree, diploma, or other qualifications.

1. Thomas Archbold, Bart de Keijzer, and Carmine Ventre (2023a). “Non-Obvious Manipulability for Single-Parameter Agents and Bilateral Trade”. In: *Proceedings of the 2023 International Conference on Autonomous Agents and Multiagent Systems, AAMAS 2023, London, United Kingdom, 29 May 2023 - 2 June 2023*. Ed. by Noa Agmon et al. ACM, pp. 2107–2115. DOI: 10.5555/3545946.3598884. URL: <https://dl.acm.org/doi/10.5555/3545946.3598884>
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And now that you don't have to be perfect, you can be good.

—John Steinbeck, *East of Eden*

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INTRODUCTION

In the study of algorithms, the algorithm designer is focused primarily on devising procedures that map inputs to their desired outputs that make efficient use of resources such as time and memory. It is assumed that the inputs to these algorithms are simply provided by nature and that they reflect the true state of the world. By contrast, in mechanism design it is assumed that the algorithm inputs are provided by a set of agents, who each possess some piece of information which must first be elicited by the mechanism in order for it to perform the computation. The challenge lies in the fact that this information is private, meaning it is in general impossible to know for sure whether or not an agent has reported their information truthfully. The agents also each derive some utility from the output of the mechanism, and we assume agents are self-interested and strategic, seeking to maximise their utility. This gives rise to the issue of incentives: since the mechanism is unable to determine whether agents are telling the truth, the mechanism designer must take care in how they elicit and use the information from the agents in order to incentivise the proper truthtelling behaviour. The goal in any mechanism design setting is therefore to produce a mechanism where it is in each agent’s best interests to report their information truthfully. If this is the case then the mechanism is said to be “incentive compatible”.

What exactly do we mean by truthtelling being in the agents’ best interests? Informally, it means that reporting truthfully is no worse than reporting dishonestly, but the exact conditions that must be satisfied to ensure this will depend on the *solution concept* that we use and, as we will introduce later, our assumptions on the cognitive abilities of the agents. Strategyproofness is a notion of incentive compatibility ubiquitous in the mechanism design literature which requires that truthtelling is a *dominant strategy* for each possible combination of reports from the other agents. In other words, an agent maximises her utility when reporting her information truthfully versus any other misreport or deviation from the truth. When

this property holds for each agent then the mechanism is said to be strategyproof, or equivalently *truthful* or *dominant strategy incentive compatible*. Many different notions of incentive compatibility exist. When a mechanism is strategyproof it is robust to manipulations by individual agents, but not necessarily coalitions of agents. Group strategyproofness, a much stronger condition than strategyproofness, requires that no coalition of agents may ever coordinate to misreport their information such that each agent in the coalition is better off. On the other hand, if a mechanism is Bayes-Nash incentive compatible then it only requires that it is a Bayes-Nash equilibrium for agents to report truthfully (that is, each agent maximises her expected utility by reporting truthfully whenever the other agents also report their information to the mechanism truthfully), hence this is a weaker notion of incentive compatibility compared to strategyproofness.

Many natural settings can be framed in the context of mechanism design, and depending on the circumstances the mechanism may use monetary transfers to realign the incentives of the agents to achieve incentive compatibility. One simple mechanism design setting with money is the single-item auction, in which a collection of agents compete for a single good which is awarded to the highest bidder. Each agent's private information is their valuation for the item, and hence they may lie about it to increase their chances of winning. In a first-price auction, the winner simply pays her bid. This payment rule is not strategyproof. To see why, take some agent i with valuation v for the item and fix the bids of the other agents, denoting the highest bid amongst these as B . Suppose $v > B$. If i bids truthfully then she wins the item and is charged her bid, resulting in a net utility of zero, while if she submits a bid lower than v but higher than B then she gets strictly positive utility (since she still wins the item but now pays less than her true valuation).

How can we design a single-item auction that is strategyproof? We still want to allocate the item to the highest bidder in order to maximise the social welfare¹, so we must instead redefine the payment rule. The much celebrated Vickrey auction (Vickrey, 1961) shows that if we award the item to the highest bidder but charge her the price of the *second highest* bid then the resulting mechanism is strategyproof. To see why, consider again some agent i with valuation v and fix the bids of the others, letting B denote the highest bid of the other agents excluding i . Assume without loss of generality that any ties are broken in i 's favour. Suppose first that $v < B$. If i bids truthfully then she loses the auction and pays nothing, resulting

¹The sum of utilities of all the agents.

in a net utility of zero. If she submits any bid $b < B$ then she of course still loses and gains utility zero, and if she overbids with $b \geq B$ then she wins the item but pays B , resulting in a utility of $v - B < 0$. Therefore when $v < B$ it is a dominant strategy to bid truthfully. Now suppose that $v \geq B$. If she bids truthfully she wins the auction and pays B , yielding utility $v - B \geq 0$. If she submits any bid $b \geq B$ then she still wins and pays B , resulting in the same utility of $v - B$ as if she had acted truthfully, and if she bids $b < B$ then she now loses the auction and gets utility zero. In both cases when $v \geq B$ the utility she gets from telling the truth is never worse than lying about her valuation, meaning it is a (weakly) dominant strategy to bid truthfully when $v \geq B$. In all cases, i maximises her utility by bidding truthfully, making the second price auction strategyproof.

Strategyproofness is the standard notion of incentive compatibility in the mechanism design literature, with numerous classical results across a range of settings beyond single-item auctions. Myerson's seminal characterisation (Myerson, 1981) of the class of allocation functions² that can be implemented as dominant-strategy incentive-compatible mechanisms in single-parameter environments (where the private information possessed by the agents can be represented by a single number, as in the single-item auction) reduces the mechanism design problem to an *algorithm design* one. The characterisation shows that payments that, when taken together with the allocation function, yield an incentive-compatible mechanism exist whenever the allocation function is *monotone* (roughly speaking, if an agent bids more then she should get more). Moreover, when this is the case, these payments are unique and are represented by an explicit formula. This greatly decreases our search space for viable mechanisms: as the mechanism designer, we need only to design monotone allocation functions, and the payments come for free thanks to Myerson's characterisation. Likewise, classic results such as the Revelation Principle (see, e.g., [Nisan, Roughgarden, et al., 2007, Chapter 7]) allow us to restrict our attention to mechanisms in which agents are queried once about their private information by the mechanism, rather than being queried about it over a number of rounds, showing that, at least in the case of truthfulness, the implementation details of the mechanism do not matter, only the outcomes they select. However, many negative results and impossibilities exist for designing strategyproof mechanisms. The Myerson-Satterthwaite theorem (Myerson and Satterthwaite, 1983) shows that the presence of private information, even in the case of bilateral trade of a single good between a buyer and a seller, is a severe impediment to efficiency,

²Functions that decide who wins the auction, and which items they are allocated.

stating that no incentive-compatible mechanism can simultaneously achieve efficiency, weak budget balance (meaning we never pay the seller more than what we receive from the buyer, hence the market does not need to be subsidised) and individual rationality (meaning it never harms an agent to participate in the mechanism). Similarly, in the context of single-winner voting, truthfulness is equally problematic: no truthful voting rule for at least three candidates exists without one of the voters having dictatorial power of the candidate selected. This applies to any collective decision-making process, be it ordinal (Satterthwaite, 1975) or otherwise (Gibbard, 1973). This extends to non-deterministic mechanisms (Gibbard, 1978; Hylland, 1980), and a similar result is shown for multi-winner elections (Duggan and Schwartz, 2000), thus showing that strategic manipulations are generally unavoidable.

Even if truthfulness is not outright impossible, then it imposes severe bounds on the performance of the mechanism, as is the case for machine scheduling. Consider a setting in which there is a set of m jobs and a set of n self-interested machines which can process these jobs. The machines possess private information about their processing speed for each job, and our goal as the mechanism designer is to allocate each machine a set of jobs to minimise either the sum of completion times for each machine, or the maximum completion time across all machines (also known as the “makespan”). [Nisan and Ronen, 2001] introduced the problem and showed that the VCG mechanism (Vickrey, 1961; Clarke, 1971; Groves, 1973), one of the most popular frameworks in mechanism design for truthful social welfare maximisation, results in a simple greedy allocation in which each job is independently allocated to the machine with minimum reported processing time for it. The makespan achieved by this is n times the optimal allocation (i.e., if we disregard truthfulness), and they conjectured that this is the best guarantee achievable by any deterministic truthful mechanism, regardless of computation time. [Christodoulou, Koutsoupias, and Kovács, 2023] show that conjecture is true: there is no truthful deterministic mechanism with approximation ratio better than n for the problem of scheduling n unrelated machines. Thus although strategyproofness seems appealing, it imposes severe information-theoretic bounds (i.e., not just limited to computation time) and considerably restricts the class of mechanisms we can consider acceptable.

Turning our attention towards rationality, mechanism design assumes that the details of the mechanism are public knowledge and that there is a central authority that will commit to running the mechanism as specified. Even though each agent may have access to all the details and might be told that the mechanism is robust to manipulation, this is not enough to

guarantee they act in the manner predicted by strategyproofness: they must also *understand* that the mechanism is strategyproof. Importantly, a mechanism’s implementation details may have a significant effect on whether or not the agents understand how to act in the mechanism, and hence are vital to the mechanism’s incentive compatibility guarantees. [Kagel, Harstad, and D. Levin, 1987] highlights that, even if two mechanisms are strategically equivalent, in practice the fact that they are strategyproof may be easier or more difficult to understand depending on how they are implemented. The ascending clock auction is strategically equivalent to the Vickrey auction (S. Li, 2017). In this auction, there is a price displayed to all bidders that steadily increases over a series of rounds, and in each round each agent can decide whether she wishes to drop out of the auction or continue. As soon as the second-to-last agent has dropped out, the final remaining bidder is declared the winner of the auction and is charged the final price displayed. Agents are more likely to act as strategyproofness predicts (that is, play their dominant strategy of reporting their true valuation for the item to the auction) in the ascending clock auction than in the Vickrey auction. Therefore, while strategyproofness may be appealing from a theoretical standpoint, it relies on the assumption that the players of the game act with perfect rationality. This assumes each player can correctly reason about the behaviour and beliefs of the other players, then formulate and solve an optimisation problem to identify an optimal strategy to play, given the strategies of the other players. This is an unrealistic expectation of how players will act in real-world scenarios, placing too high a cognitive or temporal burden to act with full rationality. As a result, a line of research investigating *bounded rationality* (also called *imperfect rationality*) in mechanism design looks at how to devise mechanisms that achieve incentive compatibility for agents with varying cognitive abilities.

1.1 BOUNDED RATIONALITY

[S. Li, 2017] introduces the notion of “obvious strategyproofness” as a solution concept for agents with bounded rationality. For agents who are cognitively limited, the fact that a mechanism is strategyproof may not be enough to encourage them to act truthfully: they must also *understand* that they will not benefit from attempting to manipulate the outcome. The motivating example is a comparison between two popular mechanisms for auctioning off a single item to a set of bidders, namely the ascending clock auction and the Vickrey auction. As discussed, these mechanisms are strategically equivalent, but agents are more likely to play their dominant strategy

in the former than in the latter. A mechanism is strategyproof if each agent weakly maximises her utility when revealing her true private information to the mechanism, meaning truthtelling is a dominant strategy for every fixed profile of actions of the other players. Obvious strategyproofness is a strengthening of strategyproofness that requires truthtelling to be an *obviously* dominant strategy: informally, this means that for every point in the mechanism’s implementation where a given agent is asked to respond to a query, the worst outcome from telling the truth is no worse than the best outcome from any other deviating strategy. In fact, obvious strategyproofness is a stronger notion than even group strategyproofness, which stipulates that there exists no possible coalition of agents that can coordinate their strategies such that each agent in the coalition improves their utility.

The notion of obvious strategyproofness is a pessimistic one, stating that agents with zero contingent reasoning skills will always attempt to manipulate the mechanism unless it is obvious for them to tell the truth. Empirical evidence shows in fact that people facing mechanisms that are not strategyproof might fail to understand that lying is beneficial. Examples include mechanisms both with and without monetary transfers. The Deferred Acceptance Algorithm, used in many countries to allocate doctors to hospitals, is known to perform well in practice and produce stable outcomes, even though hospitals could lie and hire better candidates (Roth, 1991). Uniform-price auctions are a natural generalisation of second-price auctions to the multi-unit case: with a supply of k , they charge the winners the $(k + 1)$ -th highest bid for each unit received. Variants of uniform-price auctions are used by search engines to charge for sponsored ad slots and governments to sell bonds. These auctions are famously not strategyproof and yet there is little evidence that bidders cheat in practice (Kastl, 2011). On the other hand, there are manipulable mechanisms where people can easily determine how to lie profitably: the Boston Mechanism for school choice is known to be manipulable (Pathak and Sönmez, 2008) and empirical data shows evidence of misreports (Dur, Hammond, and Morrill, 2018). Similarly, there is evidence of bid shading in pay-as-bid multi-unit auctions (Hortacısu and McAdams, 2010).

Conversely to obvious strategyproofness, one can interpret the bounded rationality of the agents much more optimistically and assume that they will tell the truth unless *lying* is obvious. This is the approach taken by [Troyan and Morrill, 2020], who introduce the notion of non-obvious manipulability to explore why certain manipulable mechanisms see agents misreporting their type less frequently than others. While two mechanisms may be ma-

nipulable in the sense that truthtelling may not be a dominant strategy for every joint action profile of the other agents, the authors argue that the extent to which manipulations are able to be recognised as beneficial can differ, meaning dishonest behaviour may be attempted less frequently in one mechanism than another. Under their relaxation, simply referred to as NOM, a mechanism is therefore allowed to be vulnerable to certain types of manipulations which are harder to identify by agents with bounded rationality. Instead of requiring the truthful action for each agent i to be a dominant strategy at each possible interaction with the mechanism, under non-obvious manipulability we need only to compare best-case utilities for truthful outcomes (i.e., the maximum utility, taken over the reports of the other agents, for i when she is truthful) to best-case utilities for dishonest outcomes, and worst-case utilities for truthful outcomes to worst-case utilities for dishonest outcomes. This is justified under the assumption that cognitively-limited agents can only identify this type of *obvious* manipulation where the benefits occur at the extremes of her utility function.

The reasoning behind NOM being defined on the extremes of an agent’s utility function is that, while an agent may not be fully informed on (or fully understand) how a given mechanism is defined, they may still be able to determine the set of outcomes of the mechanism that may arise for each given strategy. That is, while she does not know the full specification of the mechanism, she does know the range of possible outcomes that will occur conditioned on her own report. The justification for obvious manipulations to be defined in this way is strengthened by their main characterisation result that shows that these manipulations are exactly the ones which are able to be recognised by agents with these cognitive limitations. In their words, “even an agent who does not fully know how the mechanism is defined can deduce that the mechanism can be manipulated.” Whether or not this is the “correct” notion of bounded rationality in mechanism design is not the focus of this thesis.

While [Trojan and Morrill, 2020] certainly provide some convincing justification about why this particular notion of obviousness is applicable, it does seem to be a rather brittle property. Regardless, there is plenty of scope to extend this definition to other environments, such as randomised and Bayesian settings, to parameterise the degree of the agents’ rationality in a similar vein to [Ferraioli and Ventre, 2024], or even to suppose that only a portion of the population is boundedly rational, as in the literature on so-called *level- k* reasoning (R. Nagel, 1995; Stahl and Wilson, 1995) and the cognitive hierarchy (Camerer, Ho, and Chong, 2004) models of imperfect rationality. We refer the reader to [de Clippel, Saran, and Serrano, 2019;

de Clippel, Saran, and Serrano, 2023] for level- k reasoning in mechanism design, and [Wright and Leyton-Brown, 2017] for a review of various other models of boundedly-rational human behaviour in unrepeated games.

This thesis studies mechanism design for agents with the type of bounded rationality considered in [Troyan and Morrill, 2020] and explores how imposing a relaxed notion of incentive compatibility in the form of NOM can help us to avoid impossibilities faced by stronger models of bounded rationality elsewhere.

1.2 CONTRIBUTION OF THIS THESIS

This thesis contributes to the study of obvious manipulations in mechanism design. It studies Troyan and Morrill’s notion of non-obvious manipulability in several contexts. As a starting point, in Chapter 4 we study the notion of implementability. Roughly speaking, suppose we have a social choice function (voting rule, allocation function, etc.), which aggregates a profile of the agents’ preferences and returns a desirable outcome according to these preferences (for example, with respect to maximising the sum of agent utilities, minimising the maximum incurred cost, etc.) around which we want to define an incentive compatible mechanism. If we can define a payment rule that, when taken in conjunction with the social choice function, results in a mechanism that has no obvious manipulations, then the social choice function is said to be *implementable* as a NOM mechanism. We firstly explore the conditions under which certain social choice functions are implementable as NOM mechanisms using monetary transfers. This uses the technique of cycle monotonicity (Rochet, 1987), however in order to do so we first introduce a novel concept known as “(profile) labellings” which allow us to designate which bid profiles of the mechanism lead to each agent’s best- and worst-case outcomes. We then build a graph based on these given labellings in order to abstract the constraints imposed by the eventual NOM mechanism and characterise when a social choice function is able to be implemented as a NOM mechanism by considering the existence of negative weight cycles in the graph. If no such cycles exist, then there exists a NOM payment scheme based on shortest paths in the graph, and if a NOM payment scheme exists, then we show that no negative cycles must have been created in the graph. We then focus on single-parameter domains, where the agents’ private information can be represented by a single number, and give a more precise characterisation on the class of social choice functions implementable as NOM mechanisms. We find that a social choice function can be implemented as a NOM mechanism if and only if it

is “overlapping”, a property roughly describing the shape of the function that means it is possible to find a monotone restriction of bid profiles to consider with the labelling. This is analogous to the monotonicity characterisation for truthfully-implementable mechanisms. This also allows us to import known results from the literature on the exact form that payments must take in this case.

In Chapter 5 we apply these characterisations to our first setting of bilateral trade, which involves a single buyer and seller. Investigating an impossibility result due to [Trojan and Morrill, 2020] which states that every efficient and individually rational mechanism for bilateral trade that is weakly budget balanced is obviously manipulable, we relax the requirements to approximate weak budget balance and find that for a simple class of mechanisms we introduce called “single-line” mechanisms, the impossibility persists. Casting our attention to more general labelling schemes (and hence mechanisms), we find that this incompatibility is completely resolved by only considering the worst-case aspect of NOM. The same cannot be said for the best-case aspect of NOM, and we show that any best-case labelling of a mechanism breaks one of the other desired properties.

We then return to the question of implementability. In Trojan and Morrill’s original paper they introduce NOM for direct-revelation mechanisms, where each agent is assumed to report their type directly to the mechanism as in, for example, a single-item sealed-bid auction. In Chapter 6 we extend the notion of profile labellings to the more general class of indirect (or extensive-form) mechanisms. Instead of having each agent report her type directly to the mechanism, indirect mechanisms query agents about their private information over a number of rounds. After providing a definition of NOM in extensive-form mechanisms we then incorporate the implementation tree in our definition of labellings to ensure that there are no obvious manipulations available to any agent at *any* point at which they are queried by the mechanism. We show a similar characterisation on the class of social choice functions implementable by an indirect NOM mechanism holds as in Chapter 4, and use this characterisation directly in showing a partial revelation principle for NOM mechanisms. We show that for both single-parameter domains and mechanisms between two agents where one has a “two-type” domain, the class of social choice functions implementable by a direct mechanism is exactly the class of social choice functions implementable by an indirect one. We do note, however, that payments may differ between two implementations.

Finally, in Chapter 7 we turn our attention to the general settings of (goods) auctions and procurement auctions. In goods auctions we wish to

extract revenue from the agents, while in procurement auctions we allocate chores to the agents and instead must reimburse them for doing so. We study investigate the performance of NOM mechanisms along the metrics of revenue maximisation and frugality, respectively. A straightforward reading of the constraints specified by NOM gives us two properties that lead very simply to an incentive compatible mechanism, and we refer to these as “golden ticket” and “wooden spoon” profiles. The former states that for each agent i , regardless of her bid, there should be a bid profile of the other agents such that agent i is selected by the allocation function, and either wins for free (as in goods auctions) or is selected and is paid some maximum value h . The latter states in both cases that each agent should also lose the auction for some joint bid profile of the others, for every possible bid she could submit. By imposing fairly weak requirements on the approximation ratio of the allocation function against the optimal social welfare, and the guarantees of the payment function in terms of extracting revenue or frugal reimbursement, we show that these conditions are also necessary.

Overall this thesis provides several useful results towards the study of obvious manipulations in mechanism design, extending the work initiated by [Troyan and Morrill, 2020] and continued in [Aziz and Lam, 2021; Josué Ortega and Segal-Halevi, 2022; Psomas and Verma, 2022; Arribillaga and Risma, 2023; Arribillaga and Bonifacio, 2024]. Our work is the first to provide a systematic study of mechanism design using monetary transfers to achieve incentive compatibility under this new solution concept, and extends the study of mechanism design for agents with bounded rationality. More broadly speaking our work fits into the highly applicable field of simplicity in mechanism design, which aims to take into account the practical issues with using economic mechanisms for real-world agents. This work touches several different areas in the field of mechanism design, from the highly general setting of implementability (in both direct and indirect mechanisms), to the specific setups of bilateral trade and several auction and procurement auction setups of Chapter 7. We use and extend the existing techniques from the literature in these settings and adapt them to handle the unique requirements imposed by NOM. As a result we find that while NOM is indeed a significant relaxation on incentive compatibility compared to the highly popular strategyproofness, the additional flexibility that designing such mechanisms provides allows us to achieve several positive results where other notions are severely restrictive. Especially framed in the context of its twin solution concept for agents with zero contingent reasoning, obvious strategyproofness, we find that NOM can be a more optimistic alternative when designing mechanisms for agents with cognitive limitations. In many

practical scenarios the constraints imposed by OSP are simply too strong, and as a result they too heavily restrict the class of incentive compatible mechanisms available. At the same time we show that NOM is not too weak a solution concept, and while it accounts for the cognitive limitations of the agents we show that it is not simply a free pass to achieve any property we want. NOM is not a trivial notion to impose on the mechanism, and while the choice is ultimately up to the mechanism designer, this class of mechanisms deserves to be better understood in order to guide these decisions. With this work we hope to spawn interesting and relevant avenues for further research in bounded rationality in mechanism design.

LITERATURE REVIEW

Obvious strategyproofness was introduced in [S. Li, 2017] as a solution concept for agents with zero contingent reasoning skills. Loosely, a mechanism is said to be obviously strategyproof (OSP) if for each agent, the worst outcome from reporting her type truthfully to the mechanism is no worse than the best outcome from submitting any other misreport of her type. If this is the case, then truthtelling is said to be obviously dominant over misreporting. The author provides two characterisation theorems for obviously strategyproof mechanisms. The first relates to the cognitive abilities of the agents and states that a mechanism is OSP if and only if there exist no dominant strategies that can be identified by agents even when they have zero contingent reasoning skills. The second relates to the ability of the mechanism to commit to every detail of the mechanism and states that a social choice function is implementable as an OSP mechanism if and only if it can be supported by so-called “bilateral commitments”. Loosely, a bilateral commitment between the planner and an agent restricts the planner’s strategy set to only those that are measurable by the agent, meaning it is possible for the agent to tell whether the planner has committed to this strategy, or if they have deviated. Applying these characterisation theorems, the author then shows analogues to canonical results relating to implementing social choice functions as strategyproof mechanisms. Studying binary allocation problems, where a function is truthfully implementable if and only if the function is monotone in each agent’s type (Spence, 1974; Mirrlees, 1971; Myerson, 1981), and the payments guaranteeing incentive compatibility are essentially unique (Green and Laffont, 1977; Holmström, 1979), the author then shows that the OSP implementation of any given OSP-implementable social choice function is essentially a personal clock auction. The reverse direction also holds, giving a full characterisation for OSP mechanisms in this setting. Finally, the author shows that for at least three agents there is no OSP implementation for the Top Trading Cycles (TTC) algorithm (Shapley

and Scarf, 1974), which is used to assign indivisible objects amongst agents without monetary transfers.

2.1 OBVIOUS MANIPULABILITY IN MECHANISM DESIGN

Obvious strategyproofness is a rather restrictive notion of incentive compatibility and significantly strengthens the constraints required for a mechanism to be considered robust against agent manipulations. Interpreting the bounded rationality of the agents more optimistically, [Troyan and Morrill, 2020] introduce non-obvious manipulability for direct-revelation mechanisms and provide a characterisation, similarly to [S. Li, 2017], for such mechanisms as those whose profitable deviations may be recognised by a cognitively limited agent that is unable to engage in contingent reasoning. They apply this to the settings of school choice, two-sided matching, auctions, and bilateral trade and use their framework to classify mechanisms in these settings as either obviously manipulable or not obviously manipulable. For school choice they prove that no mechanism that is “stable-dominating”, meaning it always produces a matching that is either stable (meaning there exist no blocking pairs which would see a student and school bilaterally switching from their matched partners to each other) or Pareto dominates a stable matching, is obviously manipulable. For two-sided matching they show that any stable mechanism is individually rational, Pareto efficient, and not obviously manipulable. For auctions, they show that the first-price auction is obviously manipulable, as is its generalisation to multiple items, the so-called “pay-as-bid” auction, in which agents submit a vector of bids for the k units on offer and pays the sum of their winning bids. On the other hand they show that the $(k + 1)$ -th price auction, a generalisation of the second-price auction in which the k units are allocated to the k highest bids and the price of each unit is equal to the $(k + 1)$ -th highest bid, is not obviously manipulable. Finally, in the setting of bilateral trade the authors show that the double auction mechanism, in which the trade price is equal to the midpoint between the buyer’s and seller’s bid (whenever the former exceeds the latter), is obviously manipulable, and furthermore that every efficient, individually rational, budget balanced mechanism for bilateral trade is obviously manipulable.

The study of obvious manipulations as defined by [Troyan and Morrill, 2020] for strategic agents with bounded rationality is relatively new and seeing increasing interest in the algorithmic game theory literature. Broadly speaking, in addition to the settings covered in [Troyan and Morrill, 2020], it has been further studied in the context of voting, fair division and cake-

cutting, and matching. In the following sections we present recent results from the literature on the study of NOM in these settings.

2.1.1 IN VOTING

The Gibbard-Satterthwaite Theorem (Gibbard, 1973; Satterthwaite, 1975) states that every strategyproof voting mechanism for choosing between at least three candidates is dictatorial, meaning that a sole voter is able to choose the outcome. Applying non-obvious manipulability’s relaxed notion of incentive compatibility to the study of voting rules for the first time, [Aziz and Lam, 2021] study conditions under which voting rules are NOM. In this setting agents are equipped with preference relations, as opposed to cardinal utility functions, over the set of candidates. They firstly identify several classes of voting rules satisfying NOM as well as conditions under which certain rules are obviously manipulable and not obviously manipulable. For example, a voter is said to have veto power if she can report her preferences to the mechanism that guarantees a certain candidate will not be selected. They show that all obviously manipulable voting rules admit a non-dictatorial voter with veto power. As a straightforward consequence they show that no voting rule that always return an candidate that is the most preferred candidate for at least $n - 1$ voters (if it exists), where $n \geq 3$, is obviously manipulable.¹ Examples of such voting rules are *plurality with runoff* (Niou, 2001) and the *single transferable vote (STV)* (Tideman, 1995) rules, and these are therefore not obviously manipulable. Similarly, any *majoritarian* or *Condorcet extension* rule is not obviously manipulable. Interestingly, these are known (Peters, 2017) to suffer from an extreme failure of strategyproofness known as the “preference reversal paradox” (Sanver and Zwicker, 2009), where voters can manipulate the voting rule by completely reversing her true preference ranking. Turning their attention to positional scoring rules, where voters assign points to each candidate and the score of an candidate is simply the sum of points assigned to it, they provide a general sufficient condition for such rules to be obviously manipulable based only on the two highest scoring candidates relative to the number of agents. The class of k -approval rules for $0 < k < m$, where m is the number of candidates, is a subset of the class of positional scoring rules that allows voters to approve their k most-preferred candidates, or disapprove of their $m - k$ least-preferred candidates. They provide a characterisation on when k -approval is obviously manipulable that depends on the number of candi-

¹Such voting rules are said to be *almost-unanimous*, and note that if a voting rule is almost-unanimous then it is unanimous.

dates relative to both the number of agents and the number approval votes k . From this it follows that the *plurality rule* is not obviously manipulable: under plurality voting, each voter submits their most preferred candidate and the winner is simply the candidate with the largest number of votes, and it is straightforward to see that this is simply k -approval with $k = 1$.

Restricting attention to *strict* positional scoring rules, where each candidate must be assigned a unique number of points by the voters, they show that while such rules can admit voters with veto power and be obviously manipulable, there are strict positional scoring rules that admit a voter with veto power but are not obviously manipulable. One example of a voting rule in this class is the Dowdall rule (Reilly, 2002). By then considering the two halves of NOM in isolation they show that best-case obvious manipulations for a positional scoring rule exist if and only if the first $k \geq 2$ of the scoring vector are the same and the number of agents is at most $\frac{m-2}{m-k}$, and highlight a fundamental connection between the existence of obvious manipulations for best and worst cases. Namely when $m, n \geq 3$ then if a voting rule admits a best-case obvious manipulation then it also admits a worst-case obvious manipulation.

Manipulations are unavoidable for all “acceptable” voting rules by the Gibbard-Satterthwaite Theorem. However even if a voting rule is manipulable, it may be computationally intractable to compute manipulations. Under reasonable practical assumptions this suggests that agents may not submit a false report of their vote in a manipulable voting rule. For a range of voting rules including plurality voting and the Borda count, computing manipulations is computationally easy (Bartholdi, Tovey, and Trick, 1989). By contrast, computing a manipulation in STV is NP-complete (Bartholdi and Orlin, 1991). Given that several well-known rules admit obvious manipulations, [Aziz and Lam, 2021] consider the problem of computing an obvious manipulation in certain voting rules. In particular, given a set of candidates, a tie-breaking rule, and a voter with a given (true) preference over candidates, the goal is to find another preference over candidates resulting in either a greater best-case utility or greater worst-case utility. They provide a polynomial-time reduction to the following problem (Xia et al., 2009; Zuckerman, Procaccia, and Rosenschein, 2009), known as “Constructive Coalitional Unweighted Matching (CCUM)”: given a voting rule, a set of outcomes, a “distinguished candidate”, and a set of voters that have already cast their vote, does there exist a way for the remaining voters to cast their votes such that the distinguished candidate wins? Thus, if there exists a polynomial-time algorithm for computing CCUM then there exists a polynomial-time algorithm for computing obvious manipulations in posi-

tional scoring rules, and as a corollary the problem of computing obvious manipulations for k -approval can be done in polynomial time. Finally, they provide experimental results that firstly show that as the number of voters grows the proportion of preference orderings admitting an obvious manipulation tends to zero, and moreover that significantly fewer orderings admit a best-case obvious manipulation than a worst-case one. Studying the effect of k on the obvious manipulability of k -approval voting rules their experiments show that as the number of outcomes grows, so too does the number of preferences admitting an obvious manipulation.

[Arribillaga and Bonifacio, 2024] also look at obvious manipulations in the context of voting rules, and restrict attention to “tops-only” rules, which only consider each agent’s top preferences when selecting an alternative. Tops-only voting rules simple to state and hence have seen much attention in the literature. An issue with other relaxations of strategyproofness is that when focusing on tops-only voting rules they become equivalent to strategyproofness (Arribillaga, Bonifacio, and Fernandez, 2022), and are therefore unable to distinguish between different voting rules. This is not an issue faced by looking at the obvious manipulations of a tops-only voting rule, hence the solution concept can provide a more fine-grained distinction between the tops-only voting rules of interest. The main result of the authors is a characterisation of tops-only voting rules that are not obviously manipulable, and relates this class of voting rules once again with the concept of the veto power of any single agent. While [Aziz and Lam, 2021] show that a voting rule having no vetoers is a sufficient condition for it to have no obvious manipulations, the reverse is not true and there are NOM rules that admit possibly many vetoers, one of which is the “status quo” rule, which only selects a different alternative to the one assumed to be already in place if all voters select it as their most-preferred alternative. They therefore introduce a stronger notion of veto power called the *conditional veto property*, which roughly states that all possible alternatives that a given agent i has veto power over can be *strongly vetoed*, meaning that i can only veto an alternative ω by reporting any other alternative as her most preferred (but not when she lists ω as her favourite). Using this they show that a tops-only voting rule admits no obvious manipulations if and only if it satisfies this conditional veto property for each agent.

Restricting attention to subclasses of tops-only voting rules known as “generalised median voter” and “voting-by-committee” rules. For the former setting, assume the alternative set Ω is represented by the finite set of m natural numbers $\{a, a+1, a+2, \dots, b\}$ where $b = a+m-1$. In a median voter scheme for n voters there is a non-decreasing vector $\alpha = (\alpha_1, \dots, \alpha_{n-1})$

and the alternative selected by the scheme is simply the median of the votes collected and the vector α , and a generalised median voter scheme captures all non-anonymous² extensions of median voter schemes. A characterisation due to [Moulin, 1980] says that all strategyproof, tops-only, anonymous voting rules for agents with single-peaked preferences³ are generalised median voter schemes. By loosening the restriction to single-peaked domains to the universal domain of preferences, the authors show that a generalised voting scheme is not obviously manipulable if and only if its associated vector α is such that $\alpha_1 \in \{a, a+1\}$ and $\alpha_{n-1} \in \{b-1, b\}$. For the latter setting, it is now assumed that there is a set of agents N who are all part of some club and wish to select a new set of members $K \subseteq N$ to add to the club. Loosely, a social choice function that uses voting by committees selects a subset of new members if and only if there exists a committee amongst the existing members who all view it as the best alternative over adding any other possible subset. A subclass of voting by committees is known as voting-by-quotas and allocates a quota q_k to each candidate $k \in K$. [Barberà, Sonnenschein, and Zhou, 1991] shows that when agent preferences are separable⁴ then the only strategyproof voting rules are those that use voting-by-committees. A rule uses voting by quotas selects a candidate k (that is, includes it in the set of candidates to be added to the set of members) if and only if it is listed as the most preferred candidate for at least q_k agents. Relaxing the requirement that preferences be separable to the universal domain, [Arribillaga and Bonifacio, 2024] show that any anonymous rule using voting by quotas is NOM if and only if each candidate k 's quota q_k lies between 2 and $n-1$ (inclusive). For non-anonymous rules they show that any non-dictatorial scheme that uses voting by committees is not obviously manipulable if and only if there is no agent with veto power.

2.1.2 IN FAIR DIVISION

As the name suggests, fair division is concerned with fairly allocating a collection of resources amongst a set of divisions and it encompasses many practical settings including, for example, division of inheritance, divorce settle-

²A voting rule is *anonymous* if it does not use the identities of the agents to select an outcome: if f is anonymous then for any permutation of the agents $\pi : N \rightarrow N$ it holds that $f(b_1, \dots, b_n) = f(b_{\pi(1)}, \dots, b_{\pi(n)})$.

³Roughly speaking, an agent i 's preference relation is *single-peaked* if she has a single most preferred alternative and as one moves away from this alternative i 's utility monotonically decreases.

⁴A preference relation $\succ$ is *separable* if, for all subsets of elements, adding a new element to the subset is preferable over the original subset if and only if the new element is preferred to the empty set.

ments, or electronic frequency allocation. The set to be divided can contain heterogeneous or homogeneous, divisible or indivisible items, and the items can be considered “goods” (i.e., desirable), “bads” (i.e., undesirable), or a mix of both. [Josué Ortega and Segal-Halevi, 2022] apply NOM for the first time to the context of indirect mechanisms, in which agents gradually reveal their type to the mechanism over time, studying the obvious manipulability of a class of mechanisms known as Robertson-Webb mechanisms to the fair division setting of cake-cutting. In this context there is a “cake” modelled as the unit interval $[0, 1]$ and each agent has a valuation function v_i that maps subintervals of $[0, 1]$ – otherwise referred to as “pieces of the cake” – to a real number. This value depends on the agent’s type, so given an agent i with type v_i the utility i receives on the piece of cake $I \subseteq [0, 1]$ is denoted $v_i(I)$. Their mechanisms do not use payments, hence the utility agent i with valuation function v_i receives when she receives the cut $I \subseteq [0, 1]$ of the cake is simply $u_i(I; v_i) = v_i(I)$. Valuations are normalised, meaning we have $v_i(\emptyset) = 0$ and $v_i([0, 1]) = 1$ for all valuation functions v_i . Moreover agent valuations are assumed to be additive and the authors consider both “all good” and “all bad” cakes, where agent valuations map each piece of the cake to weakly positive and weakly negative values only. For general indirect mechanisms there are no restrictions on the queries agents can be posed in order to elicit their types. In [Josué Ortega and Segal-Halevi, 2022] the authors focus on a class of mechanisms for cake-cutting known as Robertson-Webb mechanisms. In the Robertson-Webb query model, first described in [Robertson and Webb, 1998] and later formalised in [Woeginger and Sgall, 2007], only two types of queries are allowed, known as “evaluation” queries and “cut” queries. In an evaluation query $\text{eval}(i, I)$ agent i is asked for her valuation $v_i(I)$ on interval $I \subseteq [0, 1]$. In a cut query $\text{cut}(i, x, \alpha)$ agent i is asked for the smallest value $y \geq x$ such that $v_i([x, y]) \geq \alpha$; in other words she is asked for the smallest cut of the cake starting at x that she values at least α .

Strategyproofness is a restrictive notion in cake-cutting. In the case of randomised mechanisms, a trivial approach is to simply select an agent uniformly at random and award her the entire cake. This is strategyproof since no queries are even posed to the agents, and (ex-ante) proportional since each agent gets utility $1/n$ in expectation.⁵ Research efforts therefore

⁵An allocation of a good among n agents is said to be *proportional* if the utility received by each agent according to her allocation (and her valuation function) is at least a $1/n$ fraction of her valuation for the entire good. Since utilities are normalised then for each agent i with allocation A_i we have $v_i(A_i) \geq 1/n$. An allocation is said to be *strongly proportional* (or *super proportional*) if this inequality is strict.

consider mechanisms that are ex-post fair with several positive results in the literature, including super proportional mechanisms that are truthful in expectation (Mossel and Tamuz, 2010) and the Constrained Serial Dictatorship of [Aziz and Ye, 2014] that is truthful in expectation, unanimous, and satisfies a stronger notion of proportionality known as robust proportionality.⁶ Deterministic mechanisms present greater challenges when strategyproofness is required, even for agents with piecewise-constant valuation functions. In a discrete setting, [Brânzei and Miltersen, 2015] show that randomised query-based models can achieve truthfulness in expectation and approximate proportionality, however in any deterministic mechanism there is always an agent who receives nothing. For two agents this implies the allocation is dictatorial, meaning one agent alone can decide the outcome and the mechanism will completely disregard the other agent's preferences. Other impossibilities for deterministic strategyproof mechanisms show that no mechanism can achieve envy-freeness (a weaker notion of fairness than proportionality) using a bounded number of queries (Kurokawa, Lai, and Procaccia, 2013), and that no mechanism can simultaneously achieve both proportionality and Pareto optimality, or both robust proportionality and non-wastefulness.⁷

A significant relaxation of strategyproofness is the concept of *maximin strategyproofness* (Steven J. Brams, Michael A. Jones, Klamler, et al., 2006; Steven J. Brams, Michael A. Jones, and Klamler, 2007; Steven J. Brams, Michael A. Jones, and Klamler, 2008), which stipulates that agents will only lie if their utility in the worst case from doing so is strictly greater than their utility in the worst case of telling the truth. Under maximin strategyproofness an agent strives to maximise her minimum utility, and does not require beliefs about the valuations of the other agents. While many cake-cutting mechanisms satisfy maximin strategyproofness, hence the authors strengthen the solution concept to non-obvious manipulability and find that these same mechanisms now fail the incentive compatibility requirements. Note that maximin strategyproofness is simply the worst-case component of NOM and is therefore a strictly weaker notion of incentive compatibility.

⁶An allocation is *unanimous* if, whenever an agent's most preferred portion of the cake of length $1/n$ is disjoint from any other agent's, then the agent in question gets this portion of the cake. An allocation is *robustly proportional* if, roughly speaking, the allocation remains proportional for all valuation functions that are ordinally equivalent to the one she reports, i.e., valuation functions that preserve the ordinal preferences over each portion of the cake.

⁷*Non-wastefulness* is a weaker notion of fairness than Pareto optimality stating that pieces of cake are only allocated to agents who want them.

The first contribution of the authors is to show that a number of Robertson-Webb mechanisms are obviously manipulable. They first consider mechanisms for two agents. In the “cut-and-choose” mechanism for two players the first player is asked to cut the cake, then the second chooses which slice to take. In the “cut-middle” mechanism for two players each player is asked to cut the cake into two halves she values equally, and a single cut is made into the cake at the midpoint of their responses. Each player is then given the slice of the cake containing her cut. They also consider mechanisms for more than two agents. In the Banach-Knaster, or “last diminisher”, mechanism (Steinhaus, 1948) agents are first assigned a fixed order and agents are then asked in turn a cut query starting from 0 that would give them a value of $1/n$. The piece is awarded to the agent with the smallest cut, and the process is repeated on the remaining agents and portion of cake. In the Dubins-Spanier, or “moving knife”, mechanism (Dubins and Spanier, 1961) a knife is moved continuously over the cake and the first agent to shout “stop” (equivalent to responding to a cut query) is allocated the piece of cake to the left, and the process is repeated for the rest of the cake.

The authors show that all of the above mechanisms are obviously manipulable. For both the cut-and-choose and cut-middle mechanisms they provide best-case obvious manipulations that break incentive compatibility. Since the two-player variant of cut-and-choose is obviously manipulable then so too is its generalisation to n players (Kuhn, 1967). In order to show obvious manipulability of the Dubins-Spanier mechanism as per their definition of obvious manipulability in extensive-form mechanisms, it must first be adapted to the discrete setting. The authors provide two such adaptations. In the first variant, the mechanism runs over a discrete set of time steps and in each time step t each agent is asked simultaneously to cut a portion of the cake that remains that would yield an equal portion of cake (according to their own valuation function) to the winner of the slice. The slice is awarded to the agent k who responded with the smallest cut, and the process is repeated on the remaining cake $[x_k, 1]$ and the remaining agents. In the second variant, the knife moves from the start of the cake 0 to the end of the cake 1 in discrete steps of size δ . After each increment, each agent is asked whether their value for the current piece of cake $[0, x]$ is at least $1/n$, where x is the knife’s current location. If this is the case for some agent then it is awarded the slice (with ties being broken according to a fixed ordering) and the process is repeated for the rest of the cake and the remaining agents.

Both of these variants of the Dubins-Spanier mechanism are obviously manipulable. The first variant is shown to be obviously manipulable for 3

agents under worst-case obvious manipulations, while the second variant is vulnerable to best-case obvious manipulations even when $n = 2$. Likewise the Banach-Knaster mechanism is obviously manipulable for 2 agents, and is susceptible to obvious manipulations in both the best and worst case.

They then provide a third variant of the Dubins-Spanier mechanism, called the “leftmost-leaves” mechanism (Bhardwaj, Kumar, and Josue Ortega, 2020; Kyropoulou, Josué Ortega, and Segal-Halevi, 2022), which is *not* obviously manipulable. This mechanism is similar to their first discretised variant of the Dubins-Spanier mechanism in that in each time period t each agent i without a piece of cake is asked a cut query starting from the leftmost uncut piece of cake x^{t-1} , however they differ in the value parameter α in the query. The first variant has $\alpha = 1/n$ regardless of the time step of the mechanism, meaning agents are always asked to produce a cut that yields a proportional share of the remaining cake with respect to the entire cake and population of agents. In leftmost-leaves, at step t each agent is asked to cut the cake in order to share the remaining cake proportionally *amongst the remaining players*. In time step t the leftmost end of the cake that has not yet been assigned is denoted x^{t-1} , and each agent i that has not yet been assigned a slice is asked the query $\text{cut}(i, x^{t-1}, \frac{v_i([x^{t-1}, 1])}{n-t+1})$. Interestingly, this shows that mechanisms which are outcome-equivalent (that is, return the same outcome for every input to the mechanism) may differ with respect to their (obvious) incentive properties as a result of the mechanisms’ implementations. A similar result is covered in [S. Li, 2017] for obvious strategyproofness, where it is shown that the ascending clock auction is obviously strategyproof but the second-price auction is not.

The authors note that leftmost-leaves is not the only proportional NOM mechanism for cake-cutting, and provide an adaptation of another proportional mechanism due to [Even and Paz, 1984] that is NOM. These mechanisms are NOM for essentially the same reasons: in each period when telling the truth it is possible for each agent to both win the entire remaining cake, and when deviating from telling the truth it is possible to be allocated a slice that is worth less than $1/n$, thus breaking proportionality. A characterisation of proportional NOM mechanisms remains an open problem.

Leftmost-leaves satisfies neither envy-freeness, where each agent prefers her own slice (according to her own utility function) to any other agent’s, nor Pareto optimality, where the only way to improve an agent’s allocation is to make another agent worse off. As yet there is no Robertson-Webb mechanism for cake-cutting that is both NOM and either envy-free or Pareto optimal. The authors highlight that while the Selfridge-Conway mechanism for three agents (Steven J. Brams and Taylor, 1996) is envy-free it is obviously

manipulable. Casting their attention towards direct-revelation mechanisms, interestingly they find that best-case and worst-case NOM mechanisms are guaranteed by Pareto optimality and proportionality, respectively. That is, if a direct mechanism always returns a Pareto optimal allocation then it satisfies BNOM, and if it always returns a proportional allocation then it satisfies WNOM. As a result, the “Nash-optimal mechanism” that, given a profile of valuation functions returns an allocation maximising their product, is NOM, since the allocations it returns are known to be envy-free (and therefore proportional) and Pareto optimal (Segal-Halevi and Sziklai, 2019).

Finally, the authors provide a simple adaptation of leftmost-leaves which, in each time step, selects a cut of *maximum* size and show that this mechanism is NOM for “all bad” cakes. Currently there are no known NOM mechanisms for mixed cakes, with the authors conjecturing that no Robertson-Webb mechanism for cake-cutting can always produce a connected⁸ and proportional allocation without being obviously manipulable.

Mechanisms for cake cutting consider how to fairly divide a single divisible resource (i.e., the cake) amongst a set of strategic agents. Looking towards the fair division of multiple indivisible items, [Psomas and Verma, 2022] study NOM mechanisms for fairly allocating a set of indivisible goods among agents with additive valuation functions. Here strategyproofness presents similar impossibilities: even for two agents a serial dictatorship is the unique rule achieving truthfulness and Pareto optimality (Klaus and Miyagawa, 2002), and although EF1 and Pareto optimality are compatible in general due to [Caragiannis et al., 2019] it is not possible to achieve with any deterministic truthful mechanism, even for additive agents (Amanatidis, Birmpas, et al., 2017). Disregarding strategic considerations, an EF1 allocation is known to always exist (Budish, 2010) and can be found by running the sequential “Round Robin” algorithm (Caragiannis et al., 2019), where each agent is asked sequentially to select an item from the set remaining to be assigned. Additionally, the “Probabilistic Serial” algorithm of [Bogomolnaia and Moulin, 2001] satisfies ex-ante envy-freeness. On whether the two properties can be achieved simultaneously, [Freeman, Shah, and Vaish, 2020] show that there always exist ex-ante envy-free distributions over deterministic allocations that satisfy EF1, and provide a polynomial time algorithm in their “Recursive Probabilistic Serial” procedure that achieves such ex-ante envy-free and ex-post EF1 allocations. Building on this, the “Probabilistic Serial Lottery” algorithm of [Aziz, 2020] satisfies both properties without relying on separation oracles and additionally satisfies “ordinal efficiency”

⁸An allocation is “connected” if no agent gets more than one piece of cake, meaning each agent is allocated a single contiguous interval.

(Bogomolnaia and Moulin, 2001; Bogomolnaia and Moulin, 2002), a fairness notion in between ex-post and ex-ante Pareto efficiency that considers an allocation efficient if it is Pareto optimal against *all* cardinal utility profiles compatible with the given profile of ordinal preferences.

In the context of non-obvious manipulability, [Psomas and Verma, 2022] show that the Round-Robin mechanism has no obvious manipulations and therefore that NOM is compatible with EF1. By showing that the Probabilistic Serial Lottery algorithm of [Aziz, 2020] is NOM they highlight a stronger advantage of the solution concept in that it is compatible with the “best of both worlds” fairness criterion of ex-ante envy-freeness and ex-post EF1. They then study the extent to which NOM is compatible with maximising the utilitarian social welfare, egalitarian social welfare, and Nash social welfare.⁹ They present a negative picture for mechanisms that maximise egalitarian social welfare and Nash social welfare, showing that every optimal mechanism is obviously manipulable, even with only 3 agents and 4 items. Likewise, any mechanism optimising the utilitarian social welfare is obviously manipulable for $n = 2$ agents, however for $n \geq 3$ they note that the optimal mechanism, which simply returns the allocation that maximises the social welfare with respect to the reported valuation profile, is NOM when used with a carefully defined tie-breaking rule. Crucially this mechanism is also not dictatorial.

In optimising the Nash social welfare of an allocation, the allocation is automatically Pareto efficient and EF1. Since all mechanisms that maximise the Nash social welfare of an allocation are obviously manipulable, they look to other Pareto efficient NOM mechanisms that return EF1 allocations. To this end they provide a black box reduction from the problem of designing Pareto efficient, EF1, and NOM *mechanisms* to the problem of designing Pareto efficient and EF1 *algorithms*. Thus, given an algorithm that returns Pareto efficient and EF1 allocations, it is simple to design NOM mechanisms for strategic agents which always output Pareto efficient, EF1 allocations when agent valuations are additive.¹⁰ Using the fractionally Pareto efficient and EF1 algorithm of [Murhekar and Garg, 2021] therefore results in a fractionally Pareto efficient NOM mechanism that always outputs EF1 allocations for additive agents, and by additionally imposing the weak requirement that allocations are ex-ante proportional, this mech-

⁹The sum of agent utilities, the minimum agent utility, and the geometric mean of agent utilities, respectively.

¹⁰Additionally, the Pareto efficient, EF1 algorithm is required to output so-called “clean” and “non-wasteful” allocations: the former condition requires each agent has strictly positive valuation for every good assigned to her, while the latter requires that no agent has a positive valuation for any unallocated item.

anism is the best possible.¹¹ Finally, the authors show limitations in what can be achieved by NOM mechanisms: it is impossible to achieve ex-ante proportionality and ex-post Pareto efficiency while ex-post maximising the number of agents with positive utility. As a result they recover the fact that ex-ante proportional allocations that ex-post maximise the Nash social welfare do not exist, and show that ex-ante proportional allocations are impossible to achieve by simply randomising over allocations that are either ex-post Pareto efficient and ex-post egalitarian, or ex-post leximin.

2.1.3 IN MATCHING

[Troyan, 2022] studies obvious manipulability in the assignment problem. In this setting there is a set of agents and a set of objects, and each object has some capacity denoting the maximum number of agents to whom the object can be assigned. The *rank* of an object j with respect to an agent i 's ordinal preference relation $\succ_i$ is the order in which j appears in $\succ_i$, with her most preferred object having rank one, her second-most having rank two, and so on. Troyan focuses specifically on the “rank-minimising” mechanism, which given an ordinal preference profile of the n agents simply returns (a set of) assignments minimising the average rank of the object received by each agent, where for technical reasons related to the extreme-case analysis required by NOM the mechanism is set-valued rather than returning a single allocation of objects. [Featherstone, 2013] shows that no ordinal assignment mechanism can achieve both *rank-efficiency* (a refinement of Pareto efficiency introduced by Featherstone to incorporate the rank distribution of the resulting assignment) and strategyproofness, hence the relaxation of strategyproofness to non-obvious manipulability. The main result of the paper is that the rank-minimising mechanism is not obviously manipulable. As the author points out, this result formalises a point made by [Cerrone, Hermstrüwer, and Kesten, 2024; Josué Ortega and Klein, 2022] stating that there is evidence to suggest that a mechanism being strategyproof can have less sway in actually getting agents to act truthfully than the complexity of the mechanism and its ability to be understood as manipulable.

[Troyan, Delacrétaz, and Kloosterman, 2020] study one-sided matching markets. Their context is assigning students to schools, where each student must submit an ordinal preference profile of reflecting their private preferences, and each school has a publically known ordinal preference relation

¹¹[Freeman, Shah, and Vaish, 2020] show that there exist instances where no randomised algorithm can output an allocation that is always ex-ante proportional, ex-post EF1, and ex-post fractionally Pareto optimal.

over the students in addition to a maximum capacity of students who can be assigned to it. Only the students are strategic. The authors propose a solution to the following incompatibility of fairness and efficiency in this setting: while the Deferred Acceptance algorithm (Gale and Shapley, 1962) produces the most efficient matching among all those which are fair, there may be matchings which Pareto dominate it. Hence the algorithm is not Pareto efficient. They relax the notion of stability to *essential stability*, arguing that the former notion disqualifies many matchings unnecessarily. The main contribution of the paper as it relates to obvious incentive compatibility in mechanism design is as follows. Given that the set of stable matchings forms a lattice, and therefore admits both a student-optimal and a student-pessimal matching, they show that the student-pessimal is the unique essentially stable matching. Taken together with a result of [Troyan and Morrill, 2020] stating that no stable-dominating mechanism admits any obvious manipulations, this implies that no essentially stable matching is obviously manipulable, suggesting that studying the non-strategyproof class of essentially stable Pareto efficient (ESPE) mechanisms may be particularly salient for practical scenarios given that agents may not recognise profitable deviations in manipulable mechanisms.

In [Arribillaga and Risma, 2023] the authors study obvious manipulations in the context of matching mechanisms. In their setting there are two sides to the market, referred to as “doctors” and “hospitals”, and the goal is to compute a many-to-one matching between doctors and hospitals, meaning each doctor can be assigned to at most a single hospital, while hospitals can accept multiple doctors. There is additionally a set of contracts, which describe the working conditions (such as salary, workload, schedules) of the given doctor which are *not* fixed beforehand. It is assumed that the doctors have private preferences over the hospitals and that the hospitals have substitutable preferences over the doctors which are public knowledge, where substitutability roughly means that a hospital gaining a new contract can never make it want to take back a contract it has already rejected. The authors focus on matching rules that find stable allocations, roughly meaning that no pair of agents can unilaterally deviate from an existing contract and sign a new one amongst themselves. Hence the matchings ought to be sustainable over time, assuming agent preferences remain unchanged. [Hatfield and Milgrom, 2005] show that there always exists a stable matching when hospital preferences are substitutable, and that an extreme matching – a unanimously most preferred stable allocation for either the doctors (doctor-optimal) or the hospitals (hospital-optimal) can be found via a generalised version of the Deferred Acceptance algorithm (Gale and Shapley,

1962). [Roth, 1982] shows that in the absence of contracts in the one-to-one matching model the only stable and strategyproof rule is the doctor-optimal one, while in the many-to-one model with substitutable preferences this rule could be manipulable in both the presence (Hatfield and Milgrom, 2005) and absence (Martínez et al., 2004) of contracts. Given that in the presence of substitutable preferences there is no guarantee that a strategyproof mechanism exists (in other words, all matching rules are manipulable), Arribillaga and Risma relax the incentive compatibility requirements to allow only non-obvious manipulations. The first of their main results prove that in the general context of many-to-one matching models with contracts and substitutable preferences for hospitals, the doctor-optimal rule is not obviously manipulable. Hence, while there may be no rules that are robust against all manipulations, there exists a rule that has no obvious manipulations and hence is robust against agents with the bounded rationality assumed by [Trojan and Morrill, 2020]. Meanwhile, for the many-to-one model *without* contracts and with substitutable hospital preferences the hospital-optimal rule is similarly not obviously manipulable. Conversely, the hospital-optimal rule is obviously manipulable for doctors even in the one-to-one matching model with contracts, showing that presence or absence of contracts can significantly influence the strategic considerations for the agents. Finally the authors turn attention towards so-called “quantile-stable” matching rules (Chen et al., 2021), which are effectively a means of interpolating stability between either side of the market,¹² and show that the doctor-optimal is the only quantile-stable matching rule which with no obvious manipulations when doctors are strategic.

2.2 OBVIOUS STRATEGYPROOFNESS IN MECHANISM DESIGN

This thesis finds itself situated in the wider study of bounded rationality in mechanism design. While NOM is a newer solution concept and is seeing growing interest in the research community, significant attention in this regard has been already been paid to the study of OSP mechanisms. We mention here briefly the numerous domains in which the study of OSP mechanisms has been applied.

[Ferraioli and Ventre, 2021; Ferraioli, Meier, et al., 2019b] study OSP mechanisms for scheduling related machines, with the latter showing a tight approximation bound of $\sqrt{n}$ for OSP mechanisms, even when all agents

¹²For some quantile $q \in (0, 1]$ the q -quantile stable mechanism maps the input preference profile to the $\lceil kq \rceil$ -best stable matching for one side of the market, where k denotes the number of stable matchings admitted by the profile.

have a three-value domain. The latter also considers OSP mechanisms for facility location, showing a tight approximation bound of $n - 1$ to the optimum social cost. The authors additionally show that the limitations in the settings of both scheduling related machines and facility location can be somewhat mitigated if one assumes some level of verification power from the mechanism (Ventre, 2014; Fotakis, Krysta, and Ventre, 2018). Also looking at OSP for machine scheduling and facility location, [Ferraioli and Ventre, 2022] introduce an interpolation between strategyproofness and OSP, in which agents can perform contingent reasoning on a subset of the outcomes of the mechanism, showing in these settings that better approximation ratios are possible only when the agents have large contingent reasoning sets. In other words, more rational agents only slightly improves the approximation guarantee. [de Keijzer, Kyropoulou, and Ventre, 2020] apply OSP to the setting of single-minded combinatorial auctions. In a combinatorial auction there is a set M of items and each agent i is allocated a bundle $A_i \subseteq 2^M$. Agents are single-minded, meaning that each agent wants only a particular bundle S_i , and receive zero utility for being allocated any other bundle. The authors consider whether the desired bundles are known or unknown to the mechanism: in the case of the former they show that OSP can be as powerful as strategyproofness, and that one can implement the truthful $\sqrt{m}$ -approximation algorithm (Lehmann, O’Callaghan, and Shoham, 2002) as an OSP mechanism. This is essentially optimal unless $\text{NP} = \text{ZPP}$ (Halldórsson, 2000). When declaration domains have two values they prove a tight approximation bound of 2, and then present a randomised mechanism that is universally OSP and achieves a finite approximation of the optimal social welfare when agents have finite domains. When the desired bundles are unknown then the authors provide an OSP mechanism whose approximation ratio is equal to the size of the largest desired set, applying OSP for the first time to multidimensional domains. In [Ferraioli, Meier, et al., 2019a] the authors study OSP in the context of set system problems, in which there is some ground set E and the feasible set of outcomes is simply some subset of 2^E . They characterise the set of instances which admit optimal OSP mechanisms, using the ascending-descending auction paradigm first introduced in [Ferraioli, Meier, et al., 2019b]. Additionally, the set of conditions required for an instance to admit an OSP mechanism can be verified in polynomial time, yielding the first results of automated OSP mechanism design. Studying similarly OSP from a computational perspective, [Golowich and S. Li, 2021] investigate the complexity of deciding whether a given social choice function admits a strategyproof, or obviously strategyproof, mechanism. For both strategyproofness and OSP they find that when the function

is specified as a table then the question is decidable in polynomial time. If the functions are represented as Boolean circuits both questions become coNP -complete. The connection between OSP and the ascending-descending auction approach of [Ferraioli, Meier, et al., 2019b; Ferraioli, Meier, et al., 2019a] is further developed in [Ferraioli, Penna, and Ventre, 2021], in which the authors show (roughly) that for binary allocation problems, OSP can be achieved if and only if the algorithm uses a paradigm known as “two-way greedy”. These algorithms use two functions to rank the bids of the players and iteratively include and exclude them from the solution in a greedy manner until one feasible solution remains. They then give several bounds on the approximation guarantee of OSP mechanisms, showing that the approximation ratio is unbounded when minimising social cost in set systems, and is $\Omega(\sqrt{n})$ when maximising social welfare in set systems. They also provide an upper bound of $\sqrt{n-1}$ for so-called “asymmetric knapsack auctions” when bidders have three-value domains. Further tight approximation bounds on OSP mechanisms for related machine scheduling and set systems are given in [Ferraioli, Meier, et al., 2023].

[Pycia and Troyan, 2019] study OSP mechanisms without transfers and provide a characterisation roughly stating that a mechanism without transfers is OSP if and only if it is a so-called “millipede game”. They introduce a strengthening of OSP known as *strong obvious strategyproofness (SOSP)* and use this to characterise the Random Priority mechanism (see, e.g., [Bogomolnaia and Moulin, 2001]) as the unique mechanism that is efficient, fair, and simple to play, and thus providing an explanation for its widespread practical use compared to other OSP (but not SOSP) mechanisms based on more complex millipede games. [Troyan, 2019] also explores reasons for which strategyproof mechanisms may not be used in practice, and they formalise the intuition that TTC is difficult to understand by showing that, while it can be implemented as an OSP (not just strategyproof) mechanism, the conditions under which it can be are unlikely to hold in practical scenarios. This “weak acyclicity” condition roughly means that TTC becomes equivalent to a “dual dictatorship”, where there exist at most two agents who can entirely dictate the outcome by cooperating amongst themselves.

[Bade and Gonczarowski, 2017] present a “gradual revelation principle” for OSP mechanisms. In a gradual revelation mechanism, agents gradually reveal more about their private information, and the authors then show that any social choice function that is implementable by an OSP mechanism is also implementable by an OSP gradual revelation mechanism. For the class of single-peaked preferences, they also characterise the set of unanimous social choice functions implementable by an OSP mechanism as “dictator-

ships with safeguards against extremism”, in which a dictator may choose the outcome but the other agents can prevent an outcome that is too extreme. [Mackenzie, 2020] also considers a form of revelation principle for OSP mechanisms, presenting an algorithm that converts any probabilistic OSP implementation to a so-called “randomised round table implementation”, in which agents take turns making public announcements about their private information. In each step, the agent making the announcement declares a set which contains her type, selecting this set to declare from a partition of the set she has already revealed. Therefore, in these circumstances the mechanism designer may safely restrict attention to these randomised round table mechanisms. [Arribillaga, Massó, and Neme, 2020] also consider OSP mechanisms in the domain of single-peaked preferences and show that a generalised median voter schemes (of the previous section) is obviously strategy-proof if and only if it satisfies the increasing intersection property. They additionally provide an algorithm that, for each generalised median voter scheme that satisfies the property, defines an extensive-form game that implements the social choice function as an OSP mechanism.

OSP is of course an even more restrictive notion than strategyproofness. In addition to the negative results and bounds on the performance of OSP mechanisms proved in the works above, [Bade and Gonczarowski, 2017] show that for $m \geq 2$ items, no welfare-maximising social choice function in which losers are charged nothing is implementable by an OSP mechanism, even for additive valuation functions. Focusing on approximate welfare maximisation in combinatorial auctions, [Ron, 2024] shows that no individually rational OSP mechanism that makes no positive transfers can approximate the social welfare better than $\min\{m, n\}$ for unit-demand bidders. As a consequence, this impossibility holds for any class of valuation functions encompassing unit-demand, including submodular, subadditive, and arbitrary monotone valuations. They show that the same bound on welfare of $\min\{m, n\}$ also holds for additive valuations. On a positive note, there exists a welfare maximising OSP mechanism for additive bidders with three-value domains (i.e., each agent’s valuation for an item is in the set $\{0, \ell, h\}$). Switching attention to unknown single-minded bidders, it is again impossible for any individually rational OSP mechanism with no positive transfers to approximate welfare better than $\{m, n\}$. Finally, in a multi-unit auction there are m identical items and each agent’s valuation function maps quantities of items to values. With known single-minded bidders, no OSP mechanism can do better than $\Omega(\sqrt{\log n})$ -approximate social welfare (Ferraioli, Penna, and Ventre, 2021), though there are OSP mechanisms that $O(\min\{\log m, \log n\})$ -approximate social welfare (Dütting, Gkatzelis, and Roughgarden, 2017; Christodoulou,

Gkatzelis, and Schoepflin, 2022). However, similar impossibilities exist for unknown single-minded bidders: no individually rational OSP mechanism that makes no positive transfers can approximate social welfare better than $\min\{m, n\}$, and this impossibility extends to arbitrary monotone valuations.

OSP is therefore an interesting notion of simplicity in the study of bounded rationality in mechanism design. Although it is rather restrictive and significantly impacts the approximation guarantees one can achieve versus even strategyproof mechanisms, it is also useful to describe the behaviour seen in practice and provides us with mechanisms that are robust to manipulation even against the cognitively limited agent who may not know better. Given that OSP mechanisms restrict the class of incentive compatible mechanisms so severely, by relaxing the incentive compatibility requirements with NOM we can hopefully provide mechanisms for cognitively limited agents that will not be manipulated in practice and can provide some more positive guarantees with regards to aspects such as efficiency, fairness, and other desiderata.

2.3 SIMPLICITY IN MECHANISM DESIGN

Both NOM and OSP find themselves in a line of research on simplicity in algorithmic mechanism design. To finish this chapter we briefly mention some other work from the literature that study mechanism simplicity and take into account the cognitive limitations of the agents interacting with them. An overview of recent advancements in the study of designing simple mechanisms is given in [S. Li, 2024]. [Pycia and Troyan, 2023] study extensive-form mechanisms for agents who can only plan for a subset of the future decisions they may be called by the mechanism to make, called their “planning horizon”, and this may be updated as the mechanism progresses. They then introduce a class of simplicity standards based on the number of moves ahead that each agent can plan, encapsulating OSP, one-step simple (OSS), and SOSP, and use these to characterise simple mechanisms in a range of settings, arguing why, for example, posted price mechanisms may be simpler to follow than ascending price auctions. [Ferraioli and Ventre, 2024] use the notion of “lookahead” (Shannon, 1950; de Groot, 1978; Pearl, 1984) to introduce a weaker notion of OSP that allows us to interpolate between the absent contingent reasoning of OSP and the perfect rationality of strategyproofness. Specifically, in “ k -step” OSP, agents decide their next move by only planning for the subsequent k steps. They provide an algorithmic characterisation of this class of mechanisms for single-parameter domains and binary allocation problems, then use this to show roughly that mech-

anisms based on reverse greedy algorithms, such as deferred acceptance, are more robust to imperfect rationality than those that use greedy algorithms. They show that, essentially, increasing the “degree of rationality” of the agents only slightly improves approximation guarantees of incentive compatible mechanisms for machine scheduling and facility location.

Introducing an interpolation between OSP and SOSP known as “ k -step OSP”, where agents decide their next move by only planning for the subsequent k steps, [Ferraioli and Ventre, 2024]

[Gonczarowski, Heffetz, and Thomas, 2023] investigate how to *present* a mechanism as strategyproof to the agents. They introduce the notions of menu descriptions, one-side-proposing, low memory, and local bookkeeping as measures of a mechanism’s simplicity, and examine the tradeoffs between describing the outcome to the agents and describing the choices each agent has to make. In a similar vein, [L. Nagel and Saitto, 2023] introduce a measure of simplicity for strategyproof mechanisms, extending the ranking of mechanism simplicity beyond OSP and SOSP, and provide rankings of the simplicity of certain mechanisms that are consistent with experimental evidence. For example, the Ausubel auction [Ausubel, 2004] is the simplest implementation of the VCG outcome, achieving strictly lower complexity than the VCG auction itself.

[Camara, 2022] studies cognitive limitations arising from a lack of computational power from the agents. The author models the decision maker as a Turing Machine and uses this to justify ruling out decision making behaviour from the agents when it is computationally hard. Roughly, they show that under this model expected utility maximisation is intractable unless the utility function is additively separable, meaning $u(x_1, \dots, x_n) = \sum_i^n u_i(x_i)$ for some collection of functions $u_1, u_2, \dots, u_n$, whenever the agents’ choice rules are rational, tractable, and symmetric. Then dropping the assumption of symmetry, they show that any rational and tractable choice rule is equivalent to “dynamic choice bracketing”, a behaviour that assumes choices are made via dynamic programming, where the agent’s current decision depends only on a small number of choices that she has not yet considered. They also prove a “choice trilemma” between rationality, approximate optimality, and tractability, meaning that all three cannot be achieved simultaneously. For example, if an agent’s choice rule is rational and tractable, then it cannot guarantee a non-zero fraction of the optimal payoff. In other words, it may be beneficial for the agent to take seemingly irrational behaviour in order to arrive at an objectively better outcome than had she been strictly rational to the outside observer.

Finally, [J. Li and Dworczak, 2021] study simplicity from the mecha-

nism designer's point of view, addressing the question: given some set of strategically unsophisticated agents who lack the ability to predict other agents' strategies (strategyproof), lack contingent reasoning (OSP), or lack foresight (SOSP), is it optimal for the mechanism to offer a mechanism from the corresponding class of incentive compatible mechanisms for the type of cognitive limitation being considered? They show that when the mechanism offered to the agents is not simple, although it may make it harder to predict how the agents will behave, it may lead to a better outcome for the mechanism designer, thus advocating for less simple and more complex mechanisms to be used as a means of confusing players in order to achieve a better outcome.

PRELIMINARIES

General setup. For a natural number $k \in \mathbb{N}$ we denote by $[k]$ the set $\{1, 2, \dots, k\}$, and for a set S and element $e \in S$ let $S - e$ denote $S \setminus \{e\}$. We consider the standard mechanism design setup in which there is a set $N = [n]$ of agents, and each agent $i \in N$ has a piece of private information t_i known as her type that comes from a domain D_i of possible types. The type of an agent describes the utility she gets from an abstract set of “states of the world”, also referred to as alternatives. We denote this set of alternatives as Ω . For an agent i we consider her type as the function $t_i : \Omega \rightarrow \mathbb{R}$ that maps alternatives to real-valued utilities. Let $D = \times_{i \in N} D_i$ denote the set of type profiles (also known as the domain or the type space of the agents). For some agent $i \in N$ we denote by D_{-i} the set of all types profiles excluding i . For a type profile $\mathbf{t} = (t_1, t_2, \dots, t_n) \in D$ and some $b_i \in D_i$ we use the notation $(b_i, \mathbf{t}_{-i})$ to denote the type profile obtained from $\mathbf{t}$ by replacing its i th coordinate with b_i .

Social choice functions. A social choice function is a means of selecting a single element from Ω and takes into account the types of the agents in order to select an outcome that is “best” for the population. For example, in a single-winner election it may represent a set of candidates, while in auction settings it may represent the set of feasible allocations. Formally a social choice function is any function $f : D \rightarrow \Omega$ that maps type profiles to alternatives.

Eliciting the type profile. In algorithm design we assume we have access to the true state of the world, and our main focus is to map type profiles to alternatives efficiently. In contrast, mechanism design assumes that before we may select an alternative, a type profile must be elicited from the agents, and doing so requires a model of how agent’s communicate to the mechanism. To this end, each agent i is equipped with a set of actions A_i . In general

an indirect mechanism will determine the type profile of the agents over a series of rounds, at each round querying an agent who responds by taking some action in order to *signal* her type. The mechanism then proceeds to query other agents (potentially returning to agents it has already queried) until it determines the full type profile of the agents. The exact procedure taken by an indirect mechanism to elicit the type profile (including the queries it poses to the agents, the order in which the agents are queried, and the actions they may take in response) is specified by the mechanism's *implementation*. We formalise this in Definition 1. A subset of indirect mechanisms are direct revelation mechanisms, in which each of the agents are queried simultaneously and exactly once. In this setting it is therefore assumed that agents are able to report their type directly to the mechanism, and therefore for each $i \in N$ the set of actions A_i coincides exactly with her domain D_i . Regardless of how the mechanism determines the type profile of the agents, we will refer to the type that the mechanism determines for a given agent i as the type *reported* by agent i . Additionally, since an agent's type is private information they may misreport it to the mechanism if it may lead to a better outcome (that is, a higher utility for the agent), we refer to the type reported by each agent i to the mechanism as her bid, and henceforth refer to the type profile elicited from the agents as a bid profile. We denote the bid of agent i as b_i and the bid profile elicited by the mechanism as $\mathbf{b} = (b_1, b_2, \dots, b_n)$.

Definition 1 (Implementation tree). An *implementation tree* is a directed rooted tree $T = (V, E)$ in which each node $u \in V$ is labelled with a set of type profiles $D^{(u)} \subseteq D$ that are said to be *compatible with* u . Each internal node u of the tree is additionally labelled with a player $\pi(u) \in N$, representing the agent queried by the mechanism at u . Agent $i = \pi(u)$ then has a set of actions they may take in response to the mechanism's query at u , each of which is said to *signal* a particular type of agent i to the mechanism. Each of these actions is represented with an edge in the tree, and each edge (u, v) is labelled with a set of types $\sigma_i(u, v)$ signalled by taking the action corresponding to edge (u, v) . We write the set of type profiles compatible with u as $D^{(u)} = (D_i^{(u)}, D_{-i}^{(u)})$ and note that $D_i^{(u)}$ reflects the set of types compatible with u for player i . Additionally for some type $b_i \in D_i^{(u)}$ compatible for i at u we denote by $D^{(u)}(b_i) \subseteq D^{(u)}$ the set of all type profiles compatible with i signalling b_i at node u . In other words, if when i signals b_i at node u and the tree proceeds to child node v in the tree, then we have $D^{(u)}(b_i) = D^{(v)}$.

One action may signal many types, but each type may only be signalled

by a single action at a particular node. In other words for a given node u with $i = \pi(u)$, if $b_i \in \sigma_i(u, v)$ then $b_i \notin \sigma_i(u, w)$ for any other edge (u, w) . For each pairwise disjoint set of types $\sigma_i(u, v)$ and $\sigma_i(u, w)$ that i may signal at node u , we say that i *separates* the types in $\sigma_i(u, v)$ from the types in $\sigma_i(u, w)$ at u . For some edge (u, v) in the implementation tree we say that the types in $\sigma_i(u, v)$ are *inseparable at u* .

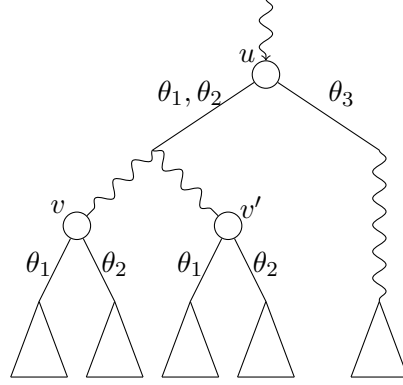


Figure 3.1: An example implementation tree. This tree depicts an extensive-form mechanism where player i has domain $D_i = \{\theta_1, \theta_2, \theta_3\}$ and is queried at nodes u , v , and v' . The wedges represent the set of type profiles compatible with i signalling these types at various nodes in the tree.

Mechanisms with payments. As noted, an agent's type is private information and therefore there is no guarantee that each agent will report their true type to the mechanism. This is an issue since agents may in fact have an incentive to act dishonestly when interacting with the mechanism, namely if doing so results in the mechanism selecting an alternative that leads to a higher utility than it would have done had the agent reported her type truthfully. In order to correct this problem we consider mechanisms that use monetary transfers to realign the incentives of the agents and ensure that it is in each agent's best interests to interact truthfully with the mechanism. Informally, we will say that a mechanism is incentive compatible if this is the case (in order to properly define incentive compatibility we also require a model of how the agents will respond according to the outcomes of the mechanism, which we will cover later in this section). It is assumed that the mechanism first elicits the type profile from the agents and selects an alternative from ω and then additionally returns a vector of payments describing the monetary transfers between the agents and the

mechanism in order to realign their incentives. For our purposes we assume there is some social choice function that we wish to *implement*. Formally, a mechanism is said to implement a social choice function f if for every $\mathbf{b} \in D$ the alternative selected by the mechanism is equal to $f(\mathbf{b})$. We can therefore think of a mechanism for our settings as a tuple consisting of the following three components: i) a social choice function that maps type profiles to alternatives; ii) a vector of payments describing monetary transfers between the mechanism and the agents; and iii) an implementation tree describing how the type profile is elicited from the agents. We define this formally with Definition 2.

Definition 2 (Indirect mechanism). An *indirect mechanism* is a tuple $M = (f, p; T)$ where $f : D \rightarrow \Omega$ is a social choice function, $p : D \rightarrow \mathbb{R}^n$ is a payment rule, and T is an implementation tree as outlined in Definition 1. Once the mechanism reaches some leaf node ℓ of T then it returns the alternative $f(\mathbf{b})$ and vector $p(\mathbf{b})$, where $\mathbf{b} \in D(\ell)$ is some type profile compatible with ℓ .

Strategies. We require some way to model an agent's behaviour in a given mechanism. Loosely, a strategy for player i dictates which actions they will take (and hence which types they will signal) in the implementation tree, for every node at which they are queried by the mechanism. We will assume that agents play a *consistent strategy* throughout the execution of the mechanism, meaning that the action selected by the agent depends only on the type the agent wishes to signal and not the node at which they are queried (see Observation 4). Hence, given an implementation tree T we denote by $s_i = (s_i^u)_{u: i=\pi(u)}$ the *strategy* of player i , where each $s_i^u : D_i \rightarrow A_i$ is a function mapping a type in i 's domain to an action signalling that type.¹ Since a given strategy s_i is simply a sequence of actions that eventually signals a given type b_i from the domain of player i to the mechanism, we will use a s_i interchangeably with b_i and hence $s = (s_i, s_{-i})$ interchangeably with $\mathbf{b}$. We also use $s_i(b_i)$ to denote a strategy of i signalling the type $b_i \in D_i$ to the mechanism. We denote the set of all strategies for player i by S_i and use $S = \times_{i \in N} S_i$ to denote the joint strategy space, or set of all joint strategy profiles.

Definition 3 (Direct mechanism). An *direct mechanism* is a tuple $M = (f, p)$ where $f : D \rightarrow \Omega$ is a social choice function and $p : D \rightarrow \mathbb{R}^n$ is a payment rule. Equivalently, it can be modelled as the indirect mechanism

¹Note that because we assume consistent strategies then each action s_i^u in player i 's strategy will signal the same type.

$(f, p; T)$ whose implementation tree T is such that each agent is queried once at the same time as every other agent, and each agent takes an action that fully separates her domain.

Incentives. Fix an indirect mechanism $M = (f, p; T)$. Given a player i whose true type is t_i we say i plays a *truthful strategy* if i plays the strategy $s_i(t_i)$ in the implementation tree. Otherwise if she plays the strategy $s_i(b_i)$ then we will say she plays a *dishonest strategy*. Given a strategy profile $s \in S$ in which agents signal the bid profile $\mathbf{b}$ we write the outcome of the mechanism as $M(s(\mathbf{b})) = M(\mathbf{b}) = (f(\mathbf{b}), p(\mathbf{b}))$. Each agent is equipped with a quasi-linear utility function that depends on her type and the outcome selected by the mechanism. For agent $i \in N$ with type $t_i \in D_i$ we denote this utility as $u_i(M(\mathbf{b}); t_i) = t_i(f(\mathbf{b})) - p_i(\mathbf{b})$. We emphasise that a positive payment $p_i(\mathbf{b})$ describes money to be *extracted* from player i . In certain settings it of course makes sense to reimburse players for the costs incurred for the mechanism selecting an alternative. When this is the case we may simply assume each element in the payment vector is non-positive. We account for this more explicitly in later chapters, particularly Chapter 5 and Sections 7.3 and 7.4 of Chapter 7.

Definition 4 (Strategyproofness). A mechanism M is *strategyproof* if for every $i \in N$, every pair of types $t_i, b_i \in D_i$, and every joint strategy profile of the other players $s_{-i}(b_{-i})$ it holds that

$$u_i(M(s_i(t_i), s_{-i}(\mathbf{b}_{-i})); t_i) \geq u_i(M(s_i(b_i), s_{-i}(\mathbf{b}_{-i})); t_i). \quad (3.1)$$

Note that (3.1) requires truthtelling to be a dominant strategy for each player, hence a mechanism that is strategyproof may also be called *truthful* or *dominant-strategy incentive-compatible (DSIC)*. If there exist a true type t_i and misreport b_i for some $\mathbf{b}_{-i}$ such that (3.1) does not hold, then the mechanism is said to be *manipulable* and b_i a *(profitable) manipulation* of the mechanism.

Imperfect rationality. Fix a player $i \in N$. In order to determine that a mechanism M is strategyproof then we must compare the utility i gets from acting truthfully with the utility she gets from acting dishonestly, for every true type and potential manipulation of the mechanism, and for each joint profile of the other bidders. Therefore in order for the agent i to recognise that M is manipulable she must perform these computations and correctly deduce whether or not truthtelling is indeed a dominant strategy. In other words, agent i can only recognise if a misreport is profitable manipulation

of M if she is *perfectly rational*. This is too strong of an assumption, and hence there has been increasing interest in the study of mechanism design for agents with *imperfect rationality*. [S. Li, 2017] introduces the notion of obvious strategyproofness, where a mechanism is only deemed incentive-compatible if truthtelling is *obviously dominant* for each agent. Loosely, a strategy is obviously dominant if the worst outcome from following it is no worse than the best outcome from following any other strategy. Therefore, under this solution concept, agents are only assumed to tell the truth if it is obvious to do so. Obvious strategyproofness, or OSP, models a strong notion of imperfect rationality in which agents have zero contingent reasoning skills. That is, they are completely unable to reason about any cases beyond their current node in the implementation tree. We refer the reader to [S. Li, 2017] for a formal definition. [Troyan and Morrill, 2020] places a similarly strong restriction on the contingent reasoning skills of the agents, although they approach the issue of incentive-compatibility for such agents from a more optimistic perspective, stipulating that agents will only *lie* if it is obvious.

Definition 5 (Non-obvious manipulability). A mechanism M is *not obviously manipulable (NOM)* if for every agent $i \in N$ and every pair of types $t_i, b_i \in D_i$ it holds that:

$$\sup_{\mathbf{b}_{-i}} \{ u_i(M(t_i, \mathbf{b}_{-i}); t_i) \} \geq \sup_{\mathbf{b}_{-i}} \{ u_i(M(b_i, \mathbf{b}_{-i}); t_i) \}, \quad (3.2)$$

$$\inf_{\mathbf{b}_{-i}} \{ u_i(M(t_i, \mathbf{b}_{-i}); t_i) \} \geq \inf_{\mathbf{b}_{-i}} \{ u_i(M(b_i, \mathbf{b}_{-i}); t_i) \}. \quad (3.3)$$

If (3.2) holds then M is *best-case not obviously manipulable (BNOM)* and if (3.3) holds then M is *worst-case not obviously manipulable (WNOM)*. If either inequality does not hold then the misreport b_i for which it does not hold is said to be an *obvious manipulation* of the mechanism for player i .

In the next chapter we warm up to studying mechanisms with payments that have no obvious manipulations. Our main goal is to tackle the question of implementability for direct-revelation mechanisms: given a social choice function, does there exist a payment scheme such that the resulting mechanism is not obviously manipulable?

IMPLEMENTABILITY

In this chapter we explore which social choice functions are implementable as NOM mechanisms. Recall that given a social choice function f a mechanism M is said to implement f if for every input profile $\mathbf{b}$ the alternative selected by M is equal to $f(\mathbf{b})$. Here we assume we are given such a function along with the domain of the agents, and our goal is to design a payment rule p such that the resulting mechanism is NOM. We begin by considering which social choice functions may be implemented as direct mechanisms under general preference domains by representing the set of incentive compatibility constraints as a weighted graph and introducing the concept of labellings, which provides a means of modifying this graph in order to select which bid profiles to compare (that is, those leading to the extremes of the agent's utility function, as dictated by NOM). We then restrict attention to the well-studied class of single-parameter agents and give a more fine-grained characterisation on the class of implementable social choice functions in this domain.

4.1 CONTRIBUTION OF THIS CHAPTER

Our overall goal is to understand how permissive NOM is as a solution concept for social choice functions. NOM requires truth-telling to yield a greater utility than misreporting for every player under the best and worst outcomes of the mechanism, taken over all actions of the other players. Therefore the constraints imposed by NOM *depend on* both the allocation and payment function since they define these maximum and minimum utilities, and the utility of every other outcome must lie between these extremes. The first contribution of this chapter is to abstract this dependency using what we call *labellings* and use these to extend the cycle monotonicity technique to handle NOM direct-revelation mechanisms. Informally a labelling is a choice of bid profiles designating the best and worst outcomes for a given player that the mechanism will define through its allocation and payments. These

induce two types of constraints: incentive-compatibility constraints, which enforce the NOM property where the extreme utilities when lying are no better than the extreme utilities from truthtelling, and labelling-induced constraints, which ensure the validity of the labels. Given an allocation function f and labelling λ for a given agent i we can then define a graph $\mathcal{G}_{i,f}^\lambda$ representing the constraints imposed by the mechanism and show that the existence of payments satisfying these constraints is related to cycles in this graph.

Main Result 1 (informal). An allocation function f is NOM implementable if and only if there exists a labelling λ such that $\mathcal{G}_{i,f}^\lambda$ has no negative-weight cycles.

This provides a useful tool for studying the allocation functions for which we can define payments leading to NOM mechanisms. We apply this to the well-studied setting of single-parameter agents where each agent's type can be expressed as a single number, and their utility for an allocation is simply the product of their type with the amount they are allocated. Negative cycles exist in the graph $\mathcal{G}_{i,f}^\lambda$ when the labels are antimonotone, roughly meaning that for two types b_i and b'_i with $b_i < b'_i$ the agent is allocated more for the label corresponding to b_i than it does for the label corresponding to b'_i . We then prove that a monotone labelling guarantees no negative cycles and that such a labelling is possible whenever the image of f_i is rich enough. Roughly speaking f is *overlapping* if for each agent i and pair of types t_i, b_i the sets $\{f_i(t_i, \mathbf{b}_{-i})\}_{\mathbf{b}_{-i}}$ and $\{f_i(b_i, \mathbf{b}_{-i})\}_{\mathbf{b}_{-i}}$ have a non-empty intersection.

Main Result 2 (informal). An allocation function f is NOM-implementable for single-parameter agents if and only if f is overlapping.

Our second contribution proves that as long as the labels define a monotone restriction of the allocation function then it is possible to define a NOM payment scheme. This uncovers a connection with strategyproofness for single-parameter agents, where the allocation must be monotone for each fixed $\mathbf{b}_{-i}$, and when this is the case it is known from [Archer and Tardos, 2001] the form the payments must take. We observe that we can take the same payments along the monotone restriction defined by the labels, while we can easily characterise the payments associated with the remaining bid profiles. This gives a complete picture on the power of NOM mechanisms for single-parameter agents.

The rest of this chapter is structured as follows: in Section 4.2 we warm up to studying obvious manipulations in general settings; in Section 4.3 we investigate which social choice functions can be implemented as direct-revelation mechanisms under general domains; and in Section 4.4 we restrict attention to single-parameter agents and provide a finer characterisation on the class of implementable social choice rules.

4.2 WARM UP

We begin by illustrating that NOM is a strictly weaker solution concept than strategyproofness: every strategyproof mechanism is NOM, but there are NOM mechanisms that are not strategyproof. It is straightforward to see the first statement: if (3.1) holds for every pair of types t_i, b_i from player i 's domain and every profile $\mathbf{b}_{-i}$ of the other players, then clearly the maximum truthful utility will be no worse than the maximum dishonest utility, and likewise for minimum utilities. Similarly it is simple to construct a mechanism that satisfies NOM but not strategyproofness.

Observation 1. If M is strategyproof then it is NOM.

Proof. Let M be an arbitrary mechanism for n players and fix player $i \in N$ with type $t_i \in D_i$. We begin by showing that if M is strategyproof then it is also BNOM. By strategyproofness we have $u_i(M(t_i, \mathbf{b}_{-i}); t_i) \geq u_i(M(b_i, \mathbf{b}_{-i}); t_i)$ for all $t_i, b_i \in D_i$ and for all joint profiles $\mathbf{b}_{-i}$ of the others. Assume for contradiction that M is obviously manipulable in the best case and therefore $\sup_{\mathbf{b}_{-i}} \{u_i(M(b_i, \mathbf{b}_{-i}); t_i)\} > \sup_{\mathbf{b}_{-i}} \{u_i(M(t_i, \mathbf{b}_{-i}); t_i)\}$ for some obvious manipulation b_i . Let $\hat{\mathbf{b}}_{-i} \in \arg \sup \{u_i(M(b_i, \mathbf{b}_{-i}); t_i)\}$ denote the bid profile leading to i 's highest utility over all $\mathbf{b}_{-i}$ when she bids b_i . Then we have:

$$\begin{aligned} u_i(M(t_i, \hat{\mathbf{b}}_{-i}); t_i) &\geq u_i(M(b_i, \hat{\mathbf{b}}_{-i}); t_i) \\ &= \sup_{\mathbf{b}_{-i}} \{u_i(M(b_i, \mathbf{b}_{-i}); t_i)\} \\ &> \sup_{\mathbf{b}_{-i}} \{u_i(M(t_i, \mathbf{b}_{-i}); t_i)\}, \end{aligned}$$

which is a contradiction, since the utility $u_i(M(t_i, \hat{\mathbf{b}}_{-i}); t_i)$ is strictly greater than the maximum possible utility i gets when bidding t_i truthfully.

A similar argument shows that M is WNOM. First assume that M is obviously manipulable in the worst case and let $\hat{\mathbf{b}}_{-i} \in \arg \inf \{u_i(M(b_i, \mathbf{b}_{-i}); t_i)\}$ be the bid profile leading to i 's worst utility when she bids $b_i \neq t_i$. Then we

have:

$$\begin{aligned} \inf_{\mathbf{b}_{-i}} \{ u_i(M(b_i, \mathbf{b}_{-i}); t_i) \} &> \inf_{\mathbf{b}_{-i}} \{ u_i(M(t_i, \mathbf{b}_{-i}); t_i) \} \\ &= u_i(M(t_i, \hat{\mathbf{b}}_{-i}); t_i) \\ &\geq u_i(M(b_i, \hat{\mathbf{b}}_{-i}); t_i), \end{aligned}$$

meaning $u_i(M(b_i, \hat{\mathbf{b}}_{-i}); t_i)$ is strictly less than the worst dishonest utility of player i . Thus if M is strategyproof then it is NOM. $\square$

We note with the following example that the reverse of Observation 1 is not true by defining a NOM mechanism that is not strategyproof.

Example 1 (NOM does not imply strategyproofness). Let $n = 2$ and take $D_1 = \{\theta_1^1, \theta_2^1\}$ and $D_2 = \{\theta_1^2, \theta_2^2\}$. Hence the domain of all type profiles is $D = D_1 \times D_2 = \{(\theta_1^1, \theta_1^2), (\theta_1^1, \theta_2^2), (\theta_2^1, \theta_1^2), (\theta_2^1, \theta_2^2)\}$. Suppose there are four alternatives $\Omega = \{\omega_{11}, \omega_{12}, \omega_{21}, \omega_{22}\}$ and we have a social choice function f such that $f(\theta_i^1, \theta_j^2) = \omega_{ij}$. Recall that all types are functions $\theta_1^1, \theta_2^1, \theta_1^2, \theta_2^2 : \Omega \rightarrow \mathbb{R}$ mapping alternatives to real-valued utilities, and suppose they are defined as in Table 4.1.

	ω_{11}	ω_{12}	ω_{21}	ω_{22}		ω_{11}	ω_{12}	ω_{21}	ω_{22}
θ_1^1	4	2	1	3	θ_1^2	4	1	2	3
θ_2^1	1	3	4	2	θ_2^2	1	4	3	2

Table 4.1: Unless enhanced with payments, the mechanism is NOM but not strategyproof for either player with the above domains.

The domains of the two players are defined such that when examining the incentive compatibility constraints, each player is comparing the same utilities (in the case of strategyproofness), or the same extreme utilities (in the case of NOM). Hence it suffices to show the claim for player 1.

Let $M = (f, p)$ where $p(\mathbf{b}) = (0, 0)$ for each $\mathbf{b} \in D$, and hence note that $u_1(M(\mathbf{b}); \theta_j^1) = \theta_j^1(f(\mathbf{b}))$ for each $\mathbf{b} \in D$, and each $j \in \{1, 2\}$. Begin by observing that M is NOM for player 1. When she has type θ_1^1 then her maximum utility when reporting truthfully is 4, compared to her best utility 3 when reporting θ_2^1 dishonestly. Likewise when she has type θ_2^1 then her maximum utility when reporting truthfully is 4, no worse than her best utility of 3 when bidding θ_1^1 dishonestly. Hence M is BNOM for player 1. Similarly for WNOM, when she reports θ_1^1 truthfully her worst utility is 2 which is no worse than her worst dishonest utility of 1, and when she has type θ_2^1 we compare her worst truthful utility of 2 with her worst dishonest utility of 1. Therefore the mechanism is WNOM, and hence NOM.

The mechanism is not strategyproof for player 1. Fix player 2's report of θ_2^2 and note that when 1 has type θ_1^1 and bids truthfully she receives utility 2, strictly less than the utility she receives when reporting θ_2^1 dishonestly, thus violating (3.1). A similar case holds for when player 1 has true type θ_2^1 .

Therefore, although the mechanism is NOM for player 1 it is not strategyproof. As already argued, we are comparing exactly the same utilities when considering player 2 (even though these utilities occur at different bid profiles), hence M is also NOM for player 2 without being strategyproof. Moreover, since we were able to define payments (in this case, $p(\mathbf{b}) = (0, 0)$ for all bid profiles $\mathbf{b}$), we can also say that f is implementable as a NOM mechanism.

We note that if player 2 has domain $D_2 = D_1 = \{\theta_1^1, \theta_2^1\}$ then the mechanism (f, p) with $p(\mathbf{b}) = (0, 0)$ for all $\mathbf{b}$ is neither BNOM nor WNOM for player 2, and hence not NOM. In Examples 2 and 3 we will study instances where the players have identical domains and show in the former that not all social choice functions are possible to implement as mechanisms without obvious manipulations, and in the latter that the payments that arise for player 2 from the shortest path payments of Theorem 3 suffice for the resulting mechanism to be NOM.

Example 2 (Not every function is implementable). Let $n = 2$ and suppose player domains are $D_1 = D_2 = \{\theta_1, \theta_2\}$ as defined in Table 4.2. Suppose $\Omega = \{\omega_{11}, \omega_{12}, \omega_{21}, \omega_{22}\}$, and let $f(\theta_i, \theta_j) = \omega_{ij}$ as in Example 1. Focus on player 1. To satisfy BNOM for type θ_1 then we need both

	ω_{11}	ω_{12}	ω_{21}	ω_{22}
θ_1	0	0	1	1
θ_2	1	1	0	0

Table 4.2: No payment scheme exists to make this instance NOM for player 1.

$p_1(\theta_2, \theta_1), p_1(\theta_2, \theta_2) \geq 1$. Then to satisfy BNOM for type θ_2 we need both $p_1(\theta_1, \theta_1), p_1(\theta_1, \theta_2) \geq 2$, but this now breaks the BNOM constraints for θ_1 . Similarly for WNOM, we must have either $p_1(\theta_2, \theta_1) \geq 1$ or $p_1(\theta_2, \theta_2) \geq 1$ for type θ_1 , then either $p_1(\theta_1, \theta_1) \geq 2$ or $p_1(\theta_1, \theta_2) \geq 2$ to fix WNOM for θ_2 . Now we have broken WNOM for θ_1 again. Hence f cannot be implemented as a NOM mechanism. In fact, it is not even implementable as either a BNOM or WNOM mechanism in isolation.

4.3 IMPLEMENTABILITY UNDER GENERAL DOMAINS

In this section we introduce the concept of the allocation graph which helps us to reason about which social choice functions can be implemented as incentive-compatible mechanisms via the cyclic monotonicity technique, first used by [Rochet, 1987]. This is an extremely flexible tool and is applicable to a wide range of settings within mechanism design. For example, for the problem of machine scheduling the cycle monotonicity framework has been applied to study truthful mechanisms for scheduling unrelated (multi-dimensional) machines (Lavi and Swamy, 2009), truthful mechanisms with verification power (Ventre, 2014), and obviously strategyproof mechanisms for related machines (Ferraoli, Meier, et al., 2019b).

We begin by restricting our attention to direct revelation mechanisms, and we will later see how he can extend the definitions to handle indirect mechanisms. We also assume for now that agents have finite domains, and hence the suprema and infima of the agents' utility functions will coincide with their maxima and minima. In this case for a given player $i \in N$ we will enumerate the types from her domain D_i as $D_i = \{\theta_1^i, \theta_2^i, \dots, \theta_d^i\}$, where $d = |D_i|$ is the cardinality of her domain. In most cases the agent i will be clear from the context hence we omit the superscript from type θ_j^i and write it simply as θ_j .

4.3.1 ENCODING MECHANISM CONSTRAINTS AS A GRAPH

One can explore which social choice functions can be implemented as incentive compatible mechanisms with monetary transfers for a given agent by examining certain “cycles” that arise from the function from the perspective of the agent's domain. Rochet (1987) is the first to study this question of implementability under general multidimensional domains (a concept they refer to instead as “rationalizability”). In this setting, a cycle for some social choice function $f : D \rightarrow \Omega$ for a given agent $i \in N$ is simply a (finite) collection of types $\theta_1, \dots, \theta_{k+1} = \theta_1$. As a starting point for studying implementability for NOM mechanisms we state the following result.

Theorem 1 (Rochet (1987)). *Social choice function f is implementable as a truthful mechanism for agent i if and only if for all finite cycles $\theta_1, \dots, \theta_{k+1} \in D_i$ and each $\mathbf{b}_{-i} \in D_{-i}$ it holds that*

$$\sum_{j=0}^k u_i(f(\theta_j, \mathbf{b}_{-i}); \theta_j) - u_i(f(\theta_{j+1}, \mathbf{b}_{-i}); \theta_j) \geq 0,$$

where $\theta_{k+1} = \theta_1$.

This result is a generalisation of the “cyclic monotonicity” condition of Rockafellar (1972) and the “single-crossing condition” of Mirrlees (1971) and Spence (1974). Moreover, the proof of the sufficiency of every cycle being non-negative provides insights into how to actually define a truthful payment scheme. Namely, by defining for each type $\theta \in D_i$ a value $V(\theta)$ that is equal to the largest total weight of a cycle starting from some arbitrary starting type θ_0 , one can then define the payments at bid profile $\mathbf{b} = (\theta, \mathbf{b}_{-i})$ to $p_i(\mathbf{b}) = u_i(f(\mathbf{b}); \theta) - V(\theta)$. These payments are guaranteed to satisfy all the constraints imposed by strategyproofness.

Our main aim is to adapt this result to the NOM setting. We emphasise the analogy to cycles by explicitly defining a graph that encodes all of the constraints imposed by an incentive compatible mechanism, and we use this graph to argue similarly that the absence of negative cycles is a necessary and sufficient condition for the existence of incentive compatible payment schemes. We begin by introducing the graph for studying which functions are implementable as *truthful* mechanisms and then, by enhancing the graph with a notion we call profile labellings, we show how a similar characterisation can be adapted to study the case of NOM-implementable social choice functions.

Consider the social choice function f mapping type profiles from D to alternatives from Ω . For each agent $i \in N$ we will define a weighted graph representing the constraints imposed by incentive compatibility (i.e., the constraints in (3.1) for strategyproofness and the constraints in (3.2) and (3.3)) and argue that the cycles in these graphs characterise the class of social choice functions that admit incentive compatible payment schemes. The set of nodes in the graph is exactly the set of all profiles from D , and edges are weighted to reflect the difference in utility agent i gets from the alternative selected by f at the source bid profile versus the destination bid profile. This difference will depend on the true type of the agent (since different types in the agent’s domain may map the same alternatives to different utilities), hence between two nodes there will be multiple edges. We differentiate between two edges between the same pair of nodes with a “type annotation” that describes the true type of the agent. More concretely, let $x, y \in D$ be two bid profiles from the domain of the social choice function f and consider some agent $i \in N$. We denote an edge between x and y with type annotation $t \in D_i$ as $(x, y; t)$, or equivalently as $\langle x \xrightarrow{t} y \rangle$. For a cycle C between k nodes $x_1, x_2, \dots, x_k$ where each edges is annotated with types $\theta_1, \theta_2, \dots, \theta_k$ respectively, we write $C = \langle x_1 \xrightarrow{\theta_1} x_2 \xrightarrow{\theta_2} \dots \xrightarrow{\theta_{k-1}} x_k \xrightarrow{\theta_k} x_1 \rangle$. We also note that

edges are directed, and just because x may be connected to y does not imply that y is connected to x by an edge. Each edge $(x, y; t)$ is weighted according to the function $w(x, y; t) = t(f(x)) - t(f(y))$, and therefore represents the difference in utility that i gets from f when she has type t and bid profile x is submitted to the mechanism, versus bid profile y . The weight of a cycle $C = \langle x_1 \xrightarrow{\theta_1} x_2 \xrightarrow{\theta_2} \dots \xrightarrow{\theta_{k-1}} x_k \xrightarrow{\theta_k} x_1 \rangle$ is simply the sum of edge weights, that is, $w(C) = \sum_{i \in [k]} w(x_i, x_{i+1}; \theta_i) = \sum_{i \in [k]} \theta_i(f(x_i)) - \theta_i(f(x_{i+1}))$, where the index $k + 1$ wraps back around to 1.

We first define the truthful allocation graph. It is slightly adapted from the one defined, for example, in [Lavi and Swamy, 2009], though the underlying graph is the same and can be used identically. Since each edge represents the difference in utility of the outcome returned by a truthful report of player i 's type versus a misreport, then we can use this to guide the specification of our truthful payment rule. Specifically, we must satisfy (3.1) for each $\mathbf{b}_{-i}$ and each pair of types t_i, b_i . If, for some $\mathbf{b}_{-i}$, there is an edge $\langle (t_i, \mathbf{b}_{-i}) \xrightarrow{t_i} (b_i, \mathbf{b}_{-i}) \rangle$ connecting two bid profiles $\mathbf{b} = (t_i, \mathbf{b}_{-i})$ and $\mathbf{b}' = (b_i, \mathbf{b}_{-i})$ with weight $\delta = t_i(f(\mathbf{b})) - t_i(f(\mathbf{b}'))$, then by simply rearranging (3.1) for the profiles $\mathbf{b}$ and $\mathbf{b}'$ we know that we must have $p_i(\mathbf{b}) - p_i(\mathbf{b}') \leq \delta$ if (f, p) is to be a strategyproof mechanism.

Definition 6 (Truthful allocation graph). Let $f : D \rightarrow \Omega$ be a social choice function and consider agent $i \in N$ with finite domain D_i . The *truthful allocation graph* $\mathcal{G}_{i,f}$ for i is a weighted direct graph $\mathcal{G}_{i,f} = (V, E, w)$ where $V = D$ and the edge set E is such that, for each $\mathbf{b}_{-i} \in D_{-i}$, for each pair of types $t_i, b_i \in D_i$ there is an edge $\langle (t_i, \mathbf{b}_{-i}) \xrightarrow{t_i} (b_i, \mathbf{b}_{-i}) \rangle$. Each edge $(x, y; t)$ is assigned the weight $w(x, y; t) = t(f(x)) - t(f(y))$.

In Figure 4.1 we give an illustration of an example truthful allocation graph on an instance with $n = 2$ for player 1. We suppose $D_1 = \{\theta_1, \theta_2, \theta_3\}$ and $D_2 = \{\circ, \square\}$. Since the graph is from the perspective of player 1, all bid profiles in a dotted boundary labelled by a type $\theta_j \in D_1$, which we will define later in Section 4.3.3 as the “ θ_j -islands” of the graph) correspond to a bid profile of the form $(\theta_j, \cdot)$. One can see that edges only occur between bid profiles of the same shape (i.e., the same partial profiles of the other players $\mathbf{b}_{-i}$ excluding the player under consideration i). Note that each of the edges is bidirectional, and their weights may be different. For example, the weight of the red edge is $\theta_1(f(\theta_1, \circ)) - \theta_1(f(\theta_2, \square))$ while the weight of the blue edge is $\theta_2(f(\theta_2, \square)) - \theta_2(f(\theta_2, \circ))$.

With the truthful allocation graph thus defined, the main result of Rochet (1987) can be equivalently stated as follows.

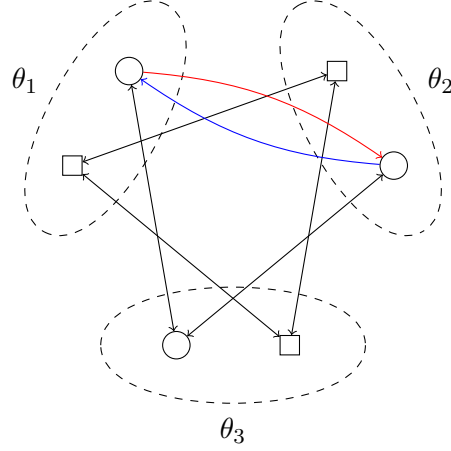


Figure 4.1: An example truthful allocation graph on an instance with $n = 2$ players where $D_1 = \{\theta_1, \theta_2, \theta_3\}$ and $D_2 = \{\bigcirc, \square\}$. According to Theorem 2, in order for f to be implementable as a truthful mechanism all cycles in this graph must be non-negative. All edges are bidirectional, but may have different weights depending on the direction (as explained above for the red and blue edges).

Theorem 2. *Social choice function f is implementable as a truthful mechanism if and only if for every $i \in N$ the truthful allocation graph $\mathcal{G}_{i,f}$ has no cycles of negative weight.*

When we have a social choice function f whose truthful allocation graph $\mathcal{G}_{i,f}$ contains no cycles of negative weight for agent i we will say that f is *cyclically monotone*, or equivalently, that f exhibits *cycle monotonicity*. Later, we will also use these terms to refer to a NOM allocation graph $\mathcal{G}_{i,f}^\lambda$ (as defined in Definition 9) that contains no negative weight cycles.

An advantage of making the connection to cycles more explicit by introducing this graph is that the truthful payments can be defined in terms of shortest paths in $\mathcal{G}_{i,f}$, which we will see in the proof of Theorem 3. Provided that the input is provided explicitly (i.e., the social choice function and agent i 's domain as a table of values) this results in truthful payments that can be computed in polynomial time. With the graph defined for helping us to decide whether a given social choice function can be implemented as a truthful mechanism, we now look to augment this graph with a notion we call profile labellings that will allow us to classify which social choice functions are implementable as NOM mechanisms. The underlying structure of this NOM allocation graph is the same as the truthful allocation graph, but by including a profile labelling we are able to effectively ignore the con-

straints imposed by NOM at certain bid profiles. The set of bid profiles we are able to ignore depends heavily on the structure of the NOM constraints. Therefore, while it is possible (as expected) to be more flexible with regards to the incentives of the agents for NOM mechanisms versus strategyproof ones, there are still circumstances in which impossibilities exist.

4.3.2 PROFILE LABELLINGS

As the previous section shows, a social choice function f is implementable as a strategyproof mechanism if for each player it holds that their allocation graph has no negative cycles. The graph effectively encodes the constraints imposed by truthfulness on the payment rule, and we can then use this graph to define the appropriate payments leading to an incentive compatible mechanism when the constraints are satisfiable. Note that strategyproofness requires comparing the utility of player i at bid profiles $\mathbf{b} = (t_i, \mathbf{b}_{-i})$ and $\mathbf{b}' = (b_i, \mathbf{b}_{-i})$ for each $t_i, b_i \in D_i$, for each joint profile $\mathbf{b}_{-i}$ of the other players. By contrast, NOM only really requires us to compare the utilities at four different bid profiles for each pair of types t_i, b_i . Namely, these correspond to comparing the maximum and minimum truthful utilities, $\max_{\mathbf{b}_{-i}} \{u_i(M(t_i, \mathbf{b}_{-i}); t_i)\}$ and $\min_{\mathbf{b}_{-i}} \{u_i(M(t_i, \mathbf{b}_{-i}); t_i)\}$, with the maximum and minimum dishonest utilities, $\max_{\mathbf{b}_{-i}} \{u_i(M(b_i, \mathbf{b}_{-i}); t_i)\}$ and $\min_{\mathbf{b}_{-i}} \{u_i(M(b_i, \mathbf{b}_{-i}); t_i)\}$, respectively.

Given that the maximum and minimum utilities are defined on both the social choice function and the payment rule, but we are only provided with the social choice function and asked if it is implementable, how can we decide which bid profiles to compare when determining whether a valid payment rule exists? We tackle this question by introducing the concept of *profile labellings*, which we use as a means of selecting which bid profiles in the domain to compare for each pair of types in the player's domain. Specifically, for each player i we will have both a best-case labelling and a worst-case labelling. The former designates for each pair of types t_i, b_i a partial profile of the other players that will eventually lead to i 's maximum utility when she bids b_i and has true type t_i , and the latter is defined analogously for minimum utilities. Formal definitions for these best-case and worst-case labellings are given in Definitions 7 and 8.

Definition 7 (Best-case labelling). Fix a mechanism M and let $i \in N$ be an agent with domain D_i . A *best-case labelling* of M for i is a function $\bar{\lambda}: D_i \times D_i \rightarrow D_{-i}$ such that for each pair of types $t_i, b_i \in D_i$ we have:

$$\bar{\lambda}(t_i, b_i) \in \arg \max_{\mathbf{b}_{-i}} \{u_i(M(b_i, \mathbf{b}_{-i}); t_i)\}. \quad (4.1)$$

Definition 8 (Worst-case labelling). Fix the social choice function f and let $i \in N$ be an agent with domain D_i . A *worst-case labelling of M for i* is a function $\underline{\lambda} : D_i \times D_i \rightarrow D_{-i}$ such that for each pair of types $t_i, b_i \in D_i$ we have:

$$\underline{\lambda}(t_i, b_i) \in \arg \min_{\mathbf{b}_{-i}} \{ u_i(M(b_i, \mathbf{b}_{-i}); t_i) \}. \quad (4.2)$$

In Definitions 7 and 8 the first argument of the labelling represents the true type of player i and the second argument the bid type. Therefore for either a best-case or worst-case labelling $\lambda \in \{\bar{\lambda}, \underline{\lambda}\}$ the profile $\lambda(t_i, t_i)$ represents a profile leading to an extreme value of i 's utility function in which i submits a truthful report of her type, while the profile $\lambda(t_i, b_i)$ represents a profile leading to an extreme value of i 's utility function in which i reports her type dishonestly, whenever $b_i \neq t_i$.

When the domain of player i has finite size, we may represent the labellings $\bar{\lambda}, \underline{\lambda}$ with $d \times d$ matrices with entries in D_{-i} , where $d = |D_i|$ is the size of i 's domain.

Given a best-case labelling $\bar{\lambda}$ and worst-case labelling $\underline{\lambda}$ we can restate the constraints imposed by a NOM mechanism (f, p) as follows. Firstly, given two types t_i and b_i then for whichever profile labelled by $\bar{\lambda}(t_i, b_i)$ it must be the case that, for all other bid profiles of the form $(b_i, \mathbf{b}_{-i})$, we have $u_i(M(b_i, \bar{\lambda}(t_i, b_i)); t_i) \geq u_i(M(b_i, \mathbf{b}_{-i}); t_i)$, since $\bar{\lambda}(t_i, b_i)$ is the profile leading to the highest utility given that i has type t_i and bids b_i . Similarly, for whichever profile labelled by $\underline{\lambda}(t_i, b_i)$ we must have $u_i(M(b_i, \mathbf{b}_{-i}); t_i) \geq u_i(M(b_i, \underline{\lambda}(t_i, b_i)); t_i)$. Combining these into one expression in (4.3) for each $t_i, b_i \in D_i$ gives us the *labelling constraints* imposed by $\bar{\lambda}$ and $\underline{\lambda}$, which enforce the validity of the labels specified by the labellings:

$$u_i(M(b_i, \bar{\lambda}(t_i, b_i)); t_i) \geq u_i(M(b_i, \mathbf{b}_{-i}); t_i) \geq u_i(M(b_i, \underline{\lambda}(t_i, b_i)); t_i). \quad (4.3)$$

We note that the above holds also for $b_i = t_i$. In this case the labelling constraints will simply correspond to enforcing the validity of the labellings for all truthful bid profiles.

Of course, we are using the labellings to encode the constraints that NOM imposes on the mechanism, hence there is another set of constraints they define. In words the *incentive compatibility constraints* state that the best (respectively, worst) outcome for player i when she has type t_i and bids t_i truthfully is no worse than her best (respectively, worst) outcome when she bids $b_i \neq t_i$ dishonestly. In symbols we have the following constraints,

for every pair of types $t_i, b_i \in D_i$, where $t_i \neq b_i$:

$$u_i(M(t_i, \bar{\lambda}(t_i, t_i)); t_i) \geq u_i(M(b_i, \bar{\lambda}(t_i, b_i)); t_i), \quad (4.4)$$

$$u_i(M(t_i, \underline{\lambda}(t_i, t_i)); t_i) \geq u_i(M(b_i, \underline{\lambda}(t_i, b_i)); t_i). \quad (4.5)$$

An important note on labellings. Before moving on to the next section, it is important to note that it is *always* possible to define a labelling λ by simply specifying any bid profile $\mathbf{b}_{-i} = \lambda(t_i, b_i)$ for each pair of types t_i, b_i from i 's domain. The purpose of a labelling is to act as a guide to help us define a payment function that results in the labelled profile $\lambda(t_i, b_i)$ truly leading to the extremes of i 's utility function when she has true type t_i and bids b_i . However, not all labellings (or any, depending on the social choice function) will lead to a satisfiable set of constraints, and therefore it will be impossible to define a payment function that results in the labelling being “true” or “valid”. The constraints that arise from a certain labelling are exactly those imposed by NOM, in addition to the labelling constraints that ensure the labels are “consistent” (in other words, that a labelled profile is truly at the extreme of i 's utility function when she has type t_i and bids b_i). Therefore, it may be useful to think of any given labelling as a “hypothetical labelling” which simply states that, if there existed a payment function resulting in the extremes of i 's utility function occurring at the bid profiles specified by the labelling (along with the labelling being consistent), then the labelled profiles would be exactly those given in (4.1) and (4.2). We stress that the payment function comes *after* specifying a labelling for a given social choice function, and of course only when the set of constraints imposed by the labelling is satisfiable. Additionally, if we are given a mechanism (f, p) then this will already have an “induced” labelling, and there is no flexibility to make (f, p) a NOM mechanism (indeed, the entire reason for introducing labellings in the first place is to guide our choice of payments so that the resulting mechanism is NOM). When f and p are already given, we may find the labelling it induces by simply finding the extremes of i 's utility function under the resulting mechanism (i.e., $\max_{\mathbf{b}_{-i}} \{u_i(M(b_i, \mathbf{b}_{-i}); t_i)\}$ and $\min_{\mathbf{b}_{-i}} \{u_i(M(b_i, \mathbf{b}_{-i}); t_i)\}$ for each t_i and b_i). For the remainder of the thesis, we use the term *valid labelling* to refer to a labelling of a social choice function that results in a set of constraints for which it is possible to define a payment function satisfying the constraints given in (4.3) to (4.5). We will see that a labelling only leads to a satisfiable set of constraints whenever the resulting NOM allocation graph (Definition 9) contains no cycles of negative weight (Theorem 3).

4.3.3 CYCLE MONOTONICITY

If we expand the expressions from (4.3) to (4.5) then we will see both the presence of social choice function f and the payment function p in the resulting expression. It is worth emphasising that the social choice function is fixed and the labellings are there to guide the design of the payment rule. A first question is, given f , does there exist a payment rule such that the resulting mechanism is NOM? And secondly, if such payments do exist, what are they? Not every social choice function will have an associated incentive compatible payment scheme (as shown in Example 2 and then in the context of labellings in Example 4), and even when it does, not every labelling will lead to a valid definition of payments. We will show how we can use these labellings both to answer the question of implementability in general domains as well as to construct the payment function when it is possible.

In the following we use $\lambda = (\bar{\lambda}, \underline{\lambda})$ to denote the “combined” labelling. Since edges are weighted in the same manner and for both labellings $\bar{\lambda}$ and $\underline{\lambda}$ the presence of an edge $(x, y; t)$ is used to encode the same thing (that the eventual utility – that is, under whatever payment scheme we end up defining – of the source profile x given type t is no less than the eventual utility of the destination profile y , then we can combine these into one labelling λ without issue. Specifically, edges will encode the constraints given in (4.3) to (4.5), that is, that the eventual utility for player i with type t on the output of mechanism when given bid profile x is no worse than when it is given bid profile y . In order to encode the analogous constraints for (4.3) for some labelling λ we include edges:

$$\langle (b_i, \bar{\lambda}(t_i, b_i)) \xrightarrow{t_i} (b_i, \mathbf{b}_{-i}) \xrightarrow{t_i} (b_i, \underline{\lambda}(t_i, b_i)) \rangle. \quad (4.6)$$

Similarly we must encode the incentive-compatibility constraints in (4.4) and (4.5). We will therefore have the following edges:

$$\langle (t_i, \bar{\lambda}(t_i, t_i)) \xrightarrow{t_i} (b_i, \bar{\lambda}(t_i, b_i)) \rangle, \quad (4.7)$$

$$\langle (t_i, \underline{\lambda}(t_i, t_i)) \xrightarrow{t_i} (b_i, \underline{\lambda}(t_i, b_i)) \rangle. \quad (4.8)$$

Taking all of these edges together describes all of the constraints imposed by a NOM mechanism. Specifically, given the social choice function f an incentive-compatible payment scheme p is possible exactly whenever the graph resulting from encoding of the above constraints contains no negative cycles.

Definition 9 (NOM allocation graph). Let $f : D \rightarrow \Omega$ be a social choice

function and fix player $i \in N$ with finite domain D_i . Moreover let $\bar{\lambda}$ and $\underline{\lambda}$ be a best-case and worst-case labelling of f for player i , respectively, and $\lambda = (\bar{\lambda}, \underline{\lambda})$ the combined labelling. The *NOM allocation graph* for i is a directed weighted multigraph $\mathcal{G}_{i,f}^\lambda = (V, E, w)$ where node set $V = D$ is equal to the set of bid profiles and E is the set of edges defined as in (4.3), (4.7) and (4.8): for each pair of types $t_i, b_i \in D_i$ we have the edges:

- $\langle (b_i, \bar{\lambda}(t_i, b_i)) \xrightarrow{t_i} (b_i, \mathbf{b}_{-i}) \xrightarrow{t_i} (b_i, \underline{\lambda}(t_i, b_i)) \rangle$
- $\langle (t_i, \bar{\lambda}(t_i, t_i)) \xrightarrow{t_i} (b_i, \bar{\lambda}(t_i, b_i)) \rangle$
- $\langle (t_i, \underline{\lambda}(t_i, t_i)) \xrightarrow{t_i} (b_i, \underline{\lambda}(t_i, b_i)) \rangle$

Each edge $(x, y; t)$ is assigned the weight $w(x, y; t) = t(f(x)) - t(f(y))$.

Figure 4.2 illustrates how one can think of a labelling as a restriction of the complete weighted multigraph that fully represents some player i 's domain. Firstly, consider the example instance between $n = 2$ players as in Figure 4.1. Again we focus on player 1 with domain $D_1 = \{\theta_1, \theta_2, \theta_3\}$. In the subgraph on the left is the complete weighted directed multigraph that entirely describes the manipulations present in the social choice function f if it were not enhanced with payments (and indeed those we may seek to eliminate with an appropriate payment function). Specifically, between each pair of nodes x, y there is an edge $(x, y; t)$ annotated by true type $t \in D_i$ with weight $t(f(x)) - t(f(y))$. By specifying a partial profile $\lambda(t_i, b_i)$ for each pair of types $t_i, b_i \in D_i$ (that is, specifying a labelling of the domain of f) then we restrict the edges of the graph on the left. As a result, we obtain the graph on the right. By carefully selecting the edges we keep and those we discard when constructing a labelling, we can avoid negative cycles that may have been present in the complete graph (see Theorem 3).

Notice that the only edges going from a bid profile $\mathbf{b} = (t_i, \cdot)$ to a bid profile $\mathbf{b}' = (b_i, \cdot)$ of a *different bid type* of player i are the incentive-compatibility edges of (4.7) and (4.8). Similarly, notice that the labelling constraint edges described by (4.6) only go between bid profiles where the bid type of player i is the same. Given a NOM allocation graph $\mathcal{G} = \mathcal{G}_{i,f}^\lambda$ and a type $b_i \in D_i$ we refer to the subgraph containing all nodes of the form $(b_i, \mathbf{b}_{-i})$ as the “ b_i -island” of $\mathcal{G}$. Hence labelling edges are all contained in a given b_i -island of $\mathcal{G}$, and those that traverse two different islands of $\mathcal{G}$ are exactly the incentive-compatibility edges.¹

¹The name “ b_i -island” comes from the fact that labelling edges for a given bid type are isolated from labelling edges for any other bid type, however we can more generally refer to any bid profiles of the form $(b_i, \mathbf{b}_{-i})$ as being from the b_i -island of player i 's domain,

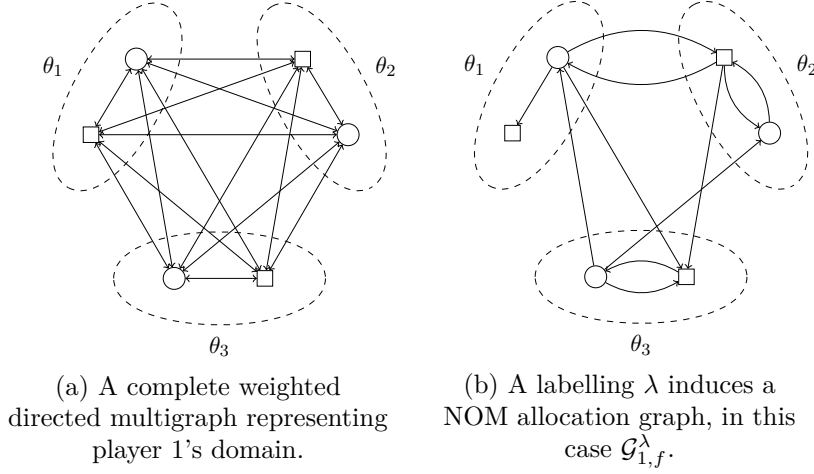


Figure 4.2: By carefully selecting the labelling, it is possible to avoid negative cycles from appearing in the induced NOM allocation graph $\mathcal{G}_{i,f}^\lambda$.

Each different labelling will give rise to different graphs. Briefly before our characterisation result on the class of social choice functions that are implementable as NOM mechanisms, we will introduce a payment scheme that results in a NOM mechanism, provided that the NOM allocation graph contains no negative cycles. We will construct a payment function p by first creating a new graph $\mathcal{G}'$ from $\mathcal{G}$ by augmenting it with an artificial source node, and then setting payments for each bid profile equal to the length of their shortest path from γ in $\mathcal{G}'$.

Definition 10 (Shortest path payments). Let $\mathcal{G} = \mathcal{G}_{i,f}^\lambda$ be some NOM allocation graph of social choice function f for player $i \in N$. Suppose $\mathcal{G}$ is finite and contains no cycles of negative weight. Let $\mathcal{G}'$ denote the graph $\mathcal{G}$ augmented with an auxiliary node γ and an edge with weight zero from γ to each node in $\mathcal{G}$. For each $\mathbf{b} \in D$ define the payments p as follows:

$$p_i(\mathbf{b}) = -\text{SP}(\gamma, \mathbf{b}), \quad (4.9)$$

where $\text{SP}(\gamma, x)$ is the shortest path in $\mathcal{G}'$ from γ to $\mathbf{b}$.

We will show that these payments result in a NOM mechanism under certain conditions of the social choice function. With the following result we characterise the set of social choice functions that admit NOM payment

as clearly there is no need to first define a labelling and NOM allocation graph to group these bid profiles together.

schemes using the structure of the graph, and in particular how this relates to the labelling of its nodes.

Theorem 3. *Social choice function f is implementable as a NOM mechanism if and only if for each player $i \in N$ there exists a labelling λ such that the graph $\mathcal{G}_{i,f}^\lambda$ has no negative weight cycles.*

Proof. ($\implies$) Suppose payments p exist such that the mechanism $M = (f, p)$ is NOM. Fix player $i \in N$ with domain $D_i = \{\theta_1, \theta_2, \dots, \theta_d\}$ and assume for contradiction that every labelling λ induces a negative weight cycle in $\mathcal{G} = \mathcal{G}_{i,f}^\lambda$. Let $C = \langle x_1 \xrightarrow{\theta_1} x_2 \xrightarrow{\theta_2} \dots \xrightarrow{\theta_{k-1}} x_k \xrightarrow{\theta_k} x_1 \rangle$ be a cycle with negative weight $w(C) = \sum_{j \in [k]} \theta_j(f(x_j)) - \theta_j(f(x_{j+1})) < 0$. Since f is NOM-implementable then the inequality encoded by each edge is satisfied:

$$\begin{aligned} \theta_1(f(x_1)) - p_i(x_1) &\geq \theta_1(f(x_2)) - p_i(x_2) \\ \theta_2(f(x_2)) - p_i(x_2) &\geq \theta_2(f(x_3)) - p_i(x_3) \\ &\vdots \\ \theta_k(f(x_k)) - p_i(x_k) &\geq \theta_k(f(x_1)) - p_i(x_1) \end{aligned}$$

The sum of each inequality must also be satisfied. Each payment term $p_i(x_j)$ appears once on each side of an inequality for $j \in [k]$, hence we are left with:

$$\theta_1(f(x_1)) - \theta_1(f(x_2)) + \dots + \theta_k(f(x_k)) - \theta_k(f(x_1)) \geq 0,$$

which is exactly the expression for the weight of the cycle C , which has negative weight by hypothesis. Hence we reach a contradiction, meaning there must be a labelling λ such that $\mathcal{G}$ contains no cycles of negative weight.

($\impliedby$) Let λ be a labelling of f for $i \in N$ such that $\mathcal{G} = \mathcal{G}_{i,f}^\lambda$ has no negative cycles. Consider the augmented NOM allocation graph $\mathcal{G}'$ obtained from $\mathcal{G}$ by introducing a node γ and edges from γ to each node in $\mathcal{G}$ with weight zero. as defined in Definition 10. Let p denote the resulting shortest path payments as given in (4.9). Notice that $\mathcal{G}'$ has no negative cycles: $\mathcal{G}$ contains no negative cycles by hypothesis, and adding γ creates no new cycles (since there are only outgoing edges from γ). Moreover, since $\mathcal{G}'$ is finite then shortest paths are well-defined. Now for any edge $(x, y; t_i)$ in $\mathcal{G}'$ then by definition of shortest paths we have $\text{SP}(\gamma, y) \leq \text{SP}(\gamma, x) + w(x, y; t_i)$ and hence $p_i(y) \geq p_i(x) - (t_i(f(x)) - t_i(f(y)))$. Therefore we have $t_i(f(x)) - p_i(x) \geq t_i(f(y)) - p_i(y)$ for any nodes x and y connected by type t_i in $\mathcal{G}'$, and therefore in $\mathcal{G}$. We will inspect the circumstances under which we have an edge $(x, y; t_i)$ and see that for any such edge we satisfy a constraint imposed by a NOM mechanism, and therefore f is implementable

with payment scheme p . Let b_i be the bid of player i in the profile x , that is, $x = (b_i, \mathbf{b}_{-i})$.

Case 1: $t_i = b_i$. In this case then $x = (t_i, \cdot)$ is a truthful bid profile for player i . The edge can either encode a labelling constraint or an incentive-compatibility constraint. Suppose it encodes the former. Then y must also be in the t_i -island of $\mathcal{G}$. Now either x corresponds to $(t_i, \bar{\lambda}(t_i, t_i))$ and y corresponds to some other profile in the t_i -island of $\mathcal{G}$, or y corresponds to $(t_i, \underline{\lambda}(t_i, t_i))$ and x corresponds to some other profile in the t_i -island of $\mathcal{G}$ (note that it can also be both). Then we have $u_i(M(t_i, \bar{\lambda}(t_i, t_i)); t_i) \geq u_i(M(t_i, \mathbf{b}_{-i}); t_i) \geq u_i(M(t_i, \underline{\lambda}(t_i, t_i)); t_i)$. Otherwise the edge encodes an incentive-compatibility constraint. In this case y has the form $y = (b'_i, \mathbf{b}'_{-i})$ for some $b'_i \neq t_i$. If x corresponds to the best-case truthful profile $(t_i, \bar{\lambda}(t_i, t_i))$ then y must be the best-case dishonest profile $(b'_i, \bar{\lambda}(t_i, b'_i))$ and we have the constraint $u_i(M(t_i, \bar{\lambda}(t_i, t_i)); t_i) \geq u_i(M(b'_i, \bar{\lambda}(t_i, b'_i)); t_i)$. Otherwise x must correspond to the worst-case truthful profile $(t_i, \underline{\lambda}(t_i, t_i))$ and y must correspond to the worst-case dishonest profile $(t_i, \underline{\lambda}(t_i, t_i))$, meaning we have the constraint $u_i(M(t_i, \underline{\lambda}(t_i, t_i)); t_i) \geq u_i(M(b'_i, \underline{\lambda}(t_i, b'_i)); t_i)$.

Case 2: $t_i \neq b_i$. In this case then the edge encodes a labelling constraint for a dishonest interaction of player i with type t_i in the mechanism. Now $y = (b_i, \cdot)$ is some profile in the b_i -island of $\mathcal{G}$. If x corresponds to the best-case dishonest profile $(b_i, \bar{\lambda}(t_i, b_i))$ then we have the constraint $u_i(M(b_i, \bar{\lambda}(t_i, b_i)); t_i) \geq u_i(M(b_i, \mathbf{b}_{-i}); t_i)$ for each $\mathbf{b}_{-i} \in D_{-i}$, and if y corresponds to the worst-case dishonest profile $(b_i, \underline{\lambda}(t_i, b_i))$ then we have the constraint $u_i(M(b_i, \mathbf{b}_{-i}); t_i) \geq u_i(M(b_i, \underline{\lambda}(t_i, b_i)); t_i)$ for each $\mathbf{b}_{-i} \in D_{-i}$.

Therefore for all edges $(x, y; t_i)$ in $\mathcal{G}'$ then by definition we satisfy either a labelling constraint or an incentive-compatibility constraint of the mechanism for some payment function p . Thus f is implementable as a NOM mechanism with p as its payment rule. $\square$

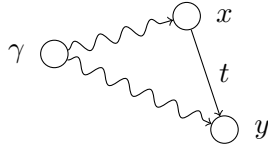


Figure 4.3: Shortest path payments as defined in Definition 10 and used in the proof of Theorem 3. The weight of edge $(x, y; t)$ above is equal to $t(f(x)) - t(f(y))$. The payment at bid profile x is set to the length of the shortest path from γ to x in the augmented graph $\mathcal{G}'$.

It is a natural question to ask whether we can efficiently decide whether there exists a labelling such that $\mathcal{G}_{i,f}^\lambda$ contains no negative cycles. We will

present this research question formally later in this section. For now we will demonstrate with the following example how the shortest path payments of Theorem 3 can be used to implement an obviously manipulable social choice function as NOM mechanism. We note that we can focus on BNOM-implementability and WNOM-implementability in isolation by only considering either the best-case labelling $\bar{\lambda}$ or worst-case labelling $\underline{\lambda}$, respectively. As we will show, it is only guaranteed that we achieve a NOM mechanism when we take both labellings simultaneously, as a BNOM payment scheme will not necessarily result in a WNOM payment scheme, and vice versa.

Example 3 (Computing NOM payment schemes). Recall the definition of the types θ_1, θ_2 from Table 4.1 and that $p_1(\mathbf{b}) = 0$ for all $\mathbf{b} \in D$ suffices for the mechanism to be NOM for player 1. Now that we are considering incentives from the perspective of player 2, note that we are comparing the maximum and minimum utilities of columns 1 and 3 with the maximum and minimum utilities of columns 2 and 4 for BNOM and WNOM, respectively. Note that $\max\{3, 2\} < \max\{1, 4\}$ hence θ_1 is an obvious manipulation (in the best case) when player 2 has type θ_1 , and $\min\{4, 1\} < \min\{2, 3\}$ hence θ_2 is an obvious manipulation (in the worst case) when player 2 has type θ_1 . We now define the following best and worst-case labellings for player 2 that result in a NOM payment scheme for f .

$$\bar{\lambda} = \begin{bmatrix} \theta_1 & \theta_1 \\ \theta_2 & \theta_1 \end{bmatrix}, \quad \underline{\lambda} = \begin{bmatrix} \theta_2 & \theta_2 \\ \theta_1 & \theta_2 \end{bmatrix}$$

Using the shortest path payments from Theorem 3 for BNOM and WNOM separately yields the payments given in Tables 4.3 and 4.4, and these result in BNOM and WNOM mechanisms in isolation. For each table entry (x, y) the value x corresponds to the utility player 2 receives on the given outcome (given her type) and the value y corresponds to the payment she is charged. Hence the final utility for each table entry is simply $x - y$. Notice that for each bid profile b (corresponding to the column) the same payment must be charged regardless of the player's type. We note that focusing only on BNOM payments may not result in a WNOM mechanism, and vice versa. Even though we achieve BNOM in Table 4.3 we do not have WNOM, and likewise we have WNOM in Table 4.4 but not BNOM. Obvious manipulations of player 2 are highlighted by red entries in the tables, and correspond to a worst-case obvious manipulation of θ_2 in Table 4.3 and a best-case obvious manipulation of θ_1 in Table 4.4. We only achieve a NOM payment scheme when we compute shortest path payments using both labellings $\bar{\lambda}$ and $\underline{\lambda}$ together, and the resulting mechanism is listed in Table 4.5.

	ω_{11}	ω_{12}	ω_{21}	ω_{22}
θ_1	(4, 0)	(2, 0)	(1, 1)	(3, 1)
θ_2	(1, 0)	(3, 0)	(4, 1)	(2, 1)

Table 4.3: BNOM payments for Example 1. Red entries represent a worst-case obvious manipulation of θ_2 , beating the worst truthful outcome in blue.

	ω_{11}	ω_{12}	ω_{21}	ω_{22}
θ_1	(4, 1)	(2, 0)	(1, 0)	(3, 2)
θ_2	(1, 1)	(3, 0)	(4, 0)	(2, 2)

Table 4.4: WNOM payments for Example 1. The red entry represents a best-case obvious manipulation of θ_1 , beating the best truthful outcome in blue.

	ω_{11}	ω_{12}	ω_{21}	ω_{22}
θ_1	(4, 2)	(2, 0)	(1, 1)	(3, 3)
θ_2	(1, 2)	(3, 0)	(4, 1)	(2, 3)

Table 4.5: NOM payments for Example 1.

As noted in Example 2 not every social choice function f admits a NOM payment scheme. In the following section we consider which cycles are guaranteed to exist in $\mathcal{G}_{i,f}^\lambda$ for any labelling λ .

4.3.4 GUARANTEED CYCLES IN THE ALLOCATION GRAPH

Suppose we have some best-case labelling $\bar{\lambda}$ for player i and let us focus on one specific b_i -island of $\mathcal{G} = \mathcal{G}_{i,f}^{\bar{\lambda}}$. Let t_i, t'_i be two types from i 's domain. Then the b_i -island of $\mathcal{G}$ must contain the profiles $(b_i, \bar{\lambda}(t_i, b_i))$ and $(b_i, \bar{\lambda}(t'_i, b_i))$, regardless of what exactly is the joint profile in either case of the other players. These represent the bid profile leading to i 's best outcome in the mechanism when she bids b_i and has type t_i or type t'_i , respectively. As per Definition 9 then we always have the two-cycle:

$$\langle (b_i, \bar{\lambda}(t_i, b_i)) \xrightarrow{t_i} (b_i, \bar{\lambda}(t'_i, b_i)) \xrightarrow{t'_i} (b_i, \bar{\lambda}(t_i, b_i)) \rangle.$$

Letting $\mathbf{b} = (b_i, \bar{\lambda}(t_i, b_i))$ and $\mathbf{b}' = (b_i, \bar{\lambda}(t'_i, b_i))$, the weight of this cycle is simply:

$$t_i(f(\mathbf{b})) - t_i(f(\mathbf{b}')) + t'_i(f(\mathbf{b}')) - t'_i(f(\mathbf{b})).$$

Note that if $\mathbf{b} = \mathbf{b}'$ then this weight is zero, and more generally if either t_i and t'_i map both $f(\mathbf{b})$ and $f(\mathbf{b}')$ to the same value, or t_i and t'_i map $f(\mathbf{b})$ to

the same value and t_i and t'_i map $f(\mathbf{b}')$ to the same value, then the weight is again zero.

More generally let us focus on the b_i -island of $\mathcal{G}$ and let $\theta_1, \theta_2, \dots, \theta_p$ be any p types from i 's domain. Write $\theta = (\theta_1, \dots, \theta_p)$. Then in the b_i -island $\mathcal{G}$ will contain the p -cycle $C_p^{\bar{\lambda}}(\theta; b_i)$:

$$C_p^{\bar{\lambda}}(\theta; b_i) = \{ \langle (b_i, \bar{\lambda}(\theta_k, b_i)) \xrightarrow{\theta_k} (b_i, \bar{\lambda}(\theta_{k+1}, b_i)) \rangle : k \in [p] \},$$

where the index k wraps around to 1 at $p + 1$. With similar reasoning we conclude that for any worst-case labelling $\bar{\lambda}$ of f for player i , for any bid b_i then the b_i -island of $\mathcal{G}$ contains the p -cycle $C_p^{\bar{\lambda}}(\theta; b_i)$:

$$C_p^{\bar{\lambda}}(\theta; b_i) = \{ \langle (b_i, \bar{\lambda}(\theta_{k-1}, b_i)) \xrightarrow{\theta_k} (b_i, \bar{\lambda}(\theta_k, b_i)) \rangle : k \in [p] \},$$

where index -1 now wraps around to p .

We can also consider edges that for some number p of types $\theta_1, \theta_2, \dots, \theta_p$ cross from the θ_k -island to the θ_{k+1} -island of $\mathcal{G}_{i,f}^{\bar{\lambda}}$ and $\mathcal{G}_{i,f}^{\lambda}$, respectively. Begin with any best-case labelling $\bar{\lambda}$ and note that for two types t_i and b_i the labelling must contain the edges:

$$\langle (t_i, \bar{\lambda}(t_i, t_i)) \xrightarrow{t_i} (b_i, \bar{\lambda}(t_i, b_i)) \xrightarrow{t_i} (b_i, \bar{\lambda}(b_i, b_i)) \xrightarrow{b_i} (b_i, \bar{\lambda}(b_i, t_i)) \xrightarrow{b_i} (t_i, \bar{\lambda}(t_i, t_i)) \rangle.$$

The first edge exists to encode the incentive-compatibility constraint when i bids t_i truthfully and b_i dishonestly (that is, her utility with true type t_i at outcome $M(t_i, \bar{\lambda}(t_i, t_i))$ is at least her utility with type t_i at outcome $M(b_i, \bar{\lambda}(t_i, b_i))$, comparing the best truthful and dishonest profiles under t_i , respectively). The second edge exists to encode the labelling constraint when i bids b_i dishonestly: since $(b_i, \bar{\lambda}(t_i, b_i))$ is the bid profile leading to her maximum utility in the mechanism when she has type t_i and bids b_i , then its associated utility must be at least the utility she receives under input $(b_i, \bar{\lambda}(b_i, b_i))$ when she has type t_i . We use similar reasoning, replacing i 's true type t_i with b_i , to argue for the existence of the third and fourth edges and thus close the cycle.

As before we can do this for any p types from i 's domain. Let $\theta = (\theta_1, \theta_2, \dots, \theta_p)$ for any p types. Then for any best-case labelling $\bar{\lambda}$ we have the cycle $C_{2p}^{\bar{\lambda}}(\theta)$:

$$C_{2p}^{\bar{\lambda}}(\theta) = \{ \langle (\theta_k, \bar{\lambda}(\theta_k, \theta_k)) \xrightarrow{\theta_k} (\theta_{k+1}, \bar{\lambda}(\theta_k, \theta_{k+1})) \xrightarrow{\theta_k} (\theta_{k+1}, \bar{\lambda}(\theta_{k+1}, \theta_{k+1})) \rangle : k \in [p] \}, \quad (4.10)$$

Similarly again for a worst-case labelling $\underline{\lambda}$ and any p types $\theta = (\theta_1, \theta_2, \dots, \theta_p)$ we have the cycle $C_{2p}^{\underline{\lambda}}$:

$$C_{2p}^{\underline{\lambda}}(\theta) = \{ \langle (\theta_k, \underline{\lambda}(\theta_k, \theta_k)) \xrightarrow{\theta_k} (\theta_{k+1}, \underline{\lambda}(\theta_k, \theta_{k+1})) \xrightarrow{\theta_{k+1}} (\theta_{k+1}, \underline{\lambda}(\theta_{k+1}, \theta_{k+1})) \rangle : k \in [p] \}, \quad (4.11)$$

Notice the type annotation of the “middle” edge in the above expression for $C_{2p}^{\underline{\lambda}}$ versus that of $C_{2p}^{\bar{\lambda}}$. Since $\underline{\lambda}$ describes worst-case profiles and edges represent a higher utility at the source node than the destination, the edge is annotated with type θ_{k+1} rather than θ_k .

Writing out the expression for the weight of the cycles $C_{2p}^{\bar{\lambda}}$ and $C_{2p}^{\underline{\lambda}}$ we notice that the sum telescopes. This happens because the type annotations are the same for the edges $\langle (\theta_k, \bar{\lambda}(\theta_k, \theta_k)) \xrightarrow{\theta_k} (\theta_{k+1}, \bar{\lambda}(\theta_k, \theta_{k+1})) \rangle$ and $\langle (\theta_{k+1}, \bar{\lambda}(\theta_k, \theta_{k+1})) \xrightarrow{\theta_k} (\theta_{k+1}, \bar{\lambda}(\theta_{k+1}, \theta_{k+1})) \rangle$ when k ranges from 1 to p , and likewise for the edges $(\theta_k, \underline{\lambda}(\theta_k, \theta_k)) \xrightarrow{\theta_k} (\theta_{k+1}, \underline{\lambda}(\theta_k, \theta_{k+1}))$ and $(\theta_{k+1}, \underline{\lambda}(\theta_k, \theta_{k+1})) \xrightarrow{\theta_{k+1}} (\theta_{k+1}, \underline{\lambda}(\theta_{k+1}, \theta_{k+1}))$. Hence the expressions for their weights are more succinctly given by:

$$w(C_{2p}^{\bar{\lambda}}) = \sum_{k \in [p]} \theta_k(f(\theta_k, \bar{\lambda}(\theta_k, \theta_k))) - \theta_k(f(\theta_{k+1}, \bar{\lambda}(\theta_{k+1}, \theta_{k+1}))), \quad (4.12)$$

$$w(C_{2p}^{\underline{\lambda}}) = \sum_{k \in [p]} \theta_k(f(\theta_k, \underline{\lambda}(\theta_k, \theta_k))) - \theta_k(f(\theta_{k+1}, \underline{\lambda}(\theta_k, \theta_{k+1}))). \quad (4.13)$$

The shorter the cycles we have to consider, the easier our job is when characterising the set of implementable social choice functions in a given setting. For other solution concepts such as strategyproofness and obvious strategyproofness, it suffices to consider two-cycles: [Saks and Yu, 2005] show that on convex domains it suffices to consider only two-cycles for strategyproofness, while [Ferraioli, Meier, et al., 2023] show a similar result for obvious strategyproofness on small domains.

We demonstrate that these guaranteed cycles are why the social choice function of Example 2 is not implementable as a NOM mechanism.

Example 4 (Not every function has a valid labelling). Recall the instance given in Example 2 where each type $\theta \in \{\theta_1, \theta_2\}$ is defined such that $\theta(f(\theta_i, \theta_j)) = 0$ if $\theta = \theta_i$, and $\theta(f(\theta_i, \theta_j)) = 1$ otherwise.

We first note that the instance does not admit BNOM payments for player 1. For any best-case labelling $\bar{\lambda}$ take the guaranteed cycle $\langle \bar{\lambda}(\theta_1, \theta_1) \xrightarrow{\theta_1} \bar{\lambda}(\theta_1, \theta_2) \xrightarrow{\theta_1} \bar{\lambda}(\theta_2, \theta_2) \xrightarrow{\theta_2} \bar{\lambda}(\theta_2, \theta_1) \xrightarrow{\theta_2} \bar{\lambda}(\theta_1, \theta_1) \rangle$. Then the first and third edges will have weight $\theta_1(f(\theta_1, \cdot)) - \theta_1(f(\theta_2, \cdot)) = \theta_2(f(\theta_2, \cdot)) - \theta_2(f(\theta_1, \cdot)) =$

$0 - 1 = -1$, while the second and fourth edges will have weight $\theta_1(f(\theta_2, \cdot)) - \theta_1(f(\theta_2, \cdot)) = \theta_2(f(\theta_1, \cdot)) - \theta_2(f(\theta_1, \cdot)) = 1 - 1 = 0$. Therefore the total weight of the guaranteed cycle is negative for any best-case labelling $\bar{\lambda}$. This is shown in Figure 4.4.

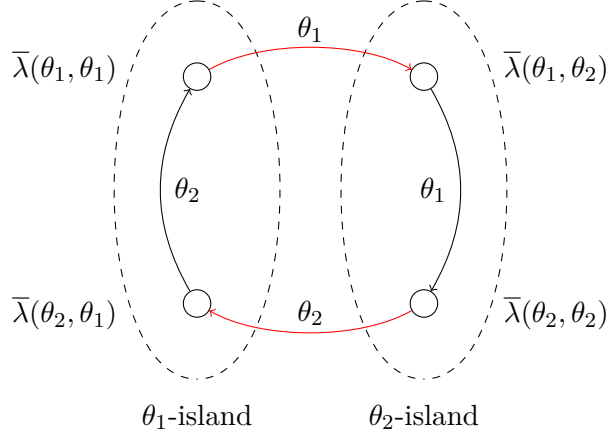


Figure 4.4: The guaranteed cycles due to any best-case labelling $\bar{\lambda}$ result in the negative cycle. Edges are annotated with type annotations. The red edges have weight -1 and the black edges weight 0 .

Likewise the social choice function does not admit a WNOM payment scheme. For any worst-case labelling $\underline{\lambda}$ take the guaranteed cycle $\langle \underline{\lambda}(\theta_1, \theta_1) \xrightarrow{\theta_1} \underline{\lambda}(\theta_1, \theta_2) \xrightarrow{\theta_2} \underline{\lambda}(\theta_2, \theta_2) \xrightarrow{\theta_2} \underline{\lambda}(\theta_2, \theta_1) \xrightarrow{\theta_1} \underline{\lambda}(\theta_1, \theta_1) \rangle$. Now the first and third edges will have weight $\theta_1(f(\theta_1, \cdot)) - \theta_1(f(\theta_2, \cdot)) = \theta_2(f(\theta_2, \cdot)) - \theta_2(f(\theta_1, \cdot)) = 0 - 1 = -1$, while the second and fourth edges will have weight $\theta_2(f(\theta_2, \cdot)) - \theta_2(f(\theta_2, \cdot)) = \theta_1(f(\theta_1, \cdot)) - \theta_1(f(\theta_1, \cdot)) = 0 - 0 = 0$. Therefore the total weight of the guaranteed cycle is negative for any worst-case labelling $\underline{\lambda}$. This is shown in Figure 4.5.

Of course, even if a function is implementable, not every labelling will lead to a valid payment scheme. Even though there may exist a labelling such that the resulting NOM allocation graph has no negative cycles, other labellings may indeed produce negative cycles, leading to an unsatisfiable set of constraints imposed by the mechanism.

Before studying implementability for single-parameter agents, we conclude this section by highlighting two conditions that are sufficient for a social choice function to be implementable as a NOM mechanism in general domains. Begin by recalling the definitions of best responses and dominant strategies.

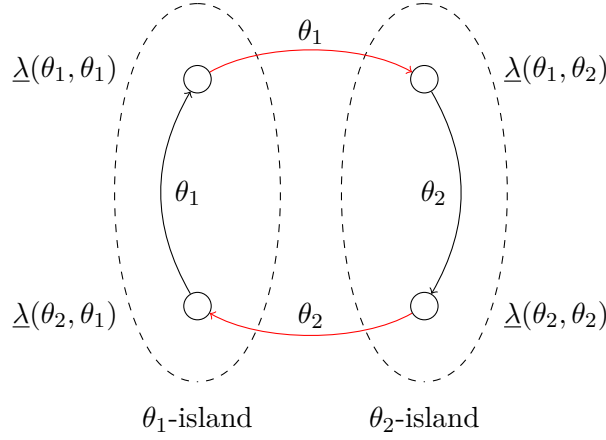


Figure 4.5: The guaranteed cycles due to any best-case labelling $\underline{\lambda}$ result in the negative cycle. Edges are annotated with type annotations. The red edges have weight -1 and the black edges weight 0 .

Definition 11 (Best response). Let $M = (f, p)$ be some mechanism and $i \in N$ be some agent with true type t_i . A *best response for agent i against bid profile $\mathbf{b}_{-i}$* is a strategy b_i^* that maximises i 's utility over all possible strategies from her domain:

$$b_i^* \in \arg \max_{b_i \in D_i} \{ u_i(M(b_i, \mathbf{b}_{-i}); t_i) \}.$$

In other words we have $u_i(M(b_i^*, \mathbf{b}_{-i}); t_i) \geq u_i(M(b_i, \mathbf{b}_{-i}); t_i)$ for each b_i in agent i 's domain.

Definition 12 (Dominant strategy). Let $M = (f, p)$ be some mechanism and $i \in N$. A strategy b_i^* is a *dominant strategy for agent i* if it is a best response against each joint profile $\mathbf{b}_{-i}$ of the other agents.

Recall that a mechanism is strategyproof if agent i reporting her true type t_i is a dominant strategy for each agent $i \in N$ and each $t_i \in D_i$. By Theorem 2 we know that in order for a social choice function to be implementable as a truthful mechanism then it must be cyclically monotone. Note that in the truthful allocation graph $\mathcal{G}_{i,f}$ edges only traverse between bid profiles of the form $(t_i, \mathbf{b}_{-i})$ and $(b_i, \mathbf{b}_{-i})$ for each pair of types t_i and b_i from i 's domain. As a result, all nodes $\mathbf{b}^{(1)}, \mathbf{b}^{(2)}, \dots \in D$ that may form a cycle in $\mathcal{G}_{i,f}$ will have the same partial profile $\mathbf{b}_{-i}$ of the other players, i.e., each $\mathbf{b}^{(j)}$ is of the form $(b_i^{(j)}, \mathbf{b}_{-i})$ where $b_i^{(j)}$ is some type from D_i and may differ between the profiles in the cycle. If a social choice function fails cycle monotonicity then it is possible this is only as a result of one

particular partial profile resulting in a negative cycle. Of course, there may be many partial profiles of the other players that result in negative cycles in the truthful allocation graph for a given player. However, as long as there exists some partial profile $\mathbf{b}_{-i}$ such that there are no negative cycles in the truthful allocation graph with its node set restricted to profiles of the form $(b_i, \mathbf{b}_{-i})$ ranging over $b_i \in D_i$ then we will see that it is possible to define a labelling of f without any negative cycles in the NOM allocation graph. We formalise this in Observation 2 and begin by defining this notion of “local cycle monotonicity” at a given partial profile.

Definition 13 (Local cycle monotonicity). Let $i \in N$ and $\mathbf{b}_{-i}$ be some joint profile of the other agents. Let $\mathcal{G}' = \mathcal{G}_{i,f}(V', E')$ be the truthful allocation graph for i as defined in Definition 6 with its node set V' (and any corresponding edges E') restricted to profiles of the form $(b_i, \mathbf{b}_{-i})$ where $b_i \in D_i$. Then f is *locally cyclically monotone for agent i at $\mathbf{b}_{-i}$* if $\mathcal{G}'$ contains no cycles of negative weight.

The idea behind the following observation is that, as long as there exists some partial profile $\mathbf{b}_{-i}$ such that f is locally monotone for agent i , it is straightforward to define a valid labelling $\lambda \in \{\bar{\lambda}, \underline{\lambda}\}$ such that $\mathcal{G}_{i,f}^\lambda$ has no negative cycles. As a result, we will have a BNOM or WNOM mechanism (depending on the labelling) in which truthtelling is a best response against this bid profile $\mathbf{b}_{-i}$. For the following observation we let $\bar{\lambda}$ -NOM stand for BNOM and $\underline{\lambda}$ -NOM for WNOM.

Observation 2. If there exists a $\mathbf{b}_{-i}$ such that social choice function f is locally monotone for agent i at $\mathbf{b}_{-i}$ then f is implementable as a λ -NOM mechanism for i , for $\lambda \in \{\bar{\lambda}, \underline{\lambda}\}$. Specifically, the labelling that sets $\lambda(t_i, b_i) = \mathbf{b}_{-i}$ results in no negative cycles in $\mathcal{G}_{i,f}^\lambda$ and hence produces payments p such that (f, p) is λ -NOM.

Proof. Let $\mathbf{b}_{-i}$ be a joint profile at which f is truthfully implementable and $\mathbf{b}'_{-i}$ a joint profile at which f is *not* truthfully implementable. Consider the truthful allocation graph $\mathcal{G}_{i,f}$. For any k types $\theta_1, \theta_2, \dots, \theta_k$, we have the k -cycles $C_{\geq 0}^k = \langle (\theta_1, \mathbf{b}_{-i}) \xrightarrow{\theta_1} (\theta_2, \mathbf{b}_{-i}) \xrightarrow{\theta_2} \dots \xrightarrow{\theta_{k-1}} (\theta_k, \mathbf{b}_{-i}) \xrightarrow{\theta_k} (\theta_1, \mathbf{b}_{-i}) \rangle$ and $C_{< 0}^k = \langle (\theta_1, \mathbf{b}'_{-i}) \xrightarrow{\theta_1} (\theta_2, \mathbf{b}'_{-i}) \xrightarrow{\theta_2} \dots \xrightarrow{\theta_{k-1}} (\theta_k, \mathbf{b}'_{-i}) \xrightarrow{\theta_k} (\theta_1, \mathbf{b}'_{-i}) \rangle$. Consider the best-case and worst-case labellings $\bar{\lambda}$ and $\underline{\lambda}$ separately. By setting $\bar{\lambda}(t_i, b_i) = \mathbf{b}_{-i}$ for each pair of types t_i, b_i observe that firstly no cycles are created in each b_i -island of $\mathcal{G}_{i,f}^{\bar{\lambda}}$, and secondly that the cycles induced by $\bar{\lambda}$ are exactly the cycles $C_{\geq 0}^k$ present in $\mathcal{G}_{i,f}$ which have non-negative weight. Hence all cycles in $\mathcal{G}_{i,f}^{\bar{\lambda}}$ have non-negative weight and by Theorem 3 f is

BNOM-implementable with labelling $\bar{\lambda}$. The same reasoning shows that f is WNOM-implementable with labelling $\underline{\lambda}(t_i, b_i) = \mathbf{b}_{-i}$ for each $t_i, b_i \in D_i$. In both cases, we may take as the NOM payment scheme p the shortest path payments as defined in Theorem 3. The resulting mechanism (f, p) is therefore λ -NOM. $\square$

While the above result is true for implementing f as a BNOM mechanism and as a WNOM mechanism separately, it may not be so simple to find a valid combined labelling, thus it may be harder to implement f as a NOM mechanism (i.e., BNOM and WNOM simultaneously). Note that simply setting $\bar{\lambda} = \underline{\lambda}$ for some combined labelling $\lambda = (\bar{\lambda}, \underline{\lambda})$ creates (many) two-cycles in each b_i -island of $\mathcal{G}_{i,f}^\lambda$, whose weights are in general not guaranteed to be non-negative. Let $\mathbf{b} = (b_i, \mathbf{b}_{-i})$ be some bid profile in the b_i -island of i 's domain and suppose $\bar{\lambda}(t_i, b_i) = \underline{\lambda}(t_i, b_i) = \mathbf{b}_{-i}$. Such a labelling creates the two-cycles $\langle \mathbf{b} \xrightarrow{t_i} (b_i, \mathbf{b}'_{-i}) \xrightarrow{t'_i} \mathbf{b} \rangle$ for each $\mathbf{b}'_{-i} \neq \mathbf{b}_{-i}$ and for each pair of “true” types $t_i, t'_i \in D_i$. Thus between $\mathbf{b}$ and each $\mathbf{b}'$ there are many parallel edges, each marked by a unique type in i 's domain. For f to be NOM-implementable then each of the two-cycles formed must have non-negative weight. Let $\delta = \min_{t_i, t'_i} \{ t_i(f(\mathbf{b})) - t_i(f(\mathbf{b}')) + t'_i(f(\mathbf{b}')) - t'_i(f(\mathbf{b})) \}$ denote the minimum weight of a two-cycle between nodes $\mathbf{b}$ and $\mathbf{b}'$ induced by λ . Clearly $\delta \leq 0$ when $t_i = t'_i$. In fact we have:

$$\begin{aligned} \delta &= \min_{t_i, t'_i} \{ t_i(f(\mathbf{b})) - t_i(f(\mathbf{b}')) + t'_i(f(\mathbf{b}')) - t'_i(f(\mathbf{b})) \} \\ &= \min_{t_i} \{ t_i(f(\mathbf{b})) - t_i(f(\mathbf{b}')) \} + \min_{t_i} \{ t_i(f(\mathbf{b}')) - t_i(f(\mathbf{b})) \} \\ &= \min_{t_i} \{ t_i(f(\mathbf{b})) - t_i(f(\mathbf{b}')) \} - \max_{t_i} \{ t_i(f(\mathbf{b})) - t_i(f(\mathbf{b}')) \}, \end{aligned}$$

which is only non-negative if $t_i(f(\mathbf{b})) = t_i(f(\mathbf{b}'))$ for all types $t_i \in D_i$.

Observation 2 says that a valid labelling is only difficult to find whenever either the social choice function f is manipulable for i at each joint profile $\mathbf{b}_{-i}$ of the other bidders, or if we want a labelling leading to a NOM mechanism (that is, BNOM and WNOM simultaneously). Given that a valid λ -NOM labelling is easy to find whenever f is locally monotone for i at some $\mathbf{b}_{-i}$, with the next observation we provide a necessary condition for a labelling to be valid when this is *not* the case and f is manipulable at every $\mathbf{b}_{-i}$. If f is not locally monotone for i at any bid profile $\mathbf{b}_{-i}$ then for every $\mathbf{b}_{-i}$ there must be some pair of types $t_i, b_i \in D_i$ for which we cannot define a payment rule p that satisfies both $u_i(M(t_i, \mathbf{b}_{-i}); t_i) \geq u_i(M(b_i, \mathbf{b}_{-i}); t_i)$ and $u_i(M(b_i, \mathbf{b}_{-i}); b_i) \geq u_i(M(t_i, \mathbf{b}_{-i}); b_i)$. Therefore by Theorem 2 the

weight of the cycle $\langle (t_i, \mathbf{b}_{-i}) \xrightarrow{t_i} (b_i, \mathbf{b}_{-i}) \xrightarrow{b_i} (t_i, \mathbf{b}_{-i}) \rangle$ in $\mathcal{G}_{i,f}$ is negative. Any valid labelling of f must therefore avoid these two-cycles, as we show below.

Observation 3. If social choice function f is not locally monotone for i at any $\mathbf{b}_{-i}$ then in order for $\bar{\lambda}$ to be a valid best-case labelling of f and $\underline{\lambda}$ to be a valid worst-case labelling of f it must be that for each $t_i, b_i \in D_i$ we have $\bar{\lambda}(t_i, t_i) \neq \bar{\lambda}(b_i, b_i)$ and $\underline{\lambda}(t_i, b_i) \neq \underline{\lambda}(b_i, t_i)$ respectively.

Proof. Take the guaranteed four-cycle between two types t_i and b_i in any labelling $\bar{\lambda}$ from (4.10). The weight of this cycle is $t_i(f(t_i, \bar{\lambda}(t_i, t_i))) - t_i(f(b_i, \bar{\lambda}(b_i, b_i))) + b_i(f(b_i, \bar{\lambda}(b_i, b_i))) - b_i(f(t_i, \bar{\lambda}(t_i, t_i)))$ in the NOM allocation graph $\mathcal{G}_{i,f}^{\bar{\lambda}}$. For each $\mathbf{b}_{-i}$, since f is not locally monotone for i then there must be a pair of types t_i, b_i such that the weight of the cycle $\langle (t_i, \mathbf{b}_{-i}) \xrightarrow{t_i} (b_i, \mathbf{b}_{-i}) \xrightarrow{b_i} (t_i, \mathbf{b}_{-i}) \rangle$ is negative in the truthful allocation graph $\mathcal{G}_{i,f}$. Note that if we have $\bar{\lambda}(t_i, t_i) = \bar{\lambda}(b_i, b_i)$ then the weight of the guaranteed four-cycle in $\mathcal{G}_{i,f}^{\bar{\lambda}}$ is equal to the weight of the two-cycle in $\mathcal{G}_{i,f}$. Thus in order for $\mathcal{G}_{i,f}^{\bar{\lambda}}$ to have no negative cycles we must have $\bar{\lambda}(t_i, t_i) \neq \bar{\lambda}(b_i, b_i)$ for each t_i, b_i when f is not locally monotone.

Similar reasoning reveals an analogous condition for valid worst-case labellings of f . The weight of the guaranteed four-cycle for $\underline{\lambda}$ in $\mathcal{G}_{i,f}^{\underline{\lambda}}$ is $t_i(f(t_i, \underline{\lambda}(b_i, t_i))) - t_i(f(b_i, \underline{\lambda}(t_i, b_i))) + b_i(f(b_i, \underline{\lambda}(t_i, b_i))) - b_i(f(t_i, \underline{\lambda}(b_i, t_i)))$. Now if we have $\underline{\lambda}(t_i, b_i) = \underline{\lambda}(b_i, t_i)$ then this weight is exactly that of the negative two-cycle in $\mathcal{G}_{i,f}$. Hence in order for $\mathcal{G}_{i,f}^{\underline{\lambda}}$ to have no negative cycles then we require $\underline{\lambda}(t_i, b_i) \neq \underline{\lambda}(b_i, t_i)$ for each t_i, b_i whenever f is not locally monotone. $\square$

Before restricting attention to single-parameter agents, we will briefly introduce the complexity of determining a valid labelling. We want to answer the following question: given $f : D \rightarrow \Omega$, does there exist a payment function $p : D \rightarrow \mathbb{R}^n$ such that for each agent $i \in N$ the mechanism (f, p) is NOM? In order to tackle this question from a computational perspective we must first specify an input representation.

Definition 14 (NOM-IMPLEMENTABILITY). Suppose we are given a social choice function $f : D \rightarrow \Omega$ and player $i \in N$ whose domain D_i is provided explicitly in a table where for each row corresponding to a type $t_i \in D_i$, there is an entry $t_i(f(\mathbf{b}))$ for each column corresponding to a bid profile $\mathbf{b} \in D$. Does there exist a payment function $p : D \rightarrow \mathbb{R}^n$ such that (f, p) is not obviously manipulable?

For the following it suffices to have an informal description of the complexity classes P, NP, and NP-complete. We refer the reader to Arora and

Barak (2009) for a comprehensive introduction to computational complexity. Roughly speaking, the class P contains all computational problems that can be solved in a number of computational steps that is polynomial in the size of the input (i.e., polynomial time). The class NP , a superset of P , contains all problems for which it is possible to verify that a solution is correct in polynomial time. A problem is said to be NP -hard if it is at least as hard (with respect to polynomial time reductions) as every problem in NP . The class NP -complete contains all problems that are both NP and NP -hard. The first problem shown to be NP -complete was SATISFIABILITY, which asks if there exists an assignment of truth values to variables such that a given Boolean formula is satisfied (Cook, 1971), and many more have been shown since (see, e.g., [Garey and Johnson, 1979] for a classic reference work on NP -completeness). If any NP -complete problem is found to have a polynomial-time algorithm, then every NP -complete problem would be polynomial-time solvable, settling an important and long-standing open problem (Cook, 1971; L. A. Levin, 1973), although it is widely believed that this is not the case.

The table in Definition 14 contains $|D_i| \cdot |D|$ entries. By Theorem 3 this is equivalent to the question: does there exist a labelling $\lambda_i = (\bar{\lambda}_i, \underline{\lambda}_i)$ such that the graph $\mathcal{G}_{i,f}^{\lambda_i}$ has no negative cycles? In order to specify a labelling λ_i we must specify a partial bid profile of the other players $\bar{\lambda}_i(t_i, b_i)$ and $\underline{\lambda}_i(t_i, b_i)$ for each pair of types t_i, b_i in i 's domain, for a total of $|D_i|^2$ bid profiles. For each of these labels $\bar{\lambda}_i(t_i, b_i)$ and $\underline{\lambda}_i(t_i, b_i)$ we can choose from $|D_{-i}|$ different profiles of the other players. Therefore the total number of possible labellings for player i is equal to $2 \cdot |D_{-i}|^{d^2}$, where $d = |D_i|$.

Given a labelling, with the input as specified in Definition 14 we can construct the graph $\mathcal{G}_{i,f}^{\lambda}$ and determine the existence of negative cycles using, for example, the well-known Bellman-Ford algorithm (see Cormen et al. (2009)). Therefore NOM -IMPLEMENTABILITY is in NP . It would appear that finding a valid labelling in a computationally efficient manner is non-trivial. We leave the following problem open.

Open problem 1. Is there a polynomial-time algorithm for deciding NOM -IMPLEMENTABILITY?

4.4 IMPLEMENTABILITY FOR SINGLE-PARAMETER AGENTS

In this section we restrict our attention to agents who are said to be “single-parameter”. In this setting the private type of each agent can be encoded as a single number and is applicable to a large variety of scenarios of interest

in mechanism design. For example, in a single-item auction each agent i has a private valuation v_i for winning the item. Therefore we can encode each alternative as a binary vector $A = (A_1, A_2, \dots, A_n) \in \{0, 1\}^n$ and represent agent i 's type t_i as the function $t_i(A_i) = v_i \cdot A_i$. This can apply to other settings than simple binary allocation auctions. If there is a single *divisible* item then we can model each allocation as a vector $(A_1, \dots, A_n) \in [0, 1]^n$, where A_i describes the proportion of the item that i is allocated. Similarly, if we have a multi-item auction that seeks to sell off some number k identical copies of a single item then we can model each allocation as a vector $(A_1, \dots, A_n) \in \{0, \dots, k\}^n$ where A_i describes the number of copies that i is allocated. In both cases, as long as each agent i 's valuation functions is linear in their allocation A_i then their type can be represented by the product $v_i \cdot A_i$.

Recall that a social choice function f maps bid profiles from D to alternatives from Ω . Throughout this section we assume that Ω is modelled by some subset of $\mathbb{R}^n$. In this case we instead refer to f as an allocation function and denote it instead as $a : D \rightarrow \mathcal{A}$ where $\mathcal{A} \subseteq \mathbb{R}^n$. We assume no externalities, meaning each agent i derives utility only from her own allocation A_i . Hence for an agent i with type t_i we write her utility on allocation $A = (A_1, \dots, A_n) \in \mathcal{A}$ given her type $t_i \in D_i$ as the product $t_i(A) = t_i \cdot A_i$. In this way we can interpret each agent's type as her "unit cost" or "unit valuation", since it describes the utility she gets per unit of whatever she is allocated under A .

Single-parameter agents are a well-studied setting within mechanism design. [Archer and Tardos, 2001] provide a characterisation for the class of allocation functions that are implementable as truthful mechanisms when agents have single-parameter private costs (though their approach applies to valuation functions too) and also provide explicit formulae for the payment schemes that achieve incentive-compatibility. They apply their characterisation to design truthful mechanisms for a number of strategic settings in combinatorial optimisation where the VCG mechanism does not apply, including the problems of scheduling related parallel machines, scheduling related machines to minimise the sum of completion times, optimising an affine function over a fixed set, and special cases of uncapacitated facility location. Other applicable settings for single-parameter agents are combinatorial auctions with single-minded bidders (Archer, Papadimitriou, et al., 2003), machine scheduling (Andelman and Mansour, 2006), and payment computation (Dobzinski and Ron, 2021).

Our goal here is to apply the machinery of the previous sections to study which allocation functions are implementable as direct NOM mechanisms

when agents are single-parameter. For the following, let $a : D \rightarrow \mathcal{A}$ be an allocation function and let $i \in N$ be an agent with domain D_i .

Definition 15 (Allocation range). Let $i \in N$. The *range of allocations to i at b_i* is the set $\mathcal{A}_i(b_i) = \{a_i(b_i, \mathbf{b}_{-i}) : \mathbf{b}_{-i} \in D_{-i}\}$ of all possible allocations to player i when she bids b_i . Moreover, the *range of allocations to i* is simply the Cartesian product $\mathcal{A}_i = \times_{b_i \in D_i} \mathcal{A}_i(b_i)$ of the range of allocations to i at each b_i .

To ease notation, for an element $\mathbf{a} \in \mathcal{A}_i$ in the allocation range to i and a bid type $\theta_j \in D_i$ denote by $\mathbf{a}_j = a_i(\theta_j, \mathbf{b}_{-i})$ the allocation corresponding to i 's bid of θ_j for some $\mathbf{b}_{-i}$. In other words, we enumerate i 's domain with subscript j and use this to index the corresponding allocation in $\mathbf{a}$. We now show some necessary conditions for labellings to induce no negative cycles in the NOM allocation graph.

4.4.1 NECESSARY CONDITIONS OF VALID LABELLINGS

Recall the definition of the truthful allocation graph $\mathcal{G} = \mathcal{G}_{i,a}$ for player i . Consider two types $t_i, b_i \in D_i$ and some partial profile $\mathbf{b}_{-i}$, and let $\mathbf{b} = (t_i, \mathbf{b}_{-i})$ and $\mathbf{b}' = (b_i, \mathbf{b}_{-i})$. Consider the two-cycle $C = \langle \mathbf{b} \xrightarrow{t_i} \mathbf{b}' \xrightarrow{b_i} \mathbf{b} \rangle$ which appears in $\mathcal{G}$. Its weight $w(C)$ is given by:

$$\begin{aligned} w(C) &= t_i(a_i(\mathbf{b})) - t_i(a_i(\mathbf{b}')) + b_i(a_i(\mathbf{b}')) - b_i(a_i(\mathbf{b})) \\ &= t_i(a_i(\mathbf{b}) - a_i(\mathbf{b}')) - b_i(a_i(\mathbf{b}) - a_i(\mathbf{b}')) \\ &= (t_i - b_i)(a_i(\mathbf{b}) - a_i(\mathbf{b}')) \end{aligned}$$

where $t_i(a_i(\mathbf{b})) = t_i \cdot a_i(\mathbf{b})$ since agents are single-parameter. Now in order for a to be truthfully implementable then $\mathcal{G}$ must have no negative cycles, hence $(t_i - b_i)(a_i(\mathbf{b}) - a_i(\mathbf{b}'))$ for all bids t_i, b_i and all joint profiles $\mathbf{b}_{-i}$. In other words, if a is implementable then either $t_i > b_i$ and $a_i(\mathbf{b}) \geq a_i(\mathbf{b}')$, or $b_i > t_i$ and $a_i(\mathbf{b}') \geq a_i(\mathbf{b})$. This is the well-known “monotonicity” condition for strategyproof mechanisms, since the allocation function must be monotone in player i 's bid for each $\mathbf{b}_{-i}$.

Casting our attention to NOM for single-parameter agents, we derive similar necessary conditions on the best-case and worst-case labellings.

Property 1 (Monotone labelling). Let $i \in N$. The labelling λ_i is said to be a *monotone labelling for i* if for all $t_i, b_i \in D_i$ it holds that:

$$(t_i - b_i)(a_i(t_i, \bar{\lambda}(t_i, t_i)) - a_i(b_i, \bar{\lambda}(b_i, b_i))) \geq 0, \quad (4.14)$$

$$(t_i - b_i)(a_i(t_i, \underline{\lambda}(b_i, t_i)) - a_i(b_i, \underline{\lambda}(t_i, b_i))) \geq 0. \quad (4.15)$$

With the following claim we show that if f is implementable as NOM mechanism for player i then the labelling for player i must be monotone.

Claim 1. Let λ_i be a labelling of f for player i . If $\mathcal{G} = \mathcal{G}_{i,f}^{\lambda_i}$ has no negative cycles then λ_i satisfies Property 1.

Proof. Let t_i, b_i be any two types from D_i . Since a is NOM-implementable then for any cycle in $\mathcal{G}$ its weight must be non-negative. The claim for both $\bar{\lambda}$ and $\underline{\lambda}$ follows immediately from the guaranteed 4-cycles given in (4.10) and (4.11) and their corresponding weights from (4.12) and (4.13). That is, we have $t_i \geq b_i$ if and only if $a_i(t_i, \bar{\lambda}(t_i, t_i)) \geq a_i(t_i, \bar{\lambda}(b_i, b_i))$ and $a_i(t_i, \underline{\lambda}(b_i, t_i)) \geq a_i(b_i, \underline{\lambda}(t_i, b_i))$. $\square$

Property 2 (Stacked labels). Let $i \in N$. The labelling λ_i is said to use *stacked labels for i* if for all $b_i \in D_i$ it holds that:

$$(t_i - t'_i)(a_i(b_i, \bar{\lambda}(t_i, b_i)) - a_i(b_i, \bar{\lambda}(t'_i, b_i))), \quad (4.16)$$

$$(t_i - t'_i)(a_i(b_i, \underline{\lambda}(t'_i, b_i)) - a_i(b_i, \underline{\lambda}(t_i, b_i))). \quad (4.17)$$

for each pair of true types $t_i, t'_i \in D_i$.

Claim 2. Let λ be a labelling of f for player i . If $\mathcal{G} = \mathcal{G}_{i,f}^{\lambda_i}$ has no negative cycles then λ_i satisfies Property 2.

Proof. Begin by noting that for any labelling λ are guaranteed the edges $(b_i, \bar{\lambda}(t_i, b_i)) \xrightarrow{t_i} (b_i, \bar{\lambda}(t'_i, b_i))$ for any pair of types $t_i, t'_i \in D_i$, encoding the constraint that the utility at the source node is no less than the utility at the destination. Since these edges exist for every pair of types t_i, t'_i then we are guaranteed two-cycles of the form $(b_i, \bar{\lambda}(t_i, b_i)) \xrightarrow{t_i} (b_i, \bar{\lambda}(t'_i, b_i)) \xrightarrow{t'_i} (b_i, \bar{\lambda}(t_i, b_i))$. It is worth noting that there is no particular reason that there is an edge from $(b_i, \bar{\lambda}(t_i, b_i))$ to $(b_i, \bar{\lambda}(t'_i, b_i))$ and vice versa, i.e., from one labelled bid profile to another, rather it simply follows from the definition of the labelled graph $\mathcal{G}$ and the fact that labelled profiles have edges outgoing to *every* other bid profile in the same b_i -island of $\mathcal{G}$. By Theorem 3 each of these cycles must be non-negative, hence $(t_i - t'_i)(a_i(b_i, \bar{\lambda}(t_i, b_i)) - a_i(b_i, \bar{\lambda}(t'_i, b_i))) \geq 0$.

The reasoning follows exactly for the worst-case labelled profiles in λ . We are guaranteed the edges $(b_i, \underline{\lambda}(t'_i, b_i)) \xrightarrow{t_i} (b_i, \underline{\lambda}(t_i, b_i))$ for each t_i, t'_i in the b_i -island of $\mathcal{G}$. Again we require the resulting two-cycles to be non-negative, hence $(t_i - t'_i)(a_i(b_i, \bar{\lambda}(t'_i, b_i)) - a_i(b_i, \bar{\lambda}(t_i, b_i))) \geq 0$. $\square$

Of course these are simply necessary conditions on valid labellings that induce NOM mechanisms and follow straightforwardly from Theorem 3, the

cycles we are guaranteed to have in $\mathcal{G}$, and the fact that agents are single-parameter. With the following we provide a more complete picture on the class of social choice functions that may be implemented as NOM mechanisms for single-parameter agents.

4.4.2 A CHARACTERISATION FOR SINGLE-PARAMETER AGENTS

As a first step to studying the class of implementable allocation functions for single-parameter agents we restrict our attention to a class of particularly simple labellings. Given a bid b_i from player i 's domain, this labelling scheme λ simply designates the same extreme-outcome bid profile regardless of the player's true type. As it happens, such simple labellings completely describe which social choice functions admit NOM payment schemes.

First notice that this simple labelling does not induce any cycles at all (and importantly, no negative cycles) in the b_i -island of $\mathcal{G}_{i,f}^\lambda$, hence (4.16) and (4.17) holds trivially and we need only to focus on satisfying (4.14) and (4.15). By considering which social choice functions admit such labelling schemes we in fact arrive at a full characterisation of the functions implementable as NOM mechanisms for single-parameter agents. With the following we show that a function must be “overlapping” if and only if it is to be implementable. When this is the case, we can in fact always find a simple labelling resulting in a NOM mechanism, which we formally define in Definition 17. It is also possible that when a function is implementable for single-parameter agents there are more general labelling schemes that lead to NOM mechanisms, as we will show in Chapter 5. For now, we introduce the notion of what it means for an allocation function to be overlapping, and hence, implementable for single-parameter agents.

At a high level the overlapping property can be described as follows. Consider all the possible allocations in the range of allocations $\mathcal{A}_i$ to player i . Ultimately our goal is to select a vector of allocations, one from each range of allocations at $\mathcal{A}_i(b_i)$ for each bid b_i , that is non-decreasing in i 's bid. This ensures that we can then place a label at a bid profile that leads to this allocation (given i 's corresponding bid of b_i) and the amount allocated to i will never decrease as her bid increases. In selecting this non-decreasing vector of allocations (if it is indeed possible to do so), we will be creating a “monotone restriction” of f with respect to i 's bid type. An allocation function is overlapping if it is always possible to find such a vector of allocations.

Definition 16 (Overlapping). The allocation function a is said to be *overlapping for i* if there exists a non-decreasing vector $\mathbf{a} \in \mathcal{A}_i$ such that for

each $\theta_j, \theta_k \in D_i$ with $\theta_j \leq \theta_k$ we have $\mathbf{a}_j \leq \mathbf{a}_k$. The allocation function a is *overlapping* if it is overlapping for each agent $i \in N$.

In the following characterisation for the class of implementable allocation functions for single-parameter agents we use the fact that a is overlapping to construct a labelling $\mathcal{G} = \mathcal{G}_{i,a}^\lambda$ that contains no negative cycles in $\mathcal{G}$. For this we will use a pseudo-inverse function $\hat{a} : D_i \times \mathcal{A}_i \rightarrow D$ which given a bid θ_j of player i and an outcome $\mathbf{a}_j \in \mathcal{A}_i(\theta_j)$ returns a bid profile $\mathbf{b} = (\theta_j, \mathbf{b}_{-i})$ such that $a_i(\mathbf{b}) = \mathbf{a}_j$. We will assume a finite domain to ease the exposition and notation, however the arguments carry over directly if we relax this assumption to countable and even uncountable domains.

Theorem 4. *Allocation function $a : D \rightarrow \mathcal{A}$ is implementable as a NOM mechanism for single-parameter domains if and only if a is overlapping.*

Proof. ($\implies$) Assume for the sake of contradiction that there are two bids $\theta_1, \theta_2 \in D_i$ with $\theta_1 < \theta_2$ such that a is not overlapping for i . That is, for any choice of $\mathbf{a}_1 \in \mathcal{A}_i(\theta_1)$ then for all $\mathbf{a}_2 \in \mathcal{A}_i(\theta_2)$ we have $\mathbf{a}_2 < \mathbf{a}_1$. Since a is implementable then there exists a labelling λ such that the graph $\mathcal{G} = \mathcal{G}_{i,a}^\lambda$ contains no cycles of negative weight by Theorem 3. By Property 1 we require $(\theta_1 - \theta_2)(a_i(\theta_1, \lambda(\theta_2, \theta_1)) - a_i(\theta_2, \lambda(\theta_1, \theta_2))) \geq 0$, so $a_i(\theta_1, \lambda(\theta_2, \theta_1)) \leq a_i(\theta_2, \lambda(\theta_1, \theta_2))$. Since $\mathbf{a}_2 < \mathbf{a}_1$ for any choice of $\mathbf{a}_1 \in \mathcal{A}_i(\theta_1)$ then we must have $a_i(\theta_1, \lambda(\theta_j, \theta_1)) > a_i(\theta_2, \lambda(\theta_k, \theta_2))$ for all $\theta_j, \theta_k \in D_i$, where θ_j and θ_k are not necessarily distinct. Thus a cannot be implementable for i , a contradiction. Therefore a must be overlapping for i .

($\impliedby$) We will use the overlaps of a for player i to define a valid worst-case labelling $\underline{\lambda}$ and note that the exact same reasoning carries over for best-case labellings $\bar{\lambda}$. Define iteratively the values:

$$\underline{\lambda}(\theta_j) = \max\{\min\{\mathbf{a}_j \in \mathcal{A}_i(\theta_j)\}, \underline{\lambda}(\theta_1), \underline{\lambda}(\theta_2), \dots, \underline{\lambda}(\theta_{j-1})\},$$

for $\theta_j \in D_i$. Note that $\underline{\lambda}(\theta_j) \geq \underline{\lambda}(\theta_k)$ if and only if $\theta_j \geq \theta_k$.

We will use $\underline{\lambda}$ to label a with $\underline{\lambda}(\theta_j, \theta_k) = \hat{a}_i(\theta_k, \underline{\lambda}(\theta_k))$ for each $\theta_j, \theta_k \in D_i$. Since for every bid θ_k we have $\underline{\lambda}(\theta_j, \theta_k) = \underline{\lambda}(\theta_\ell, \theta_k)$ for all types $\theta_k, \theta_\ell \in D_i$ we use a single type θ_k to denote this bid profile $\underline{\lambda}(\theta_k)$. We now show that no cycles that appear in the graph $\mathcal{G}$ are negative and hence $\underline{\lambda}$ is a valid worst-case labelling.

First note that since for every bid θ_j we label the worst-case outcome for i to occur at the same bid profile $\underline{\lambda}(\theta_j)$ no matter her type, then each subgraph of $\mathcal{G}$ containing only nodes $(\theta_j, \mathbf{b}_{-i})$ in which i has bid θ_j will have no cycles, and these edges represent the labelling constraints. Hence we need only focus on edges which go from node $(\theta_j, \underline{\lambda}(\theta_j))$ to some $(\theta_k, \underline{\lambda}(\theta_k))$,

which represent the incentive-compatibility constraints. Recall that the type annotation on each of these edges is always equal to the bid of player i at the source node, so every edge has the form $(\theta_u, \underline{\lambda}(u)) \xrightarrow{\theta_u} (\theta_v, \underline{\lambda}(\theta_v))$. Consider any cycle C in $\mathcal{G}$ with weight $w(C) = \sum_{(u,v;\theta_u) \in C} \theta_u(a_i(\underline{\lambda}(\theta_u)) - a_i(\underline{\lambda}(\theta_v)))$. We will refer to negative weight edges in C as “uphill” edges and positive weight edges as “downhill” edges. We can safely ignore edges of zero weight. Note that uphill edges occur when the output of a for player i is lower at the source than the destination profile, and since $\underline{\lambda}(\theta_j) \geq \underline{\lambda}(\theta_k)$ if and only if $\theta_j \geq \theta_k$ then all such edges go from a node in which i bids a smaller type to one in which she bids a larger one. The opposite is true for downhill edges. For a given bid θ_j we will refer to the types $\max\{\theta_k \in D_i : \theta_k < \theta_j\}$ and $\min\{\theta_k \in D_i : \theta_k > \theta_j\}$ as the adjacent type to θ_j , and adjacent nodes in $\mathcal{G}$ will be those in which i submits adjacent types as bids to the mechanism. We will show that the weight of C is bounded below by the weight of a (non-simple) cycle C' which only contains edges between adjacent nodes, whose weight is in turn always non-negative.

In the following we will use simply the label $\underline{\lambda}(\theta_j)$ as shorthand for the bid profile $(\theta_j, \underline{\lambda}(\theta_j))$. First consider a downhill edge $e^+ = \langle \underline{\lambda}(\theta_j) \xrightarrow{\theta_j} \underline{\lambda}(\theta_{j-k}) \rangle$ going directly from $\underline{\lambda}(\theta_j)$ to $\underline{\lambda}(\theta_{j-k})$ that “skips” $k-1$ adjacent nodes. Since $\theta_j > \theta_{j-k}$ then $\underline{\lambda}(\theta_j) = a_i(\theta_j, \underline{\lambda}(\theta_j)) > a_i(\theta_{j-k}, \underline{\lambda}(\theta_{j-k})) = \underline{\lambda}(\theta_{j-k})$. The weight $w(e^+)$ of such an edge is therefore:

$$\begin{aligned} w(e^+) &= \theta_j[a_i(\underline{\lambda}(\theta_j)) - a_i(\underline{\lambda}(\theta_{j-k}))] \\ &= \theta_j \sum_{\ell=j-k}^{j-1} a_i(\underline{\lambda}(\theta_{\ell+1})) - a_i(\underline{\lambda}(\theta_\ell)) \\ &= \sum_{\ell=j-k}^{j-1} \theta_j[a_i(\underline{\lambda}(\theta_{\ell+1})) - a_i(\underline{\lambda}(\theta_\ell))] \\ &\geq \sum_{\ell=j-k}^{j-1} \theta_\ell[a_i(\underline{\lambda}(\theta_{\ell+1})) - a_i(\underline{\lambda}(\theta_\ell))] \end{aligned}$$

The second equality follows from writing the first line as a telescoping sum, meaning all but the “outer” two terms cancel out. For the third equality we simply distribute θ_j across the each of the terms in the sum for easier comparison with the final line, where the inequality holds because each of the θ_ℓ terms at most θ_j and each term $a_i(\underline{\lambda}(\theta_{\ell+1})) - a_i(\underline{\lambda}(\theta_\ell))$ is non-negative.

Following the same reasoning we can show that the weight of an uphill edge $e^- = \langle (\theta_j, \underline{\lambda}(\theta_j)) \xrightarrow{\theta_j} (\theta_{j+k}, \underline{\lambda}(\theta_{j+k})) \rangle$, in which we skip $k-1$ adjacent

nodes, can be similarly lower bounded:

$$\begin{aligned}
w(e^+) &= \theta_j [a_i(\underline{\lambda}(\theta_j)) - a_i(\underline{\lambda}(\theta_{j+k}))] \\
&= \theta_j \sum_{\ell=j}^{j+k-1} [a_i(\underline{\lambda}(\theta_\ell)) - a_i(\underline{\lambda}(\theta_{\ell+1}))] \\
&= \sum_{\ell=j}^{j+k-1} \theta_j [a_i(\underline{\lambda}(\theta_\ell)) - a_i(\underline{\lambda}(\theta_{\ell+1}))] \\
&\geq \sum_{\ell=j}^{j+k-1} \theta_\ell [a_i(\underline{\lambda}(\theta_\ell)) - a_i(\underline{\lambda}(\theta_{\ell+1}))]
\end{aligned}$$

We again rewrite the expression for the weight of the edge as a telescoping sum. The final inequality now holds because each of the θ_ℓ terms are at least θ_j , and each term $a_i(\underline{\lambda}(\theta_\ell)) - a_i(\underline{\lambda}(\theta_{\ell+1}))$ is non-positive.

Notice in both cases that the right hand side of the final inequality is the expression for the weight of a path between two non-adjacent nodes that only traverses edges between adjacent nodes. Now if we take any cycle C and replace all such non-adjacent edges with a path comprising only adjacent edges then we will be left with a non-simple cycle that is composed of many two-cycles (see Figure 4.6). We know from the labelling monotonicity property of our labelling $\underline{\lambda}$ that all two-cycles are non-negative, hence so is their sum. This non-negative value is a lower bound on $w(C)$, thus there must be no negative cycles induced by labelling $\underline{\lambda}$ in $\mathcal{G}$ and by Theorem 3 then a is implementable and the proof is complete. $\square$

Figure 4.6 shows how we can decompose a cycle that uses edges between labelled nodes of two adjacent types into one that only goes between adjacent types, as used in the proof of Theorem 4. The left figure shows a downhill edge that goes between nodes of two non-adjacent bid-islands. After doing this for every edge in a cycle, as in the right figure, we can lower bound the weight of the original cycle, and the resulting cycle composed of these resulting two-cycles we know to be non-negative as this is a consequence of f being overlapping. Therefore the labelling produces a NOM mechanism.

We illustrate in Figure 4.7 what it means for an allocation function a to be overlapping for player i . In the figure suppose that the square and circle profiles correspond to the partial bid profiles $\mathbf{b}_{-i}$ and $\mathbf{b}'_{-i}$, respectively. In order to be implementable as a truthfully implementable then the allocation function a must be monotone for each fixed partial profile. However, notice that a is decreasing between bids θ_1 to θ_3 for $\mathbf{b}_{-i}$, and decreasing between

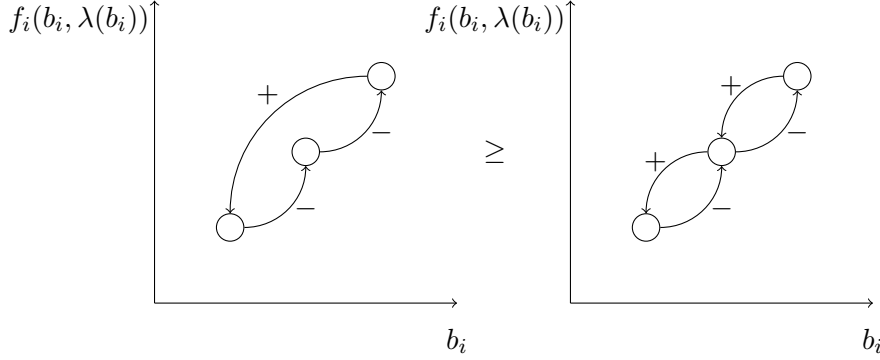


Figure 4.6: The left figure shows a labelling and a cycle with a longer edge going between nodes of two non-adjacent types. The right figure shows a cycle with (multi-)edges going only between adjacent types. The weight of the right cycle is a lower bound on the weight of the left, and we know this to be non-negative.

bids θ_2 to θ_4 for $\mathbf{b}'_{-i}$, therefore a is not implementable as a strategyproof mechanism. On the other hand, a is overlapping and hence implementable as a NOM mechanism.

Since the allocation function a in Figure 4.7 is overlapping then for each $t_i \in D_i$ we can set $\lambda(t_i, \theta)$ to the partial profile connected to the red dashed line, for each bid type $\theta \in \{\theta_1, \dots, \theta_4\}$. This labelling is the same simple type of labelling specified in the proof of Theorem 4, and as shown it is enough to guarantee that $\mathcal{G}$ contains no negative cycles. With this type of labelling we automatically guarantee that for each b_i -island of $\mathcal{G}$ there will be no negative (or indeed, any) cycles, hence all labelling constraints are satisfied for free and the only cycles we must consider are those induced by the incentive-compatibility edges. We formalise this with the following definition of this “simple” type of labelling.

Definition 17 (Single-line labelling). Fix $i \in N$ and mechanism M , and let $\lambda \in \{\bar{\lambda}, \underline{\lambda}\}$ be any labelling. Then λ is *single-line* for player i if for each bid $b_i \in D_i$ there is some $\mathbf{b}_{-i}$ such that $\lambda(t_i, b_i) = \mathbf{b}_{-i}$ for every $t_i \in D_i$. The labelling λ is *single-line* if it is single-line for every $i \in N$.

In other words, if i bids b_i then there is a single bid profile $\mathbf{b}_{-i}$ that leads to an extreme outcome of her utility function regardless of her true type. Since the profile leading to the extreme utility is the same regardless of player i 's true type in a single-line mechanism for each given bid then we will denote the labelled profile $\lambda(t_i, b_i)$ as simply $\lambda(b_i)$ for every $t_i \in D_i$.

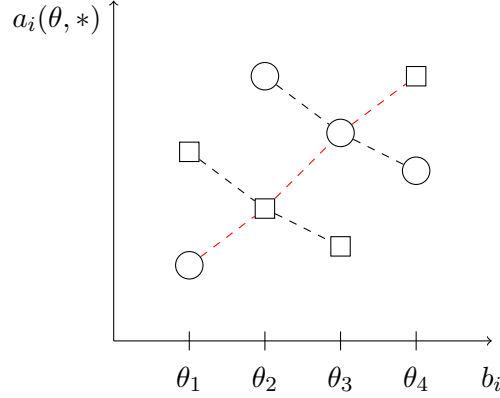


Figure 4.7: This allocation function is overlapping and hence implementable for player i . Let the squares denote some partial bid profile $\mathbf{b}_{-i}$ and the circles some other $\mathbf{b}'_{-i}$. Notice that a is not monotone in b_i for each fixed partial profile (since the graphs of a with respect to the profile are both decreasing) and hence not truthfully implementable.

4.4.3 EXPLICIT PAYMENTS FOR SINGLE-LINE MECHANISMS

[Archer and Tardos, 2001] show that a work function that allocates chores (where types describe *costs* as opposed to valuations) admits truthful payments if and only if it is non-increasing and characterise these payments using the area under the allocation function. [Apt and Heering, 2022] recently provided an elementary proof of the uniqueness of these payment functions using a little-known result from the theory of real functions. It is straightforward to derive a similar result in our setting for single-line mechanisms: we show that NOM payments exist if and only if one can find a non-decreasing collection of outcomes for the best- and worst-cases of the function. Fixing such a labelling λ , we may treat f as a function only in player i 's bid b_i and derive the following payments by solving the resulting differential equations from [Archer and Tardos, 2001] in the same way.

Theorem 5. *The social choice function $f : D \rightarrow \mathcal{A}$ is implementable if and only if f is overlapping. When this is the case and the mechanism (f, p) is single-line then for each player $i \in N$ with labelling λ_i and each bid $b_i \in D_i$, the payments take the following form:*

$$p_i(b_i, \lambda(b_i)) = h_i(\lambda_i) + b_i \cdot f_i(b_i, \lambda(b_i)) - \int_0^{b_i} f_i(z, \lambda(z)) dz, \quad (4.18)$$

where $h_i(\lambda_i)$ is any function that only depends on the labelling λ_i .

Social choice function f is truthfully implementable for single-parameter agent i for if for every fixed partial profile $\mathbf{b}_{-i}$ the function $f_i(b_i, \mathbf{b}_{-i})$ is non-decreasing in the bid b_i of player i . Theorem 4 shows a similar result for NOM mechanisms, albeit with more freedom to choose which outcomes of f to compare. That is, for each bid of player i we can choose a single $\mathbf{b}_{-i} = (b_i, \lambda_i(b_i))$ to compare with the others via a single-line labelling, and the function f needs only to be monotone with respect to these profiles in order to be implementable as a NOM mechanism. We informally refer to the set $\{(b_i, \lambda_i(b_i))\}_{b_i}$ of profiles in the single-line labelling λ_i as the *monotone restriction of f for player i* . Since f is implementable if and only if it is overlapping, and f being overlapping implies there is a single-line labelling λ with no negative cycles in the allocation graph $\mathcal{G}_{i,f}^\lambda$, then f is implementable if and only if there exists a monotone restriction of f for each player i .²

We introduce here two desiderata on a mechanism's social choice function and payment rules that allow us to further restrict the form of the payments from Theorem 5. The first condition is on the mechanism as a whole – the social choice function in conjunction with the payment rule, and roughly states that when acting truthfully an agent will not be worse off by participating in the mechanism. If this holds then the mechanism is said to be *individually rational*.

Definition 18 (Individual rationality). A mechanism $M = (f, p)$ is said to be *individually rational* if for each agent $i \in N$ and each type $t_i \in D_i$ it holds that

$$u_i(M(t_i, \mathbf{b}_{-i}); t_i) \geq 0,$$

for each partial profile $\mathbf{b}_{-i} \in D_{-i}$ of the other agents.

Equivalently referred to as voluntary participation, individual rationality ensures that as long as an agent does not submit a misreport of her type to the mechanism, then at worst she will end up with net utility zero and will not have lost anything from participating. This can equally be interpreted as ensuring that no agent is forced to bid. The next property is a slightly stronger condition on only the payment rule, and states that agents should be paid by the mechanism to participate.

Definition 19 (No positive transfers). A mechanism $M = (f, p)$ is said to make *no positive transfers* if for all agents $i \in N$ and each bid profile $\mathbf{b} \in D$

²If there is a single-line labelling of f with no negative cycles in $\mathcal{G}_{i,f}^\lambda$ then it is straightforward to see that f must be overlapping and follows almost by definition: simply take the set of labelled profiles $\{(b_i, \lambda_i(b_i))\}_{b_i}$ as the requisite monotone vector $\mathbf{a} \in \mathcal{A}_i$.

it holds that:

$$p_i(\mathbf{b}) \geq 0.$$

Requiring both of the conditions above restricts the resulting payments for single-line mechanisms. Let $\mathbf{b} = (t_i, \lambda_i(t_i))$ first be any bid profile where i bids truthfully labelled by a single-line labelling λ_i . Writing out the explicit payments of Theorem 5, we have:

$$\begin{aligned} u_i(M(\mathbf{b}); t_i) &= t_i(f_i(\mathbf{b})) - p_i(\mathbf{b}) \\ &= t_i \cdot f_i(\mathbf{b}) - \left[h_i(\lambda_i) + t_i \cdot f_i(\mathbf{b}) - \int_0^{t_i} f_i(z, \lambda_i(z)) dz \right] \\ &= -h_i(\lambda_i) + \int_0^{t_i} f_i(z, \lambda_i(z)) dz \end{aligned}$$

By individual rationality agent i 's utility on truthful bids must be non-negative, hence we have $h_i(\lambda_i) \leq \int_0^{t_i} f_i(z, \lambda_i(z)) dz$ for all t_i , and therefore $h_i(\lambda_i) \leq 0$. If the mechanism additionally makes no positive transfers then we get simply that $p_i(\mathbf{b}') \geq 0$ for all $\mathbf{b}'$, and hence for the truthful labelled profile $\mathbf{b} = (t_i, \lambda_i(t_i))$ we have $p_i(\mathbf{b}) = h_i(\lambda_i) - t_i \cdot f_i(\mathbf{b}) + \int_0^{t_i} f_i(z, \lambda_i(z)) dz \geq 0$, and therefore:

$$h_i(\lambda_i) \geq t_i \cdot f_i(\mathbf{b}) - \int_0^{t_i} f_i(z, \lambda_i(z)) dz.$$

Since f_i is monotone with respect to the labelled profiles then $\int_0^{t_i} f_i(z, \lambda_i(z)) dz$ can never exceed $t_i \cdot f_i(t_i, \lambda_i(t_i))$, so at the very least we have $h_i(\lambda_i) \geq 0$. Thus for any mechanism satisfying both individual rationality and no positive transfers we have $h_i(\lambda_i) = 0$.

As a small aside, we note that the above assumes that the agent types model valuations over the set of outcomes Ω . When instead types describe the cost incurred by an agent on each alternative then we must reimburse the agents. The payments directly correspond to those given in [Archer and Tardos, 2001], albeit only applied to the labelled profiles that give a monotone restriction of f , hence agents are now compensated with an amount equal to their cost $t_i \cdot f_i(t_i, \lambda_i(t_i))$ minus a discount factor $\int_{t_i}^{\infty} f_i(z, \lambda_i(z)) dz$, and the additional amount of $h_i(\lambda_i)$ as before. Now by individual rationality, for each type $t_i \in D_i$ and each truthful labelled profile $\mathbf{b} = (t_i, \mathbf{b}_{-i})$ we have that $u_i(M(\mathbf{b}); t_i) = p_i(\mathbf{b}) - t_i \cdot f_i(\mathbf{b}) = h_i(\lambda_i) + t_i \cdot f_i(\mathbf{b}) - \int_{t_i}^{\infty} f_i(z, \lambda_i(z)) dz - t_i \cdot f_i(\mathbf{b}) \geq 0$, hence $h_i(\lambda_i) \geq \int_{t_i}^{\infty} f_i(z, \lambda_i(z)) dz$ for each t_i . Therefore $h_i(\lambda_i) \geq \int_0^{\infty} f_i(z, \lambda_i(z)) dz$, the area under the entire work curve. We will use these in the next chapter to prove our impossibility result in the context

of NOM mechanisms for bilateral trade that are simultaneously efficient, individually rational, and approximately budget balanced.

4.5 EXTENSION TO RANDOMISED SETTINGS

Before moving on to the next chapter, we briefly make note of extending NOM to randomised and Bayesian settings, where it would seem that much of the structures introduced so far in this thesis would be readily transferrable. For now, we simply propose definitions of NOM in these two settings.

Firstly, it is straightforward to define NOM in terms of randomised mechanisms. We now simply take the extremes of the expected utility of a given player $i \in N$ acting truthfully versus dishonestly under a mechanism M .

$$\begin{aligned} \max_{\mathbf{b}_{-i}} \{ \mathbb{E}[u_i(M(t_i, \mathbf{b}_{-i}); t_i)] \} &\geq \max_{\mathbf{b}_{-i}} \{ \mathbb{E}[u_i(M(t_i, \mathbf{b}_{-i}); t_i)] \}, \\ \min_{\mathbf{b}_{-i}} \{ \mathbb{E}[u_i(M(t_i, \mathbf{b}_{-i}); t_i)] \} &\geq \min_{\mathbf{b}_{-i}} \{ \mathbb{E}[u_i(M(t_i, \mathbf{b}_{-i}); t_i)] \}. \end{aligned}$$

If both hold then the randomised mechanism M would be NOM for player i . Of course one could also study the question of implementability in such a setting, for which Myerson's characterisation would still be relevant and useful.

Additionally, we can consider the setting where each agent i 's type is drawn from a distribution $\mathcal{D}_i$. The standard consideration for studying Bayesian settings is that the bid profile of the other agents $\mathbf{b}_{-i}$ is drawn from the distribution $\mathcal{D}_{-i}$ and then we must ensure incentive compatibility (i.e., greater expected utility for truthtelling versus dishonesty) by taking i 's expected utility over $\mathcal{D}_{-i}$. An issue with this is that by taking the extremes of the expected utility function in this case then we are losing information about the distribution: if there is some partial profile $\mathbf{b}_{-i}$ that is realised with very little probability but still affects the extremes of i 's utility function then it is included as if it occurred with probability one. Instead, a weaker notion would be to take the expectation over player i 's own distribution $\mathcal{D}_i$. This could lend itself to an interpretation of bounded rationality in which agents do not have a clear understanding of their own type, however further discussion on this setting is outside our current scope. In this case,

a mechanism M would be NOM for player i if the following conditions hold:

$$\begin{aligned} \max_{\mathbf{b}_{-i}} \{ \mathbb{E}_{t_i \sim \mathcal{D}_i} [u_i(M(t_i, \mathbf{b}_{-i}); t_i)] \} &\geq \max_{\mathbf{b}_{-i}} \{ \mathbb{E}_{t_i \sim \mathcal{D}_i} [u_i(M(t_i, \mathbf{b}_{-i}); t_i)] \}, \\ \min_{\mathbf{b}_{-i}} \{ \mathbb{E}_{t_i \sim \mathcal{D}_i} [u_i(M(t_i, \mathbf{b}_{-i}); t_i)] \} &\geq \min_{\mathbf{b}_{-i}} \{ \mathbb{E}_{t_i \sim \mathcal{D}_i} [u_i(M(t_i, \mathbf{b}_{-i}); t_i)] \}. \end{aligned}$$

We leave the question of implementability with regards to randomised NOM mechanisms and mechanisms in the Bayesian setting open. In the next chapter we revisit the impossibility result for NOM mechanisms in bilateral trade due to [Troyan and Morrill, 2020] which shows that no efficient, individually rational NOM mechanism for bilateral trade exists that guarantees weak budget balance.

BILATERAL TRADE

We now apply our characterisation from the previous chapter to its first setting. We focus on bilateral trade where a single item may be sold from a seller to a buyer. We want to guarantee efficiency, where trade occurs only when the buyer’s valuation is at least the seller’s production cost; weak budget balance (WBB), for which we never pay the seller more than we charge the buyer; and individual rationality (IR), where the utilities of both buyer and seller are non-negative. An impossibility result from Troyan and Morrill tells us that NOM cannot guarantee these three properties simultaneously, so we investigate the extent to which a multiplicative subsidy of α , which guarantees that we never pay the seller more than an $\alpha \geq 1$ factor of what we receive from the buyer (i.e., α -WBB), allows us to obtain NOM, efficiency, and individual rationality simultaneously. We find that no such mechanism exists for the simple class of single-line mechanisms from Chapter 4. However, if we look instead for more general labelling and payment schemes and consider best cases and worst cases of the mechanism separately, then this incompatibility is fully resolved and we achieve exact weak budget balance with WNOM, but the impossibility is unavoidable for BNOM.

5.1 CONTRIBUTION OF THIS CHAPTER

The main motivation of this chapter is to study the following impossibility result from Troyan and Morrill’s original paper on obvious manipulability:

Theorem 6 (Troyan and Morrill (2020)). *Every efficient, individually rational, and weakly budget balanced mechanism is obviously manipulable.*

We tackle this impossibility by relaxing the notion of budget balance. Let $p_B(\mathbf{b})$ and $p_S(\mathbf{b})$ denote the amount paid by the buyer and the amount paid to the seller in a bilateral trade mechanism (note that these need not be the same amount if, for example, monetary transfers are handled by a central mediator). We say that a mechanism is α -weakly budget balanced

(α -WBB) for $\alpha \geq 1$ if $p_S(\mathbf{b}) \leq \alpha \cdot p_B(\mathbf{b})$ for every $\mathbf{b} \in D$. Although we may end up paying the seller more than the payment received from the buyer, we do not overpay by more than a factor of α . We can therefore think of this value α the multiplicative subsidy required by the mechanism in order to subsidise the market.

The approach developed in main theorems of Chapter 4 can be applied to characterise BNOM and WNOM independently. For the main result we first characterise the monotone restrictions of a given allocation function that will guarantee efficiency, individual rationality, and either BNOM or WNOM.

Main Result 1 (informal). Any single-line NOM, efficient, individually rational and α -WBB mechanism for bilateral trade has unbounded α .

For best cases, the unbounded α is implied by the incentive-compatibility constraints: using the explicit payment formulae we show that the buyer must receive the item for free while the seller must receive some positive payment in each of their respective best-case profiles. For worst cases it is implied by the labelling validity constraints.

We then look beyond using monotone restrictions of the allocation function to construct NOM mechanisms where the labelled profiles need not appear on a “single line”. Can we be more flexible in our labelling scheme to derive different payments that would satisfy individual rationality and finite subsidies for an allocation function that is overlapping and efficient? The answer is surprising: even non-single line BNOM mechanisms require infinite subsidies while non-single line WNOM mechanisms can guarantee weak budget balance (i.e., no subsidies).

Main Result 2 (informal). There is no efficient, individually rational, α -weakly budget balanced BNOM mechanism for bilateral trade with finite subsidy α .

There exists in fact a definition of WNOM labels that implement first-price auctions. We may also strengthen the notion of incentive-compatibility on either side of the market to strategyproofness – for example, the buyer is strategyproof and the seller WNOM – by combining a first-price auction on one side with a second-price auction on the other. This leaves the spread to the market designer, whereas a fully WNOM market would be budget balanced. This setting may be of interest for some classes of double-sided markets where the rationality is asymmetric and depends on the role played in the market.

Main Result 3 (informal). There is an efficient, individually rational, weakly budget balanced WNOM mechanism for bilateral trade.

We conclude by characterising individually rational NOM mechanisms that might be inefficient but require no subsidies. Intuitively, this result says that there exist trading windows defined by payment thresholds that partition the domain of buyer and seller into three sets; bids guaranteeing that the trade occurs, bids guaranteeing that the trade will not occur, and bids where the decision to trade depends on the bid at the other side of the market.

The rest of this chapter is structured as follows: in Section 5.2 we provide a brief overview of related work in the setting of bilateral trade; in Section 5.3 we introduce the new terms and notation specific to the setting of bilateral trade; in Section 5.4 we show that single-line mechanisms are too strong a requirement to achieve any efficient, individually rational, α -weakly budget mechanism with no obvious manipulations for any finite α ; in Section 5.5 we show that WNOM is compatible with all our desired properties when we drop the single-line drop, while BNOM remains impossible; and in Section 5.6 we drop the requirement for efficiency and characterise the set of mechanisms for bilateral trade that are individually rational, weakly budget balanced, and NOM.

5.2 RELATED WORK

Bilateral trade sees significant attention in the mechanism design literature. It is well-known that there exists no efficient, individually rational truthful mechanism that is weakly budget balanced due to [Myerson and Satterthwaite, 1983]. [Deng et al., 2022] positively resolved the long-standing open question on the existence of a mechanism with these properties that guarantees a constant fraction of the first-best gains-from-trade. That is, a constant fraction of the expected utility gain from trading attainable whenever buyer’s value weakly exceeds seller’s cost. Their mechanism takes a randomization over the utility-maximising fixed price mechanisms for the buyer and seller, respectively.¹ By carefully selecting the proportion with which each fixed-price mechanism is selected, this achieves a $\frac{1}{8.23} \approx 0.121$ fraction of the first-best gains from trade. With the goal of designing robust bilateral trading mechanisms, [Zhang, 2022] runs a double auction over a

¹Specifically, they randomise over the seller-pricing mechanism and the buyer-pricing mechanism, where the seller sets the price which the buyer may take or leave, and vice versa.

randomly-selected spread from a uniform distribution, where trade is executed at the midpoint between the buyer's and seller's bid only when their difference exceeds the spread. Each agent then pays half of a fixed transaction fee selected by the mechanism at the beginning. This maximises the worst case expected profit across all dominant-strategy incentive compatible and ex-post individually rational mechanisms. [Wolitzky, 2016] provides a necessary condition for the social choice function to be implementable, using this to characterise efficient trade when the agents know the other agent's expected valuation of the good. Whenever this is the case, it may be implemented by a simple double auction format.

5.3 PRELIMINARIES

In an instance of bilateral trade there is a buyer B who is interested in purchasing a product from (and which must first be produced by) a seller S . The buyer has a valuation $v \in D_B$ for the item while the seller incurs a cost $c \in D_S$ for producing it, and we want to design direct-revelation mechanisms that first elicit a bid profile $(x, y) \in D = D_B \times D_S$ from the agents and then specifies one of two alternatives, “trade” or “no trade”. We assume $D_B = D_S = [0, 1]$. We use $f : D \rightarrow \{0, 1\}$ to model these outcomes, where $f(\mathbf{b}) = 1$ if trade occurs and $f(\mathbf{b}) = 0$ otherwise.² The mechanism then returns payments that the buyer must make *to the mechanism* and that the seller receives *from the mechanism*. Therefore for each $\mathbf{b} = (x, y) \in D$ we write the agents' utilities as follows:

$$\begin{aligned} u_B(M(\mathbf{b}); v) &= v \cdot f(\mathbf{b}) - p_B(\mathbf{b}), \\ u_S(M(\mathbf{b}); c) &= p_B(\mathbf{b}) - c \cdot f(\mathbf{b}). \end{aligned}$$

This reflects that when $f(\mathbf{b}) = 1$, the buyer is given the item and hence gains a value of v while the seller must produce the item and hence incurs a cost of c . Otherwise when $f(\mathbf{b}) = 0$ then trade does not occur, the buyer does not receive value v , and the seller does not need to produce the item and hence incurs no cost.

In this setting we say the mechanism is *efficient* if $f(x, y) = 1$ if and only if $x \geq y$ for all $(x, y) \in D$. Moreover the mechanism is *weakly budget balanced* if $p_S(\mathbf{b}) \leq p_B(\mathbf{b})$ for all $\mathbf{b} \in D$. In words, this ensures we never pay the seller more than the payment we receive from the buyer. The main

²In the language of Chapter 3, the buyer's and seller's types are functions $t_B, t_S : \{0, 1\} \rightarrow \mathbb{R}$ where $t_B(0) = t_S(0) = 0$, $t_B(1) = v$, and $t_S(1) = c$. Clearly v is known only to B and c is known only to S .

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aim of this chapter is to investigate the impossibility result of Theorem 6, stating that efficiency, individual rationality, and weak budget balance are incompatible with NOM. We reiterate that we relax the notion of budget to *approximate* budget balance with a multiplicative approximation factor α on the subsidies required for the market. We say that a mechanism is α -*weakly budget balanced* (α -WBB) for $\alpha \geq 1$ if $p_S(\mathbf{b}) \leq \alpha \cdot p_B(\mathbf{b})$ for every $\mathbf{b} \in D$.

In the following section we investigate the conditions under which these properties of efficiency, individual rationality, and approximate weak budget balance can be achieved by NOM mechanisms. After showing that the simple “single-line” mechanisms of Chapter 4 induce a possibly infinite subsidy in the market, we then consider BNOM and WNOM in isolation and show that it is BNOM that provides the bottleneck, and by only focusing on WNOM we may achieve exact weak budget balance.

5.4 SINGLE-LINE MECHANISMS REQUIRE UNBOUNDED SUBSIDY

Recall that a mechanism is single-line if for each possible report she can make to the mechanism, there is a single bid profile at which she either maximises (for BNOM) or minimises (for WNOM) her utility function, regardless of her true type. We begin by writing out the explicit form taken by the payments as a result of Theorem 5 in the context of single-line mechanisms for bilateral trade. Let λ_B denote some labelling (that is, either a best-case labelling $\bar{\lambda}_B$ or worst-case labelling $\underline{\lambda}_B$) of a single-line mechanism for bilateral trade for the buyer. Note that for any labelling, the labelled profile for a given bid is simply the joint bid profile of the other players leading to an extreme value of the agent’s utility. Therefore when λ_B is single-line then for any bid $v \in D_B$ of the buyer the labelled profile $\lambda_B(v)$ simply denotes the bid of the seller that leads to an extreme of the buyer’s utility when the buyer bids v . According to Theorem 5, for any bid $v \in D_B$ and any labelled profile $(x, \lambda_B(x))$ the buyer will pay:

$$p_B(v, \lambda_B(v)) = h_B(v, \lambda_B(v)) + v \cdot f_B(v, \lambda_B(v)) - \int_0^v f_B(u, \lambda_B(u)) du.$$

Similarly, let λ_S be some labelling of a single-line bilateral trade mechanism for the seller. Now for any bid $c \in D_S$ of the seller, $\lambda_S(c)$ denotes the bid of the buyer leading to an extreme of the seller’s utility function when

the seller bids c . On these labelled profiles the seller must be paid:

$$p_S(\lambda_S(c), c) = h_S(\lambda_S(c), c) + c \cdot f_S(\lambda_S(c), c) - \int_0^c f_S(\lambda_S(u), u) du.$$

As in the previous chapter, notice that imposing individual rationality and no positive transfers means that the function $h_B(\lambda_B)$ is identically zero, while $h_S(\lambda_S)$ must be at least the area under the labelled allocation curve $\int_0^1 \lambda_S(\lambda_S(c), c)$.

With the following lemma we show the intuitive result that regardless of either the buyer's or the seller's true valuation or cost, respectively, their best outcomes are realised only when trade is executed and their worst outcomes are realised only when trade does not take place. In other words, the curves representing the outcomes of the allocation function corresponding to the extreme values of the agents' utility functions are constant.

We note some useful facts employed in the proofs of the statements. Let x and y be bids of the buyer and seller, respectively. For any efficient and individually rational mechanism we require $p_B(x, y) \leq x$ and $p_S(x, y) \geq y$ for all $x \geq y$, while for $x < y$ we have $p_B(x, y) = 0$. Therefore whenever trade does not take place the utility of the buyer is 0 no matter her valuation.

Since the buyer's type represents a valuation for the item and the seller's type represents a cost for producing it, then by Theorem 4 the labelled outcomes of f must form a non-decreasing line for the buyer and a non-increasing line for the seller. If a mechanism $M = (f, p)$ for bilateral trade is NOM and f takes values in $\{0, 1\}$ then there must be a threshold at which, for all bids that are at least this threshold for some player, their best- or worst-case allocation from the mechanism flips from one value to the other. We formalise this in the following lemma and show for both the buyer and the seller that these thresholds must be placed at the extremes, namely, 0 or 1.

Lemma 1. *In any efficient, individually rational, single-line BNOM mechanism the best-case outcomes always occur for both the buyer and the seller when the trade is executed, while in any efficient, individually rational, single-line WNOM mechanism the worst-case outcomes always occur for both the buyer and the seller when the trade is not executed.*

Proof. Let $M = (f, p)$ be an efficient, individually rational, single-line NOM mechanism and first consider the threshold for best-case outcomes from the buyer's side. Since f is non-decreasing let θ_B be a threshold bid of the buyer such that $f(x, \bar{\lambda}_B(x)) = 1$ for all $x \geq \theta_B$. Now take $x < \theta_B$ and let y be a bid of the seller such that $f(x, y) = 1$. By the BNOM labelling

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constraints we have $u_B(M(x, \bar{\lambda}_B(x)); v) = 0 \geq v - p_B(x, y)$ for all $v \in D_B$ so $1 \leq p_B(x, y) \leq x$, where the second inequality follows from individual rationality and efficiency. Now we have $1 \leq x < \theta_B$ where $\theta_B \in [0, 1]$, a contradiction. Hence there is no x such that $x < \theta_B$, meaning $\theta_B = 0$ and trade always occurs in the buyer's best-case outcome.

Now consider the seller. Since f must be non-increasing in her bid there again must be a threshold $\theta_S \in [0, 1]$ such that $f(\bar{\lambda}_S(y), y) = 1$ for all $y \leq \theta_S$ and 0 otherwise.

For any bid $y > \theta_S$ the seller's best-case utility is $u_S(M(\bar{\lambda}_S(y), y); c) = p_S(\bar{\lambda}_S(y), y) = h_S(\bar{\lambda}_S) + yf(\bar{\lambda}_S(y), y) - \int_0^y f(\bar{\lambda}_S(u), u) du$. Since c describes the seller's cost then $h_S(\bar{\lambda}_S)$ must be at least the area under the curve. Setting $h_S(\bar{\lambda}_S)$ to this area gives $p_S(\bar{\lambda}_S(y), y) = yf(\bar{\lambda}_S(y), y) + \int_y^1 f(\bar{\lambda}_S(u), u) du$ so her best-case utility is 0.³ Now take any x such that $f(x, y) = 1$. By the BNOM constraints we have $u_S(M(x, y); c) = p_S(x, y) - c \leq 0$ for all c . Combining this with individual rationality we have $c \geq p_S(x, y) \geq y$, which gives us a contradiction if we take $c < y$. Thus $y \leq \theta_S$ for all y and so $\theta_S = 1$, meaning trade always occurs in the seller's best-case outcome.

We can apply similar reasoning to show that trade does not occur for the buyer or seller in their worst-case outcome, no matter their bid. Since the mechanism is NOM then f must be overlapping, hence there is a threshold bid θ_B and θ_S for the buyer and seller respectively whereby for all $x \geq \theta_B$ we have $f(x, \underline{\lambda}_B(x)) = 1$ and 0 otherwise, and for all $y \leq \theta_S$ we have $f(\underline{\lambda}_S(y), y) = 1$ and 0 otherwise. First consider the buyer. Assume there is a bid $x > \theta_B$ of the buyer and y a bid of the seller such that $f(x, y) = 0$. The labelling constraints require that $v - p_B(x, \underline{\lambda}_B(x)) \leq 0$ for all $v \in D_B$, and combining this with the previous facts we get $p_B(x, \underline{\lambda}_B(x)) = x$. Now taking $v > x$ the buyer's utility at her worst-case outcome is strictly positive, a contradiction. Hence $\theta_B \geq x$ for all x and thus $\theta_B = 1$. Therefore in the buyer's worst case the trade is never executed.

Finally now focus on the worst cases of the seller, let $y < \theta_S$ and let x be a bid of the buyer such that $f(x, y) = 0$. Then $p_S(x, y) = 0$, and since trade occurs in the seller's worst case by assumption then $p_S(\underline{\lambda}_S(y), y) \leq y$.

³We can of course pay the seller more, as long as it doesn't interfere with incentives. We will be using this lemma to prove the impossibility of (bounded) α -WBB, so if we can show this when we are being frugal then the claim will hold even when using more generous payments. Thus we set $h_S(\beta)$ as small as possible – exactly the area under the curve – and this is enough to prove the unboundedness.

By (4.18) then:

$$\begin{aligned}
 p_S(\lambda_S(y), y) &\geq h_S(\lambda_S) + yf(\lambda_S(y), y) - \int_0^y f(\lambda_S(u), u) du \\
 &= h_S(\lambda_S) + y - y \\
 &\geq \int_y^\infty f(\lambda_S(u), u) du \\
 &= \theta_S,
 \end{aligned}$$

where the final inequality follows from the fact that $h_S(\lambda_S)$ must be at least the area under the curve for any individually rational mechanism that makes no positive transfers. Thus $p_S(\lambda_S(y), y) \geq \theta_S$ and $p_S(\lambda_S(y), y) \leq y$ where $y < \theta_S$, a contradiction. So $\theta_S = 1$ and the seller's worst case always occurs when the trade is not executed. $\square$

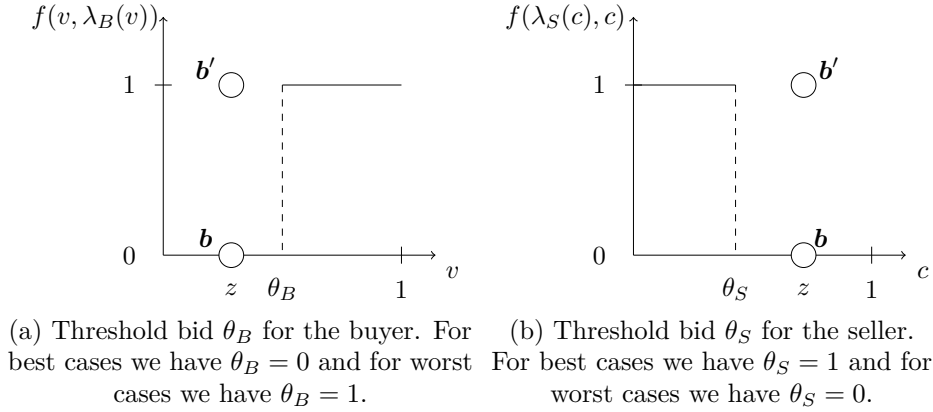


Figure 5.1: Thresholds bids used in Lemma 1.

As a corollary of Lemma 1 we have $p_B(x, \lambda_B(x)) = p_S(\lambda_S(y), y) = 0$ for all x, y since trade does not occur at the worst-case profiles. Conversely the buyer's best outcome must be when she gets the item and pays nothing, while the seller's must occur when she produces the item and is paid at least her bid. For any individually rational single-line BNOM mechanism M that makes no positive transfers the best-case payments are as in (4.18), so $p_B(x, \bar{\lambda}_B(x)) = 0$ and $p_S(\bar{\lambda}_S(y), y) \geq y + \int_y^1 f(\bar{\lambda}_S(u), u) du = 1$. The following result shows that single-line mechanisms are not flexible enough to avoid the impossibility of Troyan and Morrill (2020) without unbounded subsidy.

Theorem 7. *There is no efficient, individually rational, α -weakly budget balanced, and single-line mechanism for bilateral trade with finite subsidy α .*

Proof. We first prove the result for BNOM mechanisms. Let $M = (f, p)$ be an efficient, individually rational single-line BNOM mechanism. From Lemma 1 we know that $f(x, \bar{\lambda}_B(x)) = f(\bar{\lambda}_S(y), y) = 1$ for any $x \in D_B$ and $y \in D_S$, while $p_B(x, \bar{\lambda}_B(x)) = 0$ and $p_S(\bar{\lambda}_S(y), y) \geq y$. Since in the best case the buyer pays zero while the seller receives at least her cost, we may prove the claim by providing a bid profile that appears on the best-case curves of both the buyer and seller. Take any bid x of the buyer and let $y = \bar{\lambda}_B(x) > 0$ be a bid of the seller. Since $f(x, y) = 1$ then $p_S(x, y) \geq y$ by efficiency and individual rationality. The labelling constraints require $u_S(M(y, \bar{\lambda}_S(c)); c) \geq u_S(M(x, y); c) \geq y$ for all y and so $p_S(\bar{\lambda}_S(y), y) \geq p_S(x, y) \geq y$. Since $y = \bar{\lambda}_B(x)$ then $p_S(\bar{\lambda}_S(y), y) \geq p_S(x, \bar{\lambda}_B(x)) \geq y > 0$, while $p_B(x, \bar{\lambda}_B(x)) = 0$. Therefore at $(x, \bar{\lambda}_B(x))$ the buyer pays 0 while the seller must receive some positive amount.

Now let $M = (f, p)$ be an efficient, individually rational, single-line WNOM mechanism and consider a bid $x \in D_B$ of the buyer for which $f(x, \underline{\lambda}_B(x)) = 0$ and thus $p_B(x, \underline{\lambda}_B(x)) = u_B(M(x, \underline{\lambda}_B(x)); v) = 0$ for all v . Let $y \in D_S$ be a bid of the seller such that $f(x, y) = 1$ and note that by efficiency $x \geq y$. Since the worst utility of the buyer is 0 then $p_B(x, y) \leq v$ and thus we need $p_B(x, y) \leq \min\{x : x \geq y\} = y$. Now consider the seller with bid $y \in D_S$ such that $f(\underline{\lambda}_S(y), y) = 0$. Let $x \in D_B$ be a bid of the buyer such that $f(x, y) = 1$. By the labelling constraints we have $p_S(x, y) - c \geq p_S(\underline{\lambda}_S(y), y) \geq 0$, and since M is single-line then $p_S(x, y) \geq c$ for all c . Therefore $p_S(x, y) \geq \max\{y : x \geq y\} = y$.

Now consider the bid profile $(1, \varepsilon)$ for arbitrarily small $\varepsilon > 0$. The buyer must pay at most ε while the seller is to receive at least 1, giving the desired result. $\square$

We briefly here propose an alternative notion of approximate budget balance that instead assumes an additive subsidy δ that must be injected into the market. We say that a mechanism for bilateral trade is δ -additively budget balanced for $\delta \geq 0$ if for all bid profiles $\mathbf{b} \in D$ it holds that $p_S(\mathbf{b}) \leq p_B(\mathbf{b}) + \delta$.

Open problem 2. Does there exist a NOM mechanism for bilateral trade that is efficient, individually rational, and δ -additively budget balanced for finite δ ?

5.5 AVOIDING SUBSIDIES BY CHANGING THE LABELLING

As the previous section shows, single-line mechanisms are overly restrictive if we want to design efficient, individually rational, and α -weakly budget

balanced NOM mechanisms for bilateral trade. We will now explore if the issue of unbounded subsidy is avoided by considering more general labellings, and hence more general payment schemes. We show that while this works in a strong sense for WNOM mechanisms, where we can achieve exact (i.e., $\alpha = 1$) budget balance, surprisingly unbounded subsidies are inherent in any BNOM mechanism.

Since we require the mechanism to be efficient then trade occurs if and only if $x \geq y$, hence trade is always executed when $y = 0$. Here we will derive a WNOM payment scheme for $f(x, y) \mapsto \mathbb{1}[x \geq y]$. We first define a worst-case labelling $\underline{\lambda}_B$ of the allocation graph for the buyer and show that this induces no negative weight cycles in the labelled allocation graph $\mathcal{G}_{B,f}^{\underline{\lambda}_B}$ for the buyer. Therefore by Theorem 3 there must exist some payment function based on this labelling that yields a WNOM mechanism. By imposing individual rationality on the mechanism constraints encoded by the edges of the graph, we then show that the payments produced by this labelling are exactly first-price payments. Hence this mechanism is WNOM from the buyer's perspective. Likewise, first-price payments also ensure WNOM for the seller. As a result the mechanism is WNOM for both parties, and since the first-price payments extracted from the buyer always exceed the bid of the seller then we achieve exact budget balance. With Theorem 7 in mind, we can conclude (and indeed we shall see in (5.1)) that the first-price auction in the context of bilateral trade does not use single-line payments.

Consider the following worst-case labelling for the buyer,

$$\underline{\lambda}_B(v, x) = \begin{cases} 1 & \text{if } v \geq x \\ 0 & \text{otherwise} \end{cases}, \quad (5.1)$$

for each $v, x \in D_i$. Note by efficiency that $\underline{\lambda}_B(v, x) = 0$ means that trade occurs for any x , and $\underline{\lambda}_B(v, x) = 1$ means that trade does not occur unless $x = 1$. Now fix a given bid x of the buyer and observe that every cycle induced by $\underline{\lambda}_B$ in $\mathcal{G}_{i,f}^{\underline{\lambda}_B}$ has non-negative weight. The x -island contains exactly two nodes, $(x, 0)$ and $w(x, 1)$, and for distinct any pair of true types v, v' with $v \geq x > v'$ there will be a two-cycle with weight:

$$(v - v')(f_i(x, \underline{\lambda}_B(v', x)) - f_i(x, \underline{\lambda}_B(v, x))).$$

Note that for any two types $v, v' \in D_B$ such that $v' < x \leq v < 1$ we have $f_i(x, \underline{\lambda}_B(v, x)) = 0$ and $f_i(x, \underline{\lambda}_B(v', x)) = 1$. Therefore the value of $f_i(x, \underline{\lambda}_B(v', x)) - f_i(x, \underline{\lambda}_B(v, x))$ is identically 1 for any two such types v and v' . The smallest weight of such a two-cycle occurs when $v = x$ and $v' = x - \varepsilon$

where $\varepsilon > 0$, resulting in a weight of $x - (x - \varepsilon) = \varepsilon > 0$. Therefore, no matter the pair of types v and v' chosen, the weight of their associated two-cycle induced by the labelling $\underline{\lambda}_B$ in any given x -island of $\mathcal{G}_{B,f}^\lambda$ is always non-negative.

It now remains to show that $\underline{\lambda}_B$ induces no negative cycles anywhere else in the graph. Recall that incentive compatibility edges are the only ones to traverse between the different islands of the allocation graph, and note with the following observation that incorporating these incentive compatibility edges into a cycle of labelling edges does not create any negative cycles.

Take two types x, x' and assume that $x < x'$. By the definition of $\underline{\lambda}_B$ there is the cycle $\langle (x, 1) \xrightarrow{x} (x', 0) \xrightarrow{x'} (x', 1) \xrightarrow{x'} (x, 1) \rangle$, and this cycle's weight is minimised when $x' = x + \varepsilon$ for some $\varepsilon > 0$. This is the shortest such cycle formed by the labelling $\underline{\lambda}_B$, and it is straightforward to verify that these are the only types of cycle induced by the labelling. Moreover, it is simple to show that that increasing the length of the cycle will only increase its weight. The weights of this class of cycles induced by the labelling are therefore always non-negative. Thus f admits a WNOM payment scheme by Theorem 3. It now remains to derive the payments produced by this labelling.

Claim 3. The worst-case labelling given in (5.1) induces first-price payments for the buyer.

Proof. Fix a bid $x < 1$ of the buyer. Since $\underline{\lambda}_B(x, x) = 1$ then trade does not occur and we have $u_B(M(x, 1); v)$ for all v . Since $\underline{\lambda}_B(v, x) = 0$ for $v < x$ then trade occurs and we have $v - p_B(x, 0) \leq u_B(M(x, 1); v) = 0$. Thus for all $v < x$ we have $p_B(x, 0) \geq v$ and so $p_B(x, 0)$ tends to x . Combining this with individual rationality we have $x \leq p_B(x, 0) \leq x$ and therefore when the trade occurs the buyer pays her bid, and 0 otherwise. $\square$

Now consider payments for the seller where she gets paid her own bid when trade occurs. It is straightforward to verify that these payments for the seller are WNOM. Under this combined first-price payment scheme, when trade occurs the payments are $p_B(x, y) = x$ and $p_S(x, y) = y$, and 0 otherwise. Since trade occurs only when $x \geq y$ then the payment given to the seller never exceeds the amount extracted from the buyer, hence we achieve (exact) weak budget balance. We present the resulting simple first-price mechanism for bilateral trade in Mechanism 1.

Theorem 8. *Mechanism 1 is an efficient, individually rational, weakly budget balanced WNOM mechanism.*

Mechanism 1 First-price bilateral trade.

Input: Bid profile $(x, y) \in [0, 1]^2$.

Output: Outcome $f(x, y) \in \{0, 1\}$, payments $(p_B, p_S) \in \mathbb{R}_{\geq 0}^2$.

Let $f(x, y) = \mathbb{1}[x \geq y]$ and payments $p = (p_B, p_S)$ such that:

$$p_B(x, y) = \begin{cases} x & x \geq y \\ 0 & \text{otherwise} \end{cases}, \quad p_S(x, y) = \begin{cases} y & x \geq y \\ 0 & \text{otherwise} \end{cases}.$$

return $(f(x, y), p)$.

This result shows that when considering more general labelling schemes than single-line, WNOM is compatible with all of efficiency, individual rationality, and α -weak budget balance, even with $\alpha = 1$. Naturally we next investigate whether the same can be said for BNOM. With the following we show that moving away from single-line mechanisms does not avoid the issue of unbounded subsidies for BNOM as it does for WNOM. The proof is by case analysis on all possible labellings between two types of the buyer's domain and reveals that no labelling results in a mechanism that bypasses the negative result for single-line mechanisms, even if we restrict to these two types. Each labelling falls into one of four cases: it is single-line, for which we have already proved subsidies are unbounded; it is invalid and leads to a negative cycle, meaning NOM payments are impossible by Theorem 3; it induces no negative cycles but otherwise leads to contradiction when taken with individual rationality and efficiency; or it leaves us with the same payments as in the single-line case, meaning unbounded subsidies are unavoidable.

Theorem 9. *There is no efficient, individually rational, α -weakly budget balanced BNOM mechanism for bilateral trade with finite subsidy α .*

Proof. Let $M = (f, p)$ be an efficient mechanism for bilateral trade. We will show that for all possible best-case labellings of the mechanism for the buyer $\lambda = \bar{\lambda}_B$ lead to some violation of the desired properties. We can even restrict our attention to only two types from the buyer's domain. All such labellings will fall into one of four cases: the labelling is single-line, where we know by Theorem 7 that subsidies are unbounded; the labelling produces a negative cycle in $\mathcal{G}_{B,f}^\lambda$, and by Theorem 3 a BNOM payment scheme does not exist; the labelling produces no negative cycles in $\mathcal{G}_{B,f}^\lambda$ but otherwise leads to contradiction when imposing individual rationality and efficiency;

or the labelling results in the same payments as the single-line mechanism, and hence resulting in the same unbounded subsidies.

The first two cases are straightforward so we will focus on the latter two.

Let $D_B = [0, 1]$ and $D_S = \{0, 1\}$ and let x, z be any two types from the buyer's domain. First note that in any valid labelling λ each x -island of $\mathcal{G}_{B,f}^\lambda$ will consist only of two nodes, $(x, 0)$ and $(x, 1)$, hence for any labelling to be valid it must be that $f(x, \lambda(v, x)) \geq f(x, \lambda(z, x))$ if and only if $v \geq z$. Now suppose $x < z$ and observe that by individual rationality we have $p_B(x, 0) \leq x$ so either $\lambda(x, x) = (x, 0)$ or $\lambda(x, x) = (x, 1)$ and $p_B(x, 0) = x$.

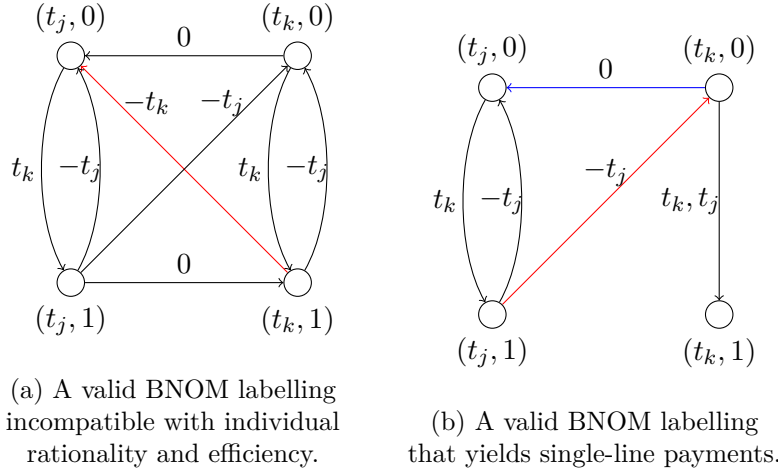


Figure 5.2: No BNOM labelling can achieve bounded subsidies in an efficient, individually rational mechanism for bilateral trade.

Case 1: λ is incompatible with individual rationality and efficiency. Consider λ where $\lambda(x, x) = (x, 1)$, $\lambda(z, x) = (x, 0)$, $\lambda(z, z) = (z, 1)$, and $\lambda(x, z) = (z, 0)$. By the incentive compatibility constraints, represented by the red edge in Figure 5.2a, we have $0 \geq z - p_B(x, 0) = z - x$, a contradiction since $x < z$.

Case 2: λ leads to single-line payments. Consider a labelling λ where $\lambda(x, x) = \lambda(z, x) = (x, 0)$, $\lambda(z, z) = (z, 0)$, and $\lambda(x, z) = (z, 1)$ and whose graph $\mathcal{G}_{B,f}^\lambda$ is given in Figure 5.2a in which edges are annotated with their weights. By incentive compatibility constraints we have $0 \geq x - p_B(z, 0)$, represented by the red edge in Figure 5.2b, and therefore $p_B(z, 0) \geq x$, as well as $z - p_B(z, 0) \geq z - p_B(x, 0)$, represented by the blue edge in Figure 5.2b) and so $p_B(z, 0) \leq p_B(x, 0)$. Recall that we have $p_B(x, 0) = x$ and therefore $x \leq p_B(z, 0) \leq p_B(x, 0) = x$, so $p_B(z, 0) = x$. Now take $x = 0$ and we see that payments in a profile in which trade occurs are identically zero. This is precisely the issue that leads to unbounded subsidies.

Each possible labelling, even when only considering two types x, z from the buyer's domain, falls into one of the four above mentioned cases. Thus the only valid labellings that are not incompatible with individual rationality and efficiency lead to exactly the same payments as we get with a single-line mechanism. Hence subsidies remain unbounded for arbitrary labellings for any efficient, individually rational, BNOM mechanism. $\square$

Therefore, where dropping the requirement for a simple-to-state single-line mechanism allows us to achieve an efficient, individually rational, and exactly budget balanced mechanism for bilateral trade when only considering worst cases by using a more general labelling scheme, the same cannot be said when we only consider best cases. This strengthens the original impossibility result of [Trojan and Morrill, 2020]. In the next section we investigate which mechanisms are NOM for bilateral trade when we drop the efficiency requirement.

5.6 TRADING WINDOWS

Now we no longer assume our allocation function to be efficient and characterise NOM mechanisms for bilateral trade that are individually rational and weakly budget balanced. We introduce some additional notation to conveniently formulate our characterisation. For a number x we denote by $\text{succ}_B(x)$ and $\text{prec}_B(x)$ respectively the least element in D_B strictly greater than x and the greatest element strictly less than x . We use $\text{succ}_S(y)$ and $\text{prec}_S(y)$ analogously for the seller's domain D_S . Let H_B denote the least element in D_B that is greater than every element in D_S , and L_B the greatest element in D_B that is less than every element in D_S . For a mechanism we may partition the buyer's type space into three sets: D_B^0 , D_B^1 , and D_B^{01} . D_B^0 consists of all bids of the buyer for which trade is guaranteed not to happen, D_B^1 consists of all bids for which trade surely happens, and D_B^{01} consists of the remaining elements of D_B , where trade depends on the seller's bid. The sets D_S^0 , D_S^1 , and D_S^{01} can be defined analogously for the seller. For Theorem 10 it is useful to keep in mind the visualisations of these sets given in Figure 5.3.

For a mechanism M define the *utility interval* $I_B(M, v, x)$ for the buyer as the two numbers $w(\ell, h)$ such that $\ell = \min\{u_B(M(x, y)) : y \in D_S\}$ and $h = \max\{u_B(M(x, y)) : y \in D_S\}$, and define $I_S(M, c, y)$ analogously for the seller. The NOM property imposes that $I_B(M, v, v) \geq I_B(M, v, x)$ for all $v, x \in D_B$ and that $I_S(M, c, c) \geq I_S(M, c, y)$ for all $c, y \in D_S$.

Theorem 10. *A deterministic bilateral trade mechanism M for domains D_B, D_S is individually rational, weakly budget balanced, and NOM if and only if there are thresholds $p_B^{\min}, p_B^{\max}, p_S^{\min}, p_S^{\max}$, where $p_B^{\min} \leq p_B^{\max}$ and $p_S^{\min} \leq p_S^{\max}$, such that:*

1. *When trade doesn't occur the buyer's and the seller's price are both zero. When trade occurs the buyer's price always exceeds the seller's price, both prices are non-negative, the buyer's price is less than their valuation, and the seller's price is at least the buyer's valuation.*
2. *The set D_B^0 contains all types in D_B less than $p_B^{\min}$, the set D_B^1 contains all types in D_B greater than $p_B^{\max}$, and D_B^{01} contains all types in D_B in between $p_B^{\min}$ and $p_B^{\max}$. If $p_B^{\min}$ is itself in D_B then $p_B^{\min}$ is in either D_B^0 or D_B^{01} . Similarly if $p_B^{\max}$ is in D_B then $p_B^{\max}$ is in either D_B^1 or D_B^{01} .*
3. *For every type $x \in D_B^{01} \cup D_B^1$ the buyer's payment in outcome $M(x, y)$ is at least $p_B^{\min}$ for all $y \in D_S$, and there exists some $y \in D_S$ such that in outcome $M(x, y)$ trade occurs and the buyer pays $p_B^{\min}$. For every type $x \in D_B^1$ the buyer's payment in outcome $M(x, y)$ is at most $p_B^{\max}$ for all $y \in D_S$, and there exists a $y \in D_S$ such that in outcome $M(x, y)$ trade occurs at price $p_B^{\max}$.*
4. *(Analogous to point 2 but for the seller) The set D_S^0 includes all types in D_S greater than $p_S^{\max}$, the set D_S^1 includes all types in D_S less than $p_S^{\min}$, and the set D_S^{01} consists of all types in D_S in between $p_S^{\min}$ and $p_S^{\max}$. If $p_S^{\min}$ itself is in D_S , then $p_S^{\min}$ is either in D_S^1 or D_S^{01} . Similarly, if $p_S^{\max}$ is in D_S then $p_S^{\max}$ is in D_S^{01} or D_S^1 .*
5. *(Analogous to point 3 but for the buyer) For every type $y \in D_S^{01} \cup D_S^1$ there is some $x \in D_B$ such that in outcome $M(x, y)$ trade occurs with seller's price $p_S^{\max}$. For every type $y \in D_S^1$ there is furthermore a $x \in D_B$ such that in outcome $M(x, y)$ trade occurs at price $p_S^{\min}$.*

Proof. ($\implies$) Assume that M is individually rational, weakly budget balanced, and NOM. Point 1 follows directly from individual rationality and weak budget balance. We will prove points 2 and 3, and since the proofs of points 4 and 5 follow analogous reasoning they are omitted. We assume for simplicity of exposition that D_B^0 , D_B^{01} , and D_B^1 are all non-empty. In case one or more of these sets are empty, the proof reduces to a simpler one.

Let $x_0^{\max}$ be the greatest type in D_B^0 and let x be an arbitrary type less than $x_0^{\max}$. We show that x is in D_B^0 . Suppose the contrary and let y be a

seller's type such that trade occurs in outcome $M(x, y)$. The buyer's price in $M(x, y)$ is at most x by individual rationality, and hence it is less than $x_0^{\max}$. This means that the second element of $I_B(M, x_0^{\max}, x)$ is positive, while $I_B(M, x_0^{\max}, x_0^{\max}) = (0, 0)$; a contradiction to NOM. We have now established that x is in D_B^0 for all x in D_B where $x \leq x_0^{\max}$.

Let $x_1^{\min}$ be the least type in D_B^1 and let x be an arbitrary type greater than $x_1^{\min}$. We show that x is in D_B^1 . Suppose the contrary; then the first element of $I_B(M, x, x)$ is 0. However note that any buyer's price charged when reporting $x_1^{\min}$ is less than $x_1^{\min}$ itself by individual rationality, and hence is less than x . This would mean that the first element of $I_B(M, x, x_1^{\min})$ is positive; a contradiction to NOM. We have now established that x is in D_B^1 for all x in D_B where $x \geq x_1^{\min}$.

The above altogether shows that D_B^0 , D_B^{01} , D_B^1 partition D_B into three contiguous ranges: D_B^0 consists of all valuations in D_B up to and including $p_0^{\max}$, D_B^{01} consists of all valuations in D_B strictly in between $p_0^{\max}$ and $p_1^{\min}$, and D_B^1 consists of all valuations in D_B from $p_1^{\min}$ and up.

Let x be a type in $D_B^{01} \cup D_B^1$, and let y be the seller's bid such that the buyer's price is minimised in outcome $M(x, y)$. Denote this price by $p(x)$. Thus we have that the second element of $I_B(M, x, x)$ is $x - p(x)$. By NOM, for any other bid x' we have that the second element of $I_B(M, x, x')$ is at most $x - p(x)$. In particular for x' in $D_B^{01} \cup D_B^1$ it holds that the second element of $I_B(M, x', x')$ is $x' - p(x')$ and thus the second element of $I_B(M, x, x')$ is $x - p(x')$. Thus, $x - p(x') \leq x - p(x)$, or equivalently $p(x) \leq p(x')$. Since this inequality holds for all x, x' in $D_B^{01} \cup D_B^1$, we have that $p(\cdot)$ is constant over $D_B^{01} \cup D_B^1$, and we define this constant as $p_B^{\min}$. Observe that $p_B^{\min}$ is at most the least type in D_B^{01} , by individual rationality (i.e., the type $\text{succ}_B(x_0^{\max})$). Furthermore $p_B^{\min} \geq x_0^{\max}$ because otherwise the second element of $I_B(M, x_0^{\max}, \text{succ}_B(x_0^{\max}))$ would equal $x_0^{\max} - p_B^{\min} \geq x_0^{\max} - x_0^{\max} = 0$, where 0 is the second element of $I_B(M, x_0^{\max}, x_0^{\max})$; a contradiction to NOM. This establishes that the value $p_B^{\min}$ lies in the interval $[x_0^{\max}, \text{succ}_B(x_0^{\max})]$, as required from points 2 and 3 in the claim.

Let x now be a type in D_B^1 , and let y be the seller's bid such that the buyer's price is maximised in outcome $M(x, y)$. Denote this price by $P(x)$. Thus, the first element of $I_B(M, x, x)$ is $x - P(x)$, for all $x \in D_B^1$. Also, this implies that the first element of $I_B(M, x, x')$ is $x - P(x')$ for all $x, x' \in D_B^1$, which in turn yields that $P(x) \leq P(x')$ for all $x, x' \in D_B^1$, and hence $P(\cdot)$ is constant on D_B^1 . Let this constant be the value of $p_B^{\max}$.

Note that $p_B^{\max}$ is at most $x_1^{\min}$, by individual rationality. Furthermore, $p_B^{\max} \geq \text{prec}_B(x_1^{\min})$, as otherwise the first element of $I_B(M, \text{prec}_B(x_1^{\min}), x_1^{\min})$ would be positive. The first element of $I_B(M, \text{prec}_B(x_1^{\min}), \text{prec}_B(x_1^{\min}))$ is

0; contradicting with NOM. This proves that points 2 and 3 of the claim hold.

($\Leftarrow$) Assume that the 5 points of the claim hold for a mechanism M . We show that M is individually rational, weakly budget balanced, and NOM. The first two properties, individual rationality and weak budget balance, follow trivially from point 1. We next show that NOM holds for the buyer. The proof for the NOM-property for the seller is analogous and is therefore omitted.

Let x be a bid in D_B^0 so that $I_B(M, x, x) = (0, 0)$. Obviously, we have $I_B(M, x, x') = (0, 0)$ for all $x' \in D_B^0$, as desired. For $x' \in D_B^{01}$ we know that the first element of $I_B(M, x, x')$ is 0 by definition of D_B^{01} . The second element of $I_B(M, x, x')$ is $x - p_B^{\min}$ by point 3, and $p_B^{\min}$ is at least x by point 2, thus $x - p_B^{\min}$ is non-positive and $I_B(M, x, x) \geq I_B(M, x, x')$ for all $x' \in D_B^{01}$, as desired. For $x' \in D_B^1$ we have $I_B(M, x, x') = (x - p_B^{\max}, x - p_B^{\min})$, and both these expressions are negative as $p_B^{\max}$ and $p_B^{\min}$ are both at least x . Hence $I_B(M, x, x) \geq I_B(M, x, x')$ for all $x' \in D_B^1$, as desired. This establishes that $I_B(M, x, x) \geq I_B(M, x, x')$ for all $x' \in D_B$ and $x \in D_B^0$.

Next, let x be a bid in D_B^{01} , so that $I_B(M, x, x) = (0, x - p_B^{\min})$. For $x' \in D_B^0$ we have that $I_B(M, x, x') = (0, 0) \leq I_B(M, x, x)$. For $x' \in D_B^{01}$ we have that $I_B(M, x, x) = I_B(M, x, x')$ by definition of D_B^{01} and the properties of $p_B^{\min}$ stated under point 3 of the claim. For $x' \in D_B^1$ we have $I_B(M, x, x') = (x - p_B^{\max}, x - p_B^{\min})$. Since $p_B^{\max} \geq x$ the expression $x - p_B^{\max}$ is non-positive, so also in this case we have $I_B(M, x, x) \geq I_B(M, x, x')$. This establishes that $I_B(M, x, x) \geq I_B(M, x, x')$ for all $x' \in D_B$ and $x \in D_B^{01}$.

Lastly let x be a bid in D_B^1 , so that $I_B(M, x, x) = (x - p_B^{\max}, x - p_B^{\min})$. For any bid $x' \in D_B^0 \cup D_B^{01}$ we have that trade may not occur, so that the first element of $I_B(M, x, x')$ is 0, which is less than or equal to $x - p_B^{\max}$. If $x' \in D_B^0$ then also the second element of $I_B(M, x, x')$ is 0, again less than $x - p_B^{\min}$. If $x' \in D_B^{01}$ then the second element of $I_B(M, x, x')$ is $x - p_B^{\min}$, equal to the second element of $I_B(M, x, x)$. Thus $I_B(M, x, x) \geq I_B(M, x, x')$ for all $x' \in D_B^0 \cup D_B^{01}$. For a bid $x' \in D_B^1$ it simply holds that $I_B(M, x, x') = (x - p_B^{\max}, x - p_B^{\min}) = I_B(M, x, x)$, by the properties of $p_B^{\max}$ and $p_B^{\min}$ in point 3 of the claim. This establishes that $I_B(M, x, x) \geq I_B(M, x, x')$ for all $x' \in D_B$ and $x \in D_B^1$. Altogether this yields that $I_B(M, x, x) \geq I_B(M, x, x')$ for all $x, x' \in D_B$, i.e., M is NOM for the buyer. $\square$

In the next chapter we return to the question of implementability, this time investigating which social choice functions may be implemented under the more general notion of indirect mechanisms. Each instance in this class of mechanisms, which encompasses the set of direct-revelation mechanisms

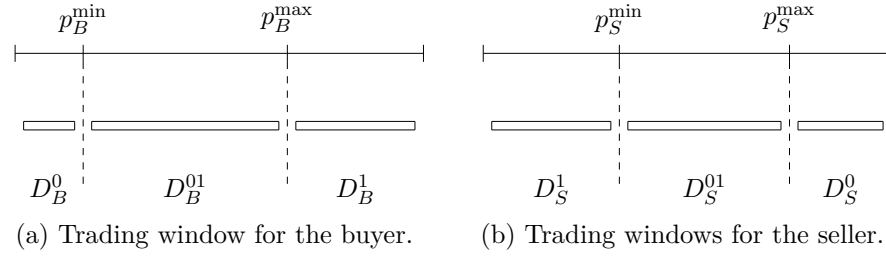


Figure 5.3: A visualisation of the trading windows used in Theorem 10. The lines at the top represent the entirety of the buyer's and seller's domain, respectively, while the bars along the bottom represent the intervals of these domains where trade is guaranteed to occur, not to occur, and where trade depends on the bid of the other player.

of Chapter 4, is based on an implementation tree which specifies how agents are queried in order to learn their type, in addition to the usual social choice function and payment rules. We explore the extent to which this more flexible definition enlarges the class of incentive compatible mechanisms.

INDIRECT MECHANISMS

The fact that the ascending clock auction is OSP while the Vickrey auction is not shows that the manner in which a given social choice function is implemented is vital to whether it is incentive compatible for agents with zero contingent reasoning. This is in contrast to the well-known revelation principle for strategyproofness,¹ which roughly states that for every social choice function that is implementable by an indirect mechanism, there exists a direct-revelation mechanism that also implements it as a truthful mechanism. Moreover, the payments are the same (Nisan, Roughgarden, et al., 2007). Considering the revelation principle for NOM mechanisms, in this chapter we extend the concept of labellings to study the classes of social choice functions that are implementable as indirect NOM mechanisms using payments. As before, given a social choice function, we are free to design both the payment rule and the implementation tree such that the NOM constraints are satisfied. We extend the machinery of Chapter 4 to the more general class of indirect mechanisms and show how these allocation graphs can be modified to take into account the point at which a given agent interacts with the mechanism. We once more characterise the set of implementable social choice functions under general domains, then use this characterisation directly in two restricted settings to show that under these domains the revelation principle, meaning that direct mechanisms are just as powerful as indirect ones.

6.1 CONTRIBUTION OF THIS CHAPTER

We begin by investigating the social choice functions that can be implemented by indirect NOM mechanisms by extending the concept of labellings

¹We do note, however, the gradual revelation principle for OSP mechanisms given in [Bade and Gonczarowski, 2017], which says that any social choice function that is implementable by an OSP mechanism is implementable by a gradual revelation OSP mechanism.

for direct mechanisms from Chapter 4 to handle extensive-form implementations. We use the well-known cycle-monotonicity condition on the graph induced by these labellings to characterise the set of social choice functions implementable by an indirect mechanism.

Main Result 1 (informal). A social choice function f is NOM-implementable in extensive-form if and only if for each i there exists a labelling λ of f such that the graph $\mathcal{G}_{i,f}^\lambda$ has no cycles of negative weight.

We use this characterisation to show two settings in which the class of social choice functions implementable in extensive-form is the same as the class of social choice functions implementable by direct-revelation mechanisms. The first setting is that of single-parameter agents, where each agent's private information is expressible as a single number. Here we can separate an agent's type from her allocation and show that an allocation function is implementable by an indirect mechanism if and only if for each agent we can find a monotone restriction (with respect to the agent's bid) of the allocation function over all allocations it returns. This generalises the characterisation of Theorem 4 for single-parameter agents and we show that in this setting the class of allocation functions implementable by direct and indirect mechanisms are the same.

Main Result 2 (informal). For any NOM indirect mechanism for single-parameter agents implementing a social choice function f there is a NOM direct mechanism implementing f .

Therefore the implementation details do not matter when designing NOM mechanisms for single-parameter agents and we may restrict focus to direct mechanisms without loss of generality. In our second setting we consider mechanisms for two agents where each agent can have an arbitrary type but one of the agents has a “two-type” domain.

Main Result 3 (informal). When there are two agents, one of which having only two possible types, for any NOM indirect mechanism there is an equivalent NOM direct mechanism.

In this setting we show that in order for the revelation principle not to hold there must be a rich set from which we can choose the labelled profiles.

The rest of the chapter is structured as follows: in Section 6.2 we adapt the definition of NOM for direct mechanisms to the setting of extensive-form mechanisms; in Section 6.3 we define profile labellings to incorporate

the implementation tree and the constraints ensuring the absence of obvious manipulations for each agent at each point in the throughout mechanism's execution; in Section 6.4 we once more use these labellings to construct a graph to investigate when the constraints induced by a NOM indirect mechanism are satisfiable; and in Section 6.5 we apply our constructions to two settings to show conditions under which direct mechanisms are just as powerful as indirect ones.

6.2 OBVIOUS MANIPULATIONS IN INDIRECT MECHANISMS

We first define NOM in the context of extensive-form mechanisms, and note that it takes a similar form as defined in [Josué Ortega and Segal-Halevi, 2022]. Roughly, for an extensive-form mechanism to be NOM, then there must be no obvious manipulations for any player at any point they are queried by the mechanism in the implementation tree (taking account for which types and hence type profiles are compatible with the current node).

Definition 20 (Non-obvious manipulability in indirect mechanisms). Let $M = (f, p; T)$ be an indirect mechanism. Then M is *not obviously manipulable* if for all players $i \in N$ and each node u in the implementation tree such that $\pi(u) = i$, the following conditions hold for every pair of separable types $t_i, b_i \in D_i^{(u)}$ at u :

$$\sup_{\mathbf{b} \in D_i^{(u)}(t_i)} \{u_i(M(\mathbf{b}); t_i)\} \geq \sup_{\mathbf{b}' \in D_i^{(u)}(b_i)} \{u_i(M(\mathbf{b}'); t_i)\}, \quad (6.1)$$

$$\inf_{\mathbf{b} \in D_i^{(u)}(t_i)} \{u_i(M(\mathbf{b}); t_i)\} \geq \inf_{\mathbf{b}' \in D_i^{(u)}(b_i)} \{u_i(M(\mathbf{b}'); t_i)\}. \quad (6.2)$$

We use the same naming convention as before, that is: a mechanism M is BNOM if (6.1) is satisfied for all i at each node at which she is queried, and WNOM if (6.2) is satisfied.

Notice that the above definition only requires us to compare the extremes of the player's utility function for each pair of separable types at each node in the tree. Moreover, given a query node u then Definition 20 only requires the agent i to signal her type truthfully at node u , meaning the eventual bid profile leading to the extreme of her utility function may not necessarily be compatible with the type signalled at u , which may occur whenever two types are not separable at a given node in the tree. This definition therefore does not require the agents to consistently signal a single type to the mechanism. This is not a problem in itself: throughout this work we want to design mechanisms in the presence of imperfectly rational

agents, and acting inconsistently throughout the execution of the mechanism is just another behaviour that may arise from such cognitively limited participants. In fact, Li’s first characterisation of obviously strategyproof mechanisms (S. Li, 2017) states that a mechanism is obviously strategyproof if and only if it can be recognised as strategyproof by an agent with zero contingent reasoning skills, meaning they cannot reason about any future states of the mechanism. At any rate, it is standard in the literature to assume that agents play a consistent strategy, and the above definition satisfies the notion of “dynamically consistent” strategies (Josué Ortega and Segal-Halevi, 2022). This roughly states that for any strategy of the players (i.e., any sequence of responses to the mechanism’s queries) there must be a possible type for which the sequence of answers is truthful, and is enforced by Definition 1 of the implementation tree.

It may be easier to restrict attention to agents who play a single type consistently throughout the mechanism’s execution. We alter the above definition of NOM in extensive-form mechanisms slightly by assuming each agent commits to signalling a single type. In Definition 21, the profile compared on the left hand side (the truthful profile when i responds at node u) now assumed to be of the form $(t_i, \mathbf{b}_{-i})$ and the profile on the right hand side (the dishonest profile when i responds at u) is assumed to be of the form $(b_i, \mathbf{b}'_{-i})$ for each query node u and each pair t_i, b_i of inseparable types at u .

Definition 21 (Non-obvious manipulability in indirect mechanisms with consistent strategies). Let $M = (f, p; T)$ be an indirect mechanism. Then M is *not obviously manipulable* if for all players $i \in N$ and each node u in the implementation tree such that $\pi(u) = i$, the following hold for every pair of separable types $t_i, b_i \in D_i^{(u)}$ at u :

$$\sup_{\mathbf{b} \in D_i^{(u)}(t_i)} \{u_i(M(\mathbf{b}); t_i)\} \geq \sup_{\mathbf{b}' \in D_i^{(u)}(b_i)} \{u_i(M(\mathbf{b}'); t_i)\}, \quad (6.3)$$

$$\inf_{\mathbf{b} \in D_i^{(u)}(t_i)} \{u_i(M(\mathbf{b}); t_i)\} \geq \inf_{\mathbf{b}' \in D_i^{(u)}(b_i)} \{u_i(M(\mathbf{b}'); t_i)\}. \quad (6.4)$$

Ultimately, as we will see in Observation 4 in the next section, restricting attention to agents who play consistent strategies does not influence the social choice functions that we can implement as NOM mechanisms. The argument uses the notions of labelling – adapted from Chapter 4 to the case of indirect mechanisms – hence we defer it to the next section. Intuitively, if the extreme (i.e., maximum or minimum) utility of the truthful outcome from the lowest level of query nodes in the implementation tree is greater

than that of each dishonest outcome compatible with the query node at which the two types are separated, then comparing outcomes arising from actions taken at query nodes higher up the implementation tree will not affect the extremes, since these profiles (and hence outcomes) must also be compatible with these higher nodes.

In the following section we recast the concept of labellings in the context of indirect mechanisms to account for the implementation tree. This provides us with another useful characterisation on the class of social choice functions that may be implemented as NOM extensive-form mechanisms, and directly allows us to show are partial revelation principle for NOM.

6.3 EXTENSIVE-FORM LABELLINGS

Given a social choice function, we want to decide whether or not it can be implemented as an indirect NOM mechanism. As before, this means we may define the payment function in order to satisfy the constraints in (6.3) and (6.4), and additionally we may now design the implementation tree in order to prevent any obvious manipulations being available to the players. As a starting point, we first assume that the implementation tree is fixed and reintroduce the concept of labellings, now incorporating the query node at which the given player is asked to respond to the mechanism, allowing us once more to designate the bid profiles to compare satisfying (6.3) and (6.4). In the following definitions, the best-case labelling $\bar{\lambda}^{(u)}$ and the worst-case labelling $\underline{\lambda}^{(u)}$ at node u for player $i = \pi(u)$ are functions mapping a true type t_i and bid type b_i from i 's domain compatible with u to a partial profile of the other players that maximises or minimises i 's utility function when i has type t_i and bids b_i in the mechanism, and the other players take actions signalling a type profile compatible with u .

Definition 22 (Best-case labelling of an indirect mechanism). Fix a mechanism M and let i be an agent with domain D_i . For each query node u such that $\pi(u) = i$ and for every pair of separable types $t_i, b_i \in D_i^{(u)}$ at node u let $\bar{\lambda}^{(u)} : D_i^{(u)} \times D_{-i}^{(u)} \rightarrow D_{-i}^{(u)}(b_i)$ denote the *best-case labelling of M for i at u* , where

$$\bar{\lambda}^{(u)}(t_i, b_i) \in \arg \max_{\mathbf{b}_{-i} \in D_{-i}^{(u)}(b_i)} \{ u_i(M(b_i, \mathbf{b}_{-i}); t_i) \}.$$

Definition 23 (Worst-case labelling of an indirect mechanism). Fix a mechanism M and let i be an agent with domain D_i . For each query node u such that $\pi(u) = i$ and for every pair of separable types $t_i, b_i \in D_i^{(u)}$ at node u let $\underline{\lambda}^{(u)} : D_i^{(u)} \times D_{-i}^{(u)} \rightarrow D_{-i}^{(u)}(b_i)$ denote the *worst-case labelling of M for i*

at u , where

$$\underline{\lambda}^{(u)}(t_i, b_i) \in \arg \min_{\mathbf{b}_{-i} \in D_{-i}^{(u)}(b_i)} \{ u_i(M(b_i, \mathbf{b}_{-i}); t_i) \}.$$

As with labellings for direct-revelation mechanisms in Chapter 4, labellings of indirect mechanisms also encode two main families of constraints. The first, labelling constraints, enforce the validity of the labelled profiles and ensure that given a true type t_i and bid type b_i of player i , the labelled profile $\lambda_i(t_i, b_i)$ truly does lead to an extreme of i 's utility function when the mechanism elicits the bid profile $(b_i, \lambda_i(t_i, b_i))$ and she has type t_i . That is, for each node in the implementation tree u and player $i = \pi(u)$, for each pair of types $t_i, b_i \in D_i^{(u)}$ that are separable at node u , the *labelling constraints for player i at node u* are as follows, for all bid profiles $\mathbf{b} \in D^{(u)}(b_i)$ compatible with the child node of u :

$$u_i(M(b_i, \bar{\lambda}^{(u)}(t_i, b_i)); t_i) \geq u_i(M(\mathbf{b}); t_i) \geq u_i(M(b_i, \underline{\lambda}^{(u)}(t_i, b_i)); t_i). \quad (6.5)$$

The second class of constraints enforces the incentive compatibility requirements for the mechanism to be NOM, meaning the best and worst truthful outcomes when responding to the mechanism's query at a given node are no worse than the best and worst dishonest outcomes from the same node, respectively. For each node u and player $i = \pi(u)$, for each pair of separable types $t_i, b_i \in D_i^{(u)}$, the *incentive-compatibility constraints for player i at node u* are as follows:

$$u_i(M(\lambda^{(u)}(t_i, t_i)); t_i) \geq u_i(M(\lambda^{(u)}(t_i, b_i)); t_i), \quad (6.6)$$

where $\lambda^{(u)}$ can be either a best-case labelling or a worst-case labelling of the mechanism for player i at node u . For a given player i we denote by $\lambda = (\lambda_i^{(u)})_{u: \pi(u)=i}$ the “full” labelling of the implementation tree for player i that incorporates the labelled profiles for player i at each node she is queried.

Recall that Definition 20 does not assume agents commit to signalling a single type consistently throughout the execution of the mechanism. This is because, even though (6.1) and (6.2) require that an agent's best and worst truthful outcomes be no worse than her best and worst dishonest outcomes when signalling her type to the mechanism at a given query node, the inequalities only need to hold at the given query node and she may signal a different type further down the tree. Compare this with (6.3) and (6.4), where the bid profile being compared at the extremes must be of the form $(b_i, \mathbf{b}_{-i})$ for some $\mathbf{b}_{-i} \in D_{-i}^{(u)}(b_i)$. In other words, if i signals b_i at node u

then she must commit to signalling b_i at every node further down in the implementation tree. This is not an issue for two types that are *separable* at u , since if i signals b_i at u then the mechanism is able to rule out that i has some type b'_i that is separated from b_i at u . However, it may be an issue if two types are inseparable at u since the final outcomes of the mechanism being compared may not reflect the type signalled by each agent at every point they are queried. An interpretation of agents interacting with the mechanism in such a manner is that they have such limited cognitive abilities that they are unable even to reason about their own future interactions with the mechanism, much less that of the other players. With the following observation we will show that imposing the assumption that agents *do* instead commit to signalling a single type consistently throughout the mechanism's execution is without loss of generality, and that if a mechanism satisfies (6.1) and (6.2) then it satisfies (6.3) and (6.4), respectively.

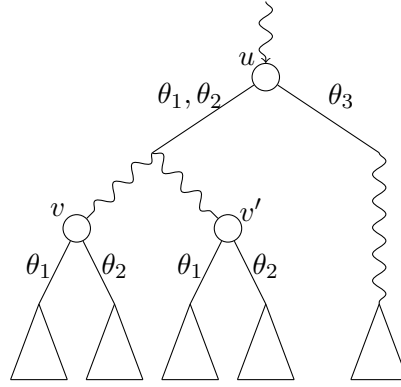


Figure 6.1: The example implementation tree from Chapter 3.

Observation 4. If a mechanism has no obvious manipulations when agents play an inconsistent strategy, then it has no obvious manipulations when they play consistently.

Proof. For simplicity we give the proof based on Figure 6.1 but emphasise that the arguments can be applied to any tree. To make the notation slightly more concise, for the following we write $\bar{\lambda}_{j,k}^{(u)}$ to denote the bid profile $\bar{\lambda}^{(u)}(\theta_j, \theta_k)$ labelled by $\bar{\lambda}$ at node u .

At node u player $i = \pi(u)$ separates types θ_1 and θ_2 from θ_3 . If she signals θ_3 then this is the last time she is queried by the mechanism, while if she signals θ_1 or θ_2 at u then the mechanism will eventually progress to node v or v' , depending on how the other players respond to their queries. In any case, at nodes v and v' player i separates θ_1 from θ_2 and these are the last

times i is queried in the left hand subtree of T . We will argue that for any best-case labelling the assumption that i plays a consistent strategy does not semantically alter the constraints required for BNOM, and analogous reasoning will apply for any worst-case labelling $\underline{\lambda}$ and WNOM.

Due to the structure of T we must designate the best-case profiles $\bar{\lambda}_{j,k}^{(u)}$ for $j, k \in \{1, 2, 3\}$ and $\bar{\lambda}_{j,k}^{(v)}, \bar{\lambda}_{j,k}^{(v')}$ for $j, k \in \{1, 2\}$. Note that for any node where i separates two types then all leaves resulting from i signalling θ_j must have the form $(\theta_j, \mathbf{b}_{-i})$ for $\mathbf{b}_{-i}$ being some bid profile of the other players compatible with said node. Therefore $\bar{\lambda}_{j,3}^{(u)}$ must be of the form $(\theta_3, \cdot)$ for $j \in \{1, 2, 3\}$. Similarly for $j \in \{1, 2\}$ the profiles $\bar{\lambda}_{j,1}^{(v)}, \bar{\lambda}_{j,1}^{(v')}$ must both be of the form $(\theta_1, \cdot)$, and $\bar{\lambda}_{j,2}^{(v')}$ must be of the form $(\theta_2, \cdot)$. Since θ_1 and θ_2 are inseparable at u then i may only play an inconsistent strategy if she signals one of these types at u . We will show that while enforcing consistent strategies will technically alter the set constraints imposed by the labelling, the outcomes these constraints encode will be the same, and hence nothing changes semantically. Fix i 's true type to be θ_1 and consider $\bar{\lambda}_{1,1}^{(u)}$ of the form $(\theta_2, \cdot)$. The labelling constraints at u ensure that the utility $u_i(M(\theta_1, \bar{\lambda}_{1,1}^{(u)}); \theta_1)$ is the maximum among all bid profiles in the set $D^{(u)}(\theta_1)$, including $\bar{\lambda}_{1,1}^{(v)}$ and $\bar{\lambda}_{1,2}^{(v)}$. Now since $\bar{\lambda}_{1,1}^{(u)}$ and $\bar{\lambda}_{1,2}^{(v)}$ both appear in the same subtree of T then in addition to the labelling constraint at u enforcing $u_i(M(\theta_1, \bar{\lambda}_{1,1}^{(u)}); \theta_1) \geq u_i(M(\bar{\lambda}_{1,2}^{(v)}); \theta_1)$, there is the labelling constraint at v stating $u_i(M(\bar{\lambda}_{1,2}^{(v)}); \theta_1) \geq u_i(M(\bar{\lambda}_{1,1}^{(u)}); \theta_1)$, hence the outcomes of f at $\bar{\lambda}_{1,1}^{(u)}$ and $\bar{\lambda}_{1,2}^{(v)}$ are equal.

Now by the incentive compatibility constraints at v for true type θ_1 we have that $u_i(M(\bar{\lambda}_{1,1}^{(v)}); \theta_1) \geq u_i(M(\bar{\lambda}_{1,2}^{(v)}); \theta_1) = u_i(M(\bar{\lambda}_{1,1}^{(u)}); \theta_1)$. Since there is a labelling constraint at u enforcing $u_i(M(\bar{\lambda}_{1,1}^{(u)}); \theta_1) \geq u_i(M(\bar{\lambda}_{1,1}^{(v)}); \theta_1)$ then we have $\bar{\lambda}_{1,1}^{(u)} = \bar{\lambda}_{1,1}^{(v)} = \bar{\lambda}_{1,2}^{(v)}$.

Now consider i playing a consistent strategy where the labelled node, denoted $\ell_{1,1}^{(u)}$, must be of the form $(\theta_1, \cdot)$. Using the same argument as before we get $\ell_{1,1}^{(u)} = \bar{\lambda}_{1,1}^{(v)}$, since both profiles appear in the same subtree of T . Since we only change $\bar{\lambda}_{1,1}^{(u)}$ to $\ell_{1,1}^{(u)}$ then we have $\bar{\lambda}_{1,1}^{(u)} = \bar{\lambda}_{1,1}^{(v)} = \bar{\lambda}_{1,2}^{(v)}$ and therefore $\bar{\lambda}_{1,1}^{(u)} = \ell_{1,1}^{(u)}$. As a final step consider the incentive compatibility constraints regardless of whether we assume consistent strategies, which require $u_i(M(\bar{\lambda}_{1,1}^{(u)}); \theta_1) \geq u_i(M(\bar{\lambda}_{1,3}^{(u)}); \theta_1)$ and $u_i(M(\ell_{1,1}^{(u)}); \theta_1) \geq u_i(M(\bar{\lambda}_{1,3}^{(u)}); \theta_1)$. Since $\bar{\lambda}_{1,1}^{(u)} = \ell_{1,1}^{(u)}$ then these constraints are exactly identical. Therefore enforcing consistent strategies does not alter any constraints semantically. $\square$

Therefore, allowing agents to play an inconsistent strategy, or, imposing that they play a consistent strategy, does not alter the constraints semantically and we may restrict attention to the latter without loss of generality. In the next section we use these extensive-form labellings to characterise the class of implementable social choice functions by once again building a graph and encoding each of the NOM constraints as an edge in the graph.

6.4 BUILDING THE INDIRECT ALLOCATION GRAPH

As in Chapter 4 we use the labellings to define a directed graph representing the constraints imposed by NOM in an indirect mechanism. For some labelling λ of an indirect mechanism we now define the graph $\mathcal{G}_{i,f}^\lambda$. As before the set of nodes is the set of bid profiles D and an edge between two nodes is denoted $(x, y; t)$ where t is some type from player i 's domain D_i . The weight of the edge is given by $w(x, y; t) = t(f(x)) - t(f(y))$.

The set of edges again encode two classes of constraints imposed by an incentive compatible mechanism. The *labelling constraint edges at node u for player $i = \pi(u)$* encode the labelling constraints in (6.5). For each pair of types $t_i, b_i \in D_i^{(u)}$ that are separable at u we have the edges:

$$\langle (b_i, \bar{\lambda}^{(u)}(t_i, b_i)) \xrightarrow{t_i} (b_i, \mathbf{b}_{-i}) \xrightarrow{t_i} (b_i, \underline{\lambda}(t_i, b_i)) \rangle \quad (6.7)$$

for each bid profile $\mathbf{b}_{-i} \in D_{-i}^{(u)}$ of the other players excluding i that is compatible with u . Likewise we have the *incentive compatibility constraint edges at node u for player $i = \pi(u)$* that reflect that player i 's best and worst outcomes from acting honestly are no worse than her best and worst outcomes from signalling her type untruthfully. They encode the constraints in (6.6). For each pair of separable types t_i, b_i at node u we have:

$$\langle (t_i, \bar{\lambda}(t_i, t_i)) \xrightarrow{t_i} (b_i, \bar{\lambda}(t_i, b_i)) \rangle \quad (6.8)$$

$$\langle (t_i, \underline{\lambda}(t_i, t_i)) \xrightarrow{t_i} (b_i, \underline{\lambda}(t_i, b_i)) \rangle. \quad (6.9)$$

The edge set of the graph $\mathcal{G}_{i,f}^\lambda$ is simply the union of the edges in (6.7) to (6.9).

We now use cycle monotonicity on this graph to characterise the class of implementable social choice functions.

Theorem 11. *Social choice function f with implementation tree T is BNOM-implementable (respectively, WNOM-implementable) as an extensive-form mechanism if and only if there exists a best-case labelling $\bar{\lambda}$ (respectively,*

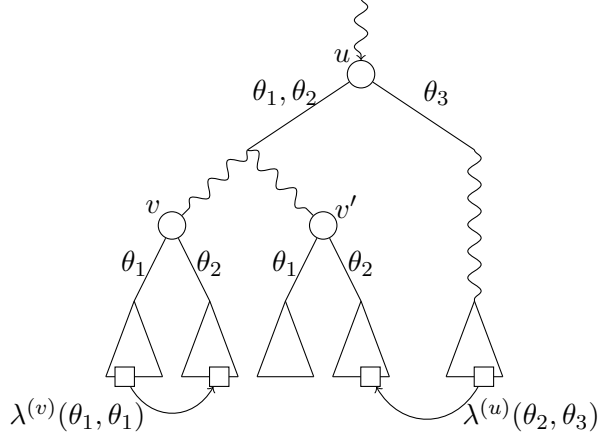


Figure 6.2: f is implementable for i if and only if we can find a labelling λ such that $\mathcal{G}_{i,f}^\lambda$ has no negative cycles.

worst-case labelling $\underline{\lambda}$ of f for player i such that $\mathcal{G}_{i,f}^{\bar{\lambda}}$ (respectively, $\mathcal{G}_{i,f}^{\underline{\lambda}}$) has no negative weight cycles, for all players $i \in N$.

The proof proceeds in exactly the same manner as in Theorem 3. To show the absence of negative weight cycles as a sufficient condition for implementability by an indirect mechanism then we again may introduce a new source node into the graph $\mathcal{G}_{i,f}^\lambda$ with weight zero and use shortest paths in the (finite) graph in order to define payments for the given player i at each bid profile. Since edges correspond directly to the NOM constraints imposed by the indirect mechanism then it is straightforward to show that each time there is an edge in the resulting graph, the shortest path payments allow us to satisfy the corresponding NOM constraint. Where this diverges slightly from the proof of Theorem 3 is that there is now an additional aspect to the labelling to account for, namely the node u at which agent i is acting, but we again show that whenever an edge exists between two pairs of nodes we are satisfying either an incentive-compatibility constraint or a labelling constraint. Without loss of generality, we only need to consider edges between two nodes at the lowest point at which they are separated in the implementation tree. If the label for some action for parent node u and child node v is different then as a consequence of Observation 4 the utility derived by the agent is the same in both outcomes associated with these nodes. To show the necessity of the condition, we again suppose that there exists a negative weight cycle for a function implementable by an indirect mechanism, then show that this leads to contradiction.

Although the reasoning behind the proof of Theorem 11 is essentially identical to Theorem 3, the difference comes in how the structure of the

implementation tree is also incorporated into the graph (at least implicitly, since the graph is based on the labelling λ and the allocation graph in turn depends on this labelling). Therefore although we have a similar result describing the class of social choice functions implementable as indirect mechanisms as we do for direct-revelation ones, it does not mean that all social choice functions implementable as an indirect mechanism are also implementable by a direct one. Note that indirect mechanisms can simulate direct mechanisms by asking each agent simultaneously to separate their entire domain, effectively getting the agents to declare their types directly. Therefore, if a social choice function is implementable by a direct mechanism then it must be by an indirect mechanism, too.

On the other hand, the revelation principle roughly states that if there exists an indirect mechanism implementing a given social choice function as an incentive-compatible mechanism (say, in dominant strategies), then there exists a direct mechanism that implements it as well. Moreover, the payments may be and are often the same. In the next section we apply the characterisation results developed here to study partial revelation principles with respect to NOM in two settings, showing that if a social choice function is implementable as a NOM mechanism by an indirect mechanism, then there is a direct mechanism that also implements it. This means that the class of social choice functions implementable by direct mechanisms is the same as the class implementable by indirect ones for these settings.

6.5 PARTIAL REVELATION PRINCIPLES

In this section we apply the characterisation in Theorem 11 to more finely classify which social choice functions are implementable as indirect mechanisms. In the first part we look at the familiar and widely studied setting of single-parameter agents, whose private information can be encoded as a number, which is applicable in a wide range of practical scenarios. In the second, we look at mechanisms for $n = 2$ agents where one agent has a domain of size two, but both agents' types themselves can be arbitrary.

6.5.1 REVELATION PRINCIPLE FOR SINGLE-PARAMETER AGENTS

A single-parameter agent is one whose type can be described by a single number: when agent i has type t_i then she values the outcome of the allocation function $f(\mathbf{b})$ as the product $t_i \cdot f_i(\mathbf{b})$ of her private type and her personal allocation. This allows us to decouple player types from the output of a mechanism's allocation function to prove structural properties about the

latter in isolation. As in Chapter 4, if f is overlapping then we may always find a “monotone restriction” of f when viewed as a (multi-valued) allocation function of player i ’s bid. In other words, there is always a set of profiles ensuring that i is allocated more as she bids more. We generalise this notion to show a similar condition for implementability by indirect mechanisms, and show that the functions implementable by indirect mechanisms is no different than those implementable by direct ones.

Let u be some node of the implementation tree T for allocation function f with $\pi(u) = i$. We denote by $\mathcal{A}_i^{(u)}(b_i) = \{f_i(b_i, \mathbf{b}_{-i}) : \mathbf{b}_{-i} \in D_{-i}^{(u)}(b_i)\}$ the set of all outcomes of f to player i when she signals type b_i at node u , taken over all possible continuations of the mechanism from the other players. Furthermore let $\mathcal{A}_i^{(u)} = \times_{b_i \in D_i^u} \mathcal{A}_i^{(u)}(b_i)$. We say that f is *overlapping for i at node u* with $i = \pi(u)$ if there exists a non-decreasing vector $\mathbf{a} \in \mathcal{A}_i^{(u)}$. This says that when i is responding to the mechanism at node u there always exists an outcome of f such that when i has a larger type she is allocated more by the mechanism. Notice that if two types b_i, b'_i are indistinguishable at node u then the ranges of f from signalling either type are the same. We characterise the class of social choice functions implementable by an indirect NOM mechanism with the following theorem.

Lemma 2. *Social choice function f is implementable as an indirect NOM mechanism if and only if there exists an implementation tree T such that f is overlapping at every node u with $\pi(u) = i$, for each player $i \in N$.*

The arguments follow identical reasoning to Lemma 2. Since the social choice function f we wish to implement is given to us then the range of outcomes $\{f_i(b_i, \mathbf{b}_{-i})\}_{\mathbf{b}_{-i}}$ of the function to player i for each bid b_i is fixed. The extra flexibility provided by an indirect mechanism over a direct-revelation implementation of f is that we may choose how to partition this set of outcomes for a given bid using the nodes of the implementation tree. The sufficiency follows from the fact that, given f which is overlapping at u for some implementation, then we can always define an extensive-form labelling ensuring no negative cycles in the graph $\mathcal{G}_{i,f}^\lambda$. For necessity, supposing that f is not overlapping at u leads to negative cycles in $\mathcal{G}_{i,f}^\lambda$ and hence contradiction by Theorem 11.

We now wish to show that the set of NOM-implementable allocation functions for direct and indirect mechanisms is the same for single-parameter agents. Our definition of an indirect mechanism is able to emulate a direct revelation mechanism: simply query each player once and get them to separate every type in their domain from one another. This amounts to asking each player to directly declare their type to the mechanism. There-

fore all direct-revelation NOM-implementable allocation functions are implementable by an indirect mechanism. This of course covers cases where agents are single-parameter. We may use Lemma 2 to show the opposite: for single-parameter agents all NOM-implementable functions by an indirect mechanism are implementable by a direct mechanism. The functions implementable by direct and indirect mechanisms is therefore the same, thus the revelation principle holds for single-parameter agents.

Theorem 12. *For every NOM indirect mechanism implementing social choice function f for single-parameter agents there is a NOM direct mechanism implementing f .*

Proof. Let f be a NOM-implementable social choice function that is implemented by indirect mechanism $(f, p; T)$. Note that by Lemma 2 f must be overlapping for $\pi(u)$ at each node u of T . We will show that restructuring the implementation to one in which player i is queried once at the root and fully separates each type in her domain, which must occur in a direct-revelation implementation of f , maintains the overlapping property of f at the root. Therefore the resulting implementation will also be NOM.

First recall the definition of $\mathcal{A}_i^{(u)}(b_i)$, which denotes the set of all outcomes possible when player i signals b_i at node u . Note that for some player $i = \pi(u)$ and two types $b_i, b'_i \in D_i^{(u)}$ that are overlapping at node u , the overlapping property of f at u may only be violated if we restrict either of the two sets $\mathcal{A}_i^{(u)}(b_i)$ and $\mathcal{A}_i^{(u)}(b'_i)$. Now fix a player i for which there are two types $b_i, b'_i \in D_i^{(v)}$ which are inseparable at u but separable at child node v of u . Clearly the set of all outcomes reachable after signalling b_i at u contains all outcomes reachable after signalling b_i at v , hence $\mathcal{A}^{(v)}(b_i) \subseteq \mathcal{A}_i^{(u)}(b_i)$. Thus modifying the tree T such that i now separates b_i from b'_i at node u rather than node v cannot violate the overlapping property of f at u (which was true by hypothesis). We may repeat this argument and continue to query i further up the tree, where she separates progressively more types from her domain. Our final modified tree T' will query i once at the root, at which point she separates each type from every other. At each point the overlapping property is maintained and thus f is also overlapping for i at the root. Hence f is implementable by this modified tree T' by Lemma 2.

As a final step we will use the overlapping property of Definition 16 to show f is implementable by a direct-revelation NOM mechanism. Note that in a direct mechanism each player is “queried” exactly once simultaneously and directly reports her type, effectively separating all types in her domain. We have reconstructed T into T' such that i is now queried once at the root and fully separates each type in her domain. Hence from i ’s perspective T'

is a direct-revelation implementation of f and since f is overlapping for i at the root in implementation T' so too is f overlapping for i in the sense of Definition 16. Note that we do not need to move each player to the root simultaneously – to show the existence of a direct-revelation mechanism it suffices to show that this can be done for each player in isolation while maintaining the overlapping property. Thus there must exist a direct-revelation mechanism that NOM-implements f . $\square$

6.5.2 REVELATION PRINCIPLE FOR TWO-TYPE DOMAINS

In this section we prove the revelation principle for a setup that is independent of agents' domains. This result says that for the revelation principle not to hold there must be a rich set of labels that the extensive-form mechanism can use. A *two-type domain* is a domain containing two possible types, where the types themselves can be arbitrary.

Theorem 13. *The revelation principle holds for NOM mechanisms when $n = 2$ and one of the two agents has a two-type domain.*

Proof. Assume without loss of generality that agent 2 has a two-value domain, $D_2 = \{t, t'\}$. Fix the social choice function f ; we will prove that if there is an extensive-form NOM mechanism M that implements f then there is a direct-revelation mechanism M' that is NOM and also implements f . Let $\mathcal{G}_{i,f}^\lambda$ and $\mathcal{H}_{i,f}^\ell$ be the graphs of interest for extensive-form and direct-revelation labellings λ and ℓ , respectively.

We start by observing that the revelation principle holds true for agent 2. Assume by contradiction that this is not true. This means that for any ℓ there is a negative-weight two-cycle in $\mathcal{H}_{i,f}^\ell$ between $(\ell(t', t), t)$ and $(\ell(t, t'), t')$ and then the outcome of the mechanism for agent 2 must differ in these two profiles (for otherwise, each edge would weigh 0). Therefore, to implement the mechanism, the extensive-form implementation must separate these profiles. Clearly, this cannot be agent 2 or we would get the same negative cycle. Thus, agent 1 must separate $\ell(t, t')$ from $\ell(t', t)$ before agent 2 separates t from t' . Reiterating the argument for all the possible labellings ℓ yields the contradiction that agent 2 can never separate the two types in her domain.

Let us now focus on the revelation principle for agent 1. Since agent 2 has a two-value domain, then any extensive-form mechanism, including M , has only one node v in each path from the root to a leaf of its implementation tree T such that $\pi(v) = 2$. Thus, the only difference between a direct-revelation mechanism M' and an extensive-form mechanism M for agent 1

is that the latter restricts the labels $\lambda^{(u)}$ at nodes u below v to belong to $D_2^{(u)}$ whereas the domain of the labels would be unrestricted for the former. Labels are unrestricted for both M and M' at nodes above v .

Therefore, if the children of v are leaves of T then M has the same labels (and then graphs) of M' , thus proving the revelation principle. Otherwise, let us focus on a type $t_1 \in D_1^{(v)}$, i.e., t_1 is still in the domain of agent 1 when 2 is queried at v . Since $|D_2| = 2$ then there are going to be labels $\lambda^{(u)}$ for t_1 at ancestors u of v and labels $\lambda^{(z)}$ for t_1 at descendants z of v that are equal. Recall that $\lambda^{(u)}$ would be a feasible choice of a label $\ell(\star, t_1)$ for a direct-revelation mechanism M' . By inspection, this means that $\mathcal{G}_{i,f}^\lambda$ contains $\mathcal{H}_{i,f}^\ell$ as a sub-graph. Since M is NOM then both $\mathcal{G}_{i,f}^\lambda$ and $\mathcal{H}_{i,f}^\ell$ have no negative-weight cycles, thus proving that M' is NOM. $\square$

In this chapter we extended the notion of profile labellings from their use in direct-revelation mechanisms in Chapter 4 to study which social choice functions can be implemented as extensive-form NOM mechanisms. We formalise the constraints required by incentive compatibility in these settings using the concept of extensive-form profile labellings and use cycle monotonicity to characterise the social choice functions that admit NOM payments. Applying this to mechanisms for single-parameter agents we find that we may restrict our attention to the simpler direct mechanisms without sacrificing incentive properties, and similarly for the setting of two players in which one has a two-value domain. An interesting direction would be to study what leverage indirect mechanisms for single-parameter agents *do* provide over direct mechanisms with respect to other desirable notions, e.g. fairness. It would also be useful to further explore the relationship between direct-revelation and extensive-form NOM mechanisms for other agent domains beyond those covered by Theorems 12 and 13. Of course, the question of whether the revelation principle holds for NOM under general domains is still open.

Open problem 3. Does the revelation principle hold for NOM in general?

In the next chapter we switch attention once again to studying the performance of NOM mechanisms compared to more restrictive solution concepts such as strategyproofness from the literature. We show that there are simple ways to modify a potentially manipulable allocation function and to define simple payment rules that will result in a NOM mechanism. These principles apply to a variety of settings, and at the most general we study goods auctions, where each agents derive some (positive) utility from their allocation, and procurement auctions, where instead agents incur a cost.

WILLY WONKA MECHANISMS

We now turn our attention to designing auctions without obvious manipulations. Imagine there are some items that we wish to allocate to a set of agents. One way of doing so is of course by running an auction, where agents declare a bid for each item on offer and an allocation is determined in order to, for example, maximise the social welfare. As we have seen, defining the appropriate payments can be a useful tool that helps us to realign the incentives of the agents in order to make the allocation rule incentive compatible. Naturally, we can also design auctions that aim to extract more revenue (i.e., sum of payments made by the agents to the mechanism) from the participants of the auction. This chapter deals with designing auctions without obvious manipulations that aim to maximise the revenue extracted from the bidders. Additionally, we consider the opposite setting where instead of allocating items to the agents, we wish to procure some items from them, and each agent incurs some associated cost for being selected by the mechanism. This is the setting of procurement auctions, and since agents incur costs (thus decreasing their utility), we as the mechanism must instead compensate them for their participation in the mechanism. Again, we can use payments firstly to ensure agents have no incentive to act dishonestly in the mechanism, and doing so will require some amount of overpayment with respect to the bids of the agents (since simply using first-price payments is obviously manipulable, and anything less than first-price payments would fail to satisfy individual rationality). Now instead of seeking to maximise the sum of payments extracted from the bidders, we are aiming to minimise the amount by which we overpay the bidders in order to achieve an incentive compatible mechanism.

Our initial motivation is to study the revenue guarantees of NOM auctions for selling digital goods. In this setting there is a infinite supply of identical goods which have zero marginal cost to produce, for example a digital music file. [Goldberg et al., 2006] introduce the notion of “competi-

tive” auctions, adapted from the analysis of online algorithms, in the setting of prior-free auctions for digital goods. An auction is said to be competitive to some benchmark if on every bid profile the revenue extracted by the mechanism is within a constant factor of that specified by the benchmark. They focus on dominant strategy incentive compatibility and show that no truthful auction in this setting is competitive to even the optimal fixed-price benchmark, much less the optimal omniscient benchmark. Specifically, let $\mathcal{T}$ represent the maximum possible revenue that a digital goods auction could extract from the bidders if it had full knowledge of the bidders’ private valuations (disregarding incentives). This benchmark is simply the sum of the bids submitted to the mechanism, and is a trivial upper bound on the revenue that a strategyproof mechanism can extract known as the “optimal omniscient benchmark”. Let $\mathcal{F}$ denote the optimal fixed-price revenue, representing the optimal revenue that a (possibly untruthful) auction can extract from the bidders when it offers the same take-it-or-leave-it price to every bidder. The authors show that $\mathcal{F}$ compares favourably to $\mathcal{T}$, with $\mathcal{F} = \Omega(\mathcal{T}/\log h)$ and $\mathcal{F} = \Omega(\mathcal{T}/\log n)$, where h denotes the highest bid and n the number of bidders. They also show that $\mathcal{F} \geq \mathcal{T}/(2\log h)$, bounding the penalty for requiring the auction to offer a single fixed price. However, for deterministic truthful auctions they show that for any constant α there exists an input such that the ratio between the revenue extracted by the mechanism and $\mathcal{F}$ is $O(1/h)$ and $\alpha h \leq \mathcal{F}$, thus showing that no deterministic truthful auction is competitive under worst-case analysis. Thus deterministic strategyproof auctions perform poorly under unlimited supply. We therefore consider revenue maximisation in deterministic auctions for the relaxed notion of incentive compatibility modelled by NOM, and from there we apply our approaches to a number of different auction settings, finding that they perform very favourably and guarantee to recover a high fraction of even the optimal omniscient revenue.

We find for both regular (goods) auctions and procurement auctions that given an allocation function that may be obviously manipulable, it is simple to design a payment function that ensures NOM incentive compatibility and negates the best-case obvious manipulations by designating a bid profile where each agent maximises her utility for all possible mechanisms that make no positive transfers. Additionally, worst-case obvious manipulations can be avoided by modifying the allocation function only slightly, to ensure that for each agent it is also possible to lose the auction. We provide a characterisation stating that every mechanism with nonzero approximation to the optimal revenue or frugality must designate both a “golden ticket” and a “wooden spoon”, corresponding to the best-case and worst-case labelled

outcomes of Chapter 4, respectively. We then show that such mechanisms when applied to both goods auctions and procurement auctions achieve good performance against the optimal.

7.1 CONTRIBUTION OF THIS CHAPTER

Starting with goods auctions in which agents are single-parameter we identify two necessary and sufficient conditions for mechanisms to be NOM in this context. These conditions state that for each agent and each valuation there must be some valuation profile of the other bidders in which she wins the auction for free and in which she loses. We refer to these bid profiles and “golden ticket” and “wooden spoon” profiles, and mechanisms where each agent has both a golden ticket and wooden spoon profile for each possible valuation as “Willy Wonka” mechanisms. For single-parameter settings this characterisation holds for any allocation and payment functions with non-zero approximation guarantee to the optimal social welfare.

Main Result 1 (informal). An auction that α -approximates social welfare and is β -competitive against social welfare for any positive α, β is not obviously manipulable if and only if it is a Willy Wonka mechanism.

We can show a slightly weaker result when generalising from binary allocation settings to more general outcome spaces. Now the golden ticket profile is any input to the mechanism where an agent receives her most valued allocation for free, and this is both necessary and sufficient whenever the allocation function is maximal-in-range. We show that the wooden spoon profile, again an input resulting in the agent losing the auction, is also sufficient to satisfy the worst-case NOM constraints.

Given black box access to an α -approximate (or maximal-in-range) allocation function we then design basic payment rules that result in high revenue for the auctioneer, which sacrifices the revenue from the lowest winning bidder in the approximately optimal allocation in order to satisfy the golden ticket property. Specifically, given an input bid profile we first compute the approximately optimal allocation and charge each winning bidder first-price payments, except the lowest winning bidder whose payment is set to zero. Of course, if we were to do this on each allocation then we might sacrifice the revenue from the only winning bidder (depending on the feasible set of allocations) and end up with zero revenue. It suffices to have only one bid profile where this occurs in order to satisfy the best-case NOM

constraints. Therefore we can choose the bid profile corresponding to an allocation that maximises the number of winners as the golden ticket and thus minimise the amount of revenue we sacrifice. For all remaining bid profiles we will simply recover the optimal revenue.

Main Result 2 (informal). There is a mechanism that is not obviously manipulable and $\alpha(1 - 1/\tau)$ -competitive to the optimal revenue, where α denotes the approximation guarantee of the allocation function and τ the maximum number of winning bidders in an optimal allocation.

Finally we turn our attention to procurement auctions where agents incur costs for being included in an allocation and agents must therefore be compensated. Now we aim to minimise the social cost, and we want to design the payments to implement the socially optimal allocation as a NOM mechanism. We want our payment rules to be frugal, meaning they should not overpay the agents to achieve incentive-compatibility. To this end we apply our golden ticket and wooden spoon properties to this setting. The former states that for each agent, on top of being compensated her bid (due to individual rationality), there must be a profile where she is paid an additional and costly sum of money that is sure to maximise her utility. The latter, again, states that each must also lose the auction. Our first mechanism is a natural analogue to the Willy Wonka mechanisms given for revenue maximisation, but we show that its performance with respect to frugality against the optimal solution can go to zero. We then amend the payment rule to modify the (costly) golden ticket profiles and show that the resulting mechanism never overpays the agents by more than a factor of two with respect to the second-lowest social cost.

Main Result 3 (informal). There is a procurement auction that is not obviously manipulable and pays at most twice the cost of the second-best solution.

We note that we change the benchmark from the optimum social cost to that of the second best social cost. For a given agent, if they can never be allocated at the same time as another agent, then the frugality ratio with respect to the optimal will go to zero if the auction is NOM, and is therefore unavoidable. We can sidestep this issue by benchmarking the mechanism against the cost of the next best solution, and show that now whenever an agent can only be the sole winner of the allocation our mechanism does not overpay at all with respect to this cost.

The rest of this chapter is structured as follows: in Section 7.2 we introduce notation and terms specific to goods auctions and procurement auctions; in Section 7.3 we investigate how to design mechanisms without obvious manipulations for revenue-maximisation; and in Section 7.4 we conversely look at how to design procurement auctions that do overpay the selected agents by too much in order to achieve incentive compatibility with respect to NOM.

7.2 PRELIMINARIES

In addition to the preliminaries and notation introduced in Chapter 3 we introduce the following terms and notation specific to our setting of goods auctions and procurement auctions. We consider direct mechanisms $M = (f, p)$ where once again $f : D \rightarrow \mathcal{A}$ is an allocation function mapping bids profiles $\mathbf{b}$ from the domain D to some feasible allocation from $\mathcal{A}$, and $p : D \rightarrow \mathbb{R}^n$ is a payment function mapping bid profiles to payment vectors which describe, depending on the setting, payments to be either extracted or reimbursed to each agent.

Goods and chores. Depending on the setting, we assume each agent receives either a non-negative or non-positive utility for each feasible allocation in $\mathcal{A}$. We refer to the former as a goods auction and the latter as a procurement auction, or equivalently, an auction in which allocations represent (bundles of) chores. In a goods auction we extract payments from the bidders and we assume our auctions make *no positive transfers*, meaning no bidder is paid money in addition to their allocation. Conversely, for a procurement auction agents incur a cost for being allocated and hence must be compensated, and here individual rationality implies that all payments to agents are non-negative. We will formally introduce these notions, including those related to the optimality of an auction's allocation and payment functions, in their respective Sections 7.3 and 7.4 as needed.

Structure of the feasible set. The structure of $\mathcal{A}$ reflects the auction setting under consideration. We provide several examples here for clarity of exposition. A set system is a tuple $(E, \mathcal{S})$ where E is referred to as the ground set and $\mathcal{S} \subseteq 2^E$ the family of subsets. In general in a *binary allocation unit-demand* setting each bidder can either be selected or not selected in each allocation A , and the feasible set of allocations $\mathcal{A}$ in these settings can be described by the set $\mathcal{S}$ in the set system $(N, \mathcal{S})$. This set $\mathcal{S}$ will vary depending on the type of auction being run, and can encompass, for

example, digital goods auctions [Goldberg et al., 2006], knapsack auctions [Aggarwal and Hartline, 2006], and spanning tree auctions [Quadir, 2017]. For these settings we can represent each allocation by the vector $A \in \{0, 1\}^n$, where $A_i = 1$ if i is allocated and $A_i = 0$ otherwise. In a multi-unit auction there are k copies of a single item and each agent may therefore be allocated up to k copies (and note that agent types may not be linear functions in the number of items they are allocated). In this case each allocation can be represented by a vector $A \in \{0, \dots, k\}^n$, where A_i describes the number of copies allocated to agent i (and clearly we require $\sum_{i \in N} A_i \leq k$ for A to be a feasible allocation). As a final example, consider a *combinatorial auction* in which there is a set S of items and each allocation distributes a (possibly empty) bundle of items to each agent. Here for each allocation A the entry $A_i \subseteq S$ describes the bundle of items allocated to i , and therefore the feasible set of allocations can be represented by the set of all A such that $A_i \subseteq S$ for each i , and $A_i \cap A_j = \emptyset$ for each distinct $i, j \in N$. Regardless of the specific setting we assume that for each player there is some feasible allocation in $\mathcal{A}$ where they do not win, otherwise clearly there is nothing to be done from the perspective of realigning incentives.

Bidding languages. In general we assume that each agent i reports her type directly to the mechanism by specifying a real number for each unique personal allocation $A_i \in \mathcal{A}_i$. To avoid cluttering the notation we will drop the i subscript from A_i . Therefore agent i 's type is represented by the vector $t_i = (t_i^A)_{A \in \mathcal{A}_i}$. In Section 7.3 we assume that for each $A \in \mathcal{A}$ we have $t_i^A \in [0, h]$ and in Section 7.4 we assume that $t_i^A \in [-h, 0]$ for some $h \in \mathbb{R}_+$. In auctions based on set systems each agent needs only to report a single number to the mechanism, and as such we refer to them in this case as being *single-parameter*. When this is not the case then agents are in general said to be *multiparameter*.

7.3 REVENUE MAXIMISATION

In this setting we consider allocation functions that (approximately) maximise the social welfare. Fix a valuation profile $\mathbf{v}$. The social welfare of an allocation A with respect to $\mathbf{v}$ is simply $\text{SW}(A, \mathbf{v}) = \sum_{i \in N} v_i(A_i)$. An allocation function a is said to α -approximate the social welfare for $\alpha \in (0, 1)$ if $\text{SW}(a(\mathbf{v}), \mathbf{v}) \geq \alpha W^*$ for every $\mathbf{v}$, where $W^* = \max_{A \in \mathcal{A}} \text{SW}(A, \mathbf{v})$ denotes the optimal social welfare of an allocation with respect to $\mathbf{v}$.

We study the revenue we can extract from the bidders using NOM mechanisms. The revenue of an auction (a, p) on bid profile $\mathbf{b}$ is simply the sum

of the payments $\sum_{i \in N} p_i(\mathbf{b})$. The performance of a mechanism's payment function will be measured with respect to some benchmark. Let $\mathcal{X}(\mathbf{b})$ be the revenue of some payment benchmark $\mathcal{X}$ when given bid profile $\mathbf{b}$. A payment function p is said to be β -competitive against $\mathcal{X}$ for $\beta \in (0, 1)$ if for all $\mathbf{b}$ it holds that $\sum_{i \in N} p_i(\mathbf{b}) \geq \beta \mathcal{X}(\mathbf{b})$.

7.3.1 BINARY ALLOCATION

We begin by characterising the class of auctions which are NOM for single-parameter agents binary allocation settings. Our characterisation shows that two properties based on the outcomes of the mechanism, known as “golden ticket” and “wooden spoon” profiles, are necessary and sufficient in order for the mechanism to be NOM. The former states that, for each bidder and each bid, it is possible, given some bid profile of the other agents, to win the auction for free. It is straightforward to see that this ensures that (3.2) is satisfied, since there is a bid profile where each agent receives her highest possible utility (assuming no positive transfers from the mechanism to the agent). On the other hand, as long as the allocation function has a positive welfare guarantee then the golden ticket property is necessary when the mechanism is BNOM, otherwise she may submit a bid below the assumed minimum payment price and get strictly greater utility in the best case than when bidding truthfully.

Lemma 3 (Golden ticket for binary allocation). *Let $M = (a, p)$ be an auction whose allocation function α -approximates the social welfare for any positive α . Then M is BNOM if and only if for every agent $i \in N$ with valuation v_i there exists a bid profile where i wins the auction for free.*

Proof. ($\implies$) Begin by observing that i must always be able to win the auction. On the bid profile $(v_i, 0, \dots, 0)$ a positive approximation to the optimal social welfare is only possible if i is allocated. Suppose for contradiction that whenever i wins the auction she pays at least some positive price p . Her best utility when bidding v_i truthfully is at most $v_i - p$. Now let b_i be any misreport such that $0 < b_i < p$ and note that $p \leq v_i$ by individual rationality. Take the profile $(b_i, 0, \dots, 0)$ and by the same argument i must win. Her utility for this misreport is at least $v_i - b_i \geq v_i - p$, breaking BNOM. Therefore i must be able to win the auction for free.

($\impliedby$) Since i can win the auction for free when bidding v_i her best truthful utility is exactly v_i . This is clearly unbeatable by any misreport if M makes no positive transfers, hence the auction is BNOM. $\square$

Note that as a corollary of Lemma 3 we have that for NOM auctions in which for each allocation the set of winning bidders is a singleton, such as single-item auctions, then the revenue guarantee against the optimal goes to zero when taken over all possible inputs. To verify that this is the case, consider the Vickrey auction, which is strategyproof and hence also NOM. The revenue obtained by this mechanism is simply the second highest bid, which is zero whenever there is a single agent i with a positive bid b_i and the remaining agents bid zero. The optimal revenue is b_i , resulting in a 0-approximation.

Analogously to the golden ticket, the wooden spoon property states that for each bidder and each bid, it is possible to lose the auction. This guarantees that (3.3) is satisfied, since now there is a bid profile where each agent receives zero utility and by individual rationality this is the lowest possible when bidding truthfully. Since this holds for each bid then the worst utility of a dishonest bid can certainly be no greater than zero. Now to prove the necessity of the wooden spoon we assume the payment function extracts any positive factor of revenue from the bidders and show that if the agent always wins the auction then it is possible to underbid and get a strictly greater worst-case utility versus a truthful bid.

Lemma 4 (Wooden spoon for binary allocation). *Let $M = (a, p)$ be an auction where the payment function guarantees to extract any positive factor $\beta > 0$ of the sum of bids in revenue from the bidders. Then M is WNOM if and only if for every agent $i \in N$ with valuation v_i there exists a bid profile such that i loses the auction.*

Proof. ($\implies$) Suppose bidder i always wins the auction. When bidding $b_i < v_i$ her worst utility is at least $v_i - b_i$. On the profile $(v_i, 0, \dots, 0)$ then i must win the auction and pay at least βv_i meaning her worst truthful utility is at most $(1 - \beta)v_i$. Taking a misreport $b_i < \beta v_i$ results in a worst-case utility of $v_i - b_i > (1 - \beta)v_i$, breaking WNOM. Therefore i must lose the auction.

($\impliedby$) Take bidder i with value v_i . Since i can lose then her worst utility when bidding v_i truthfully is at exactly 0. If i were to underbid then the same argument gives her a worst dishonest utility of 0, while submitting an overbid b_i can clearly yield a negative utility if i were to win and pay $b_i > v_i$. Hence M is WNOM. $\square$

In the proof of Lemmas 3 and 4 we use the zero bid of the other players to show that there is always a bid profile where a given agent i wins, assuming the mechanism is α -optimal for any positive α . This is of course rather

exploitative and points to a brittleness of NOM. To slightly mitigate this we note that the result holds (subject to some conditions) even if we exclude 0 from the domains of the agents.

Observation 5 (Excluding the zero bid). Lemmas 3 and 4 hold when $0 \notin D_i$ for each $i \in N$.

Proof. Take some bid profile $\mathbf{b}$. If i never wins then she does not win on $\mathbf{b}$. In general the maximum social welfare obtained by an allocation where i does not win is given by $\max_{A \in \mathcal{A}: i \notin \mu(A)} \text{SW}(A, \mathbf{b})$ while the minimum social welfare that is achieved when i is allocated is given by $\min_{A \in \mathcal{A}: i \in \mu(A)} \text{SW}(A, \mathbf{b})$. Hence for an allocation function that does not allocate i on $\mathbf{b}$ its approximation factor is at most

$$\alpha \leq \frac{\max_{A \in \mathcal{A}: i \notin \mu(A)} \text{SW}(A, \mathbf{b})}{\min_{A \in \mathcal{A}: i \in \mu(A)} \text{SW}(A, \mathbf{b})}.$$

Consider a digital goods auction modelled by the set system $(N, \mathcal{A})$ where $\mathcal{A} = 2^N$ and fix a player $i \in N$. Suppose that whenever i wins the auction she pays a price of at least $p > 0$. Then her best truthful utility is at most $v_i - p$. Now let $\alpha > (n-1)\varepsilon/\tau$ for some $\tau > 0$ and take an underbid b_i with $\tau < b_i < p$. On $\mathbf{b} = (b_i, \varepsilon, \dots, \varepsilon)$ then i must win by the approximation guarantee, but by individual rationality she can pay no more than b_i meaning her best dishonest utility is at least $v_i - b_i$, breaking BNOM. $\square$

With Lemmas 3 and 4 we have a full characterisation for NOM auctions in a range of binary allocation settings. Namely, as long the allocation function has some positive approximation guarantee to the optimal social welfare and the payment function always extracts some positive revenue from the bidders, then for each agent there must be both a bid profile where she wins the auction for free and a bid profile where she loses. This motivates the following definition for auctions that satisfy these criteria.

Definition 24 (Willy Wonka mechanism for binary allocation). A *binary allocation Willy Wonka mechanism* is a mechanism $M = (a, p)$ whose allocation and payment functions satisfy the following. For each agent $i \in N$ and each bid $b_i \in D_i$, there exists:

1. a bid profile $(b_i, \mathbf{b}_{-i})$ where i wins the auction for free.
2. a bid profile $(b_i, \mathbf{b}'_{-i})$ where i loses the auction.

Given a bid b_i of player i we refer to the *golden ticket profile for b_i* as the profile $\mathbf{b}_{-i}$ such that i wins for free on $\mathbf{b} = (b_i, \mathbf{b}_{-i})$. Likewise we refer to the

wooden spoon profile for b_i as the profile $\mathbf{b}_{-i}$ such that i loses the auction on $\mathbf{b} = (b_i, \mathbf{b}_{-i})$. Therefore as long as each player i has both a golden ticket profile and a wooden spoon profile for each bid b_i then the resulting mechanism is NOM. In the following we will assume that the wooden spoon profile is ensured by the feasible set, meaning that for a sufficiently approximately-optimal allocation function and each bid of an agent i there is a bid profile of the other agents where i must lose the auction.

Our idea is to design Willy Wonka mechanisms that sacrifice a small portion of the revenue in order to achieve incentive compatibility. Ideally, for each bidder i and each bid b_i we want to find a bid profile $\mathbf{b}_{-i}$ and an allocation A where i wins that maximises the social welfare of the bidders excluding i , since we can charge first-price payments to the winners and award i her allocation for free. This $\mathbf{b}_{-i}$ would then be the golden ticket profile for i with bid b_i . In other words for each i with bid b_i the optimal Willy Wonka mechanism would compute an allocation that obtains revenue

$$\max_{A: i \in \mu(A)} \{ \max_{\mathbf{b}_{-i}} \text{SW}(A, \mathbf{b}_{-i}) \}. \quad (7.1)$$

Suppose we have black box access to an α -approximate allocation function. In order to tackle finding a good approximation to the optimal revenue above, consider the Willy Wonka mechanism that first computes an approximately optimal allocation and then charges first-price payments to all but the lowest winning bidder, who is allocated at no charge. Then the revenue extracted by the mechanism is simply

$$\alpha \cdot \text{SW}(A, \mathbf{b}) - \min_{i \in \mu(A)} b_i.$$

A problem with this mechanism is that it specifies a golden ticket profile for every input $\mathbf{b}$. Now if there is any allocation A such that $|\mu(A)| = 1$ then we lose all of the revenue, resulting in a 0-approximation. We amend this in Mechanism 2 by only allocating the lowest bidder for free when the cardinality of the winning set is maximal over all allocations in the image of the allocation function.

We note several assumptions made by Mechanism 2 that simplifies its description. Firstly, we assume that in each bid profile corresponding to a winning set of maximum cardinality there is a distinct lowest winning bidder in the optimal allocation. If lowest winning bids are tied then in order to obtain the revenue bound of Theorem 14 we would want for only one of these to be allocated for free. We can break ties arbitrarily when this is the case, and since there is at least one profile where each player is the unique lowest

Mechanism 2 Willy Wonka auction for binary allocation.

Input: Bid profile $\mathbf{b} \in D$.

Output: Allocation $a(\mathbf{b}) \in \mathcal{A} \subseteq \{0, 1\}^n$, payments $p(\mathbf{b}) \in \mathbb{R}_{\geq 0}^n$.

Let A be an α -optimal allocation w.r.t. $\mathbf{b}$ and let $\mu = \mu(A)$.

for each bidder $i \in N$ **do**

$a_i(\mathbf{b}) \leftarrow A_i$

$p_i(\mathbf{b}) \leftarrow b_i \cdot A_i$

end for

if $A \in \arg \max\{|\mu(A')| : A' \in \mathcal{A}\}$ **then**

 Let $i \in \arg \min\{b_j : j \in \mu\}$

$p_i(\mathbf{b}) \leftarrow 0$

end if

return $a(\mathbf{b}), p(\mathbf{b})$

winning bidder for each possible then Lemma 3 ensures the mechanism is BNOM.

Again we note that this payment rule taken together with the allocation algorithm does not necessarily define any wooden spoon profiles. In most cases the wooden spoon profiles should be ensured by the welfare approximation of the allocation function being high enough and by the feasible set of allocations. In other words, if it is possible to lose the auction then WNOM is guaranteed by Lemma 4. In settings where this does not hold and it is always possible to allocate a given agent, then in order to achieve WNOM we must specify these profiles explicitly. In this special case then, again, in order to achieve the revenue bound of Theorem 14 we have to take care that the golden ticket profile of one agent does not coincide with the wooden spoon profile of another agent.

Observation 6 (Willy Wonka mechanisms are single-line). Recall the definition of profile labellings and single-line mechanisms (Definition 17) given in Chapter 4. A mechanism uses a single-line labelling if for each agent $i \in N$ and each bid b_i , the labelled profile $\lambda_i(t_i, b_i)$ is the same profile for all true types $t_i \in D_i$. We note here that if a mechanism satisfies the Willy Wonka conditions of Definition 24 then the mechanism is single-line.¹

It should be no surprise that Mechanism 2 achieves good revenue. We

¹Technically, a Willy Wonka is not necessarily single-line, since labellings are defined on bid profiles and not outcomes. Therefore, in the case that there are multiple bid profiles satisfying conditions (1) and (2) of Definition 24 then although the extreme values of an agent i 's utility function are guaranteed to occur at the same *outcome* of the mechanism, they do not necessarily occur at exactly the same bid profile. However, in addition to all such profiles leading to the same outcome, they also lead to the same payments by Theorem 5.

achieve incentive compatibility by sacrificing the potential revenue from only a single agent, and only in an allocation that maximises the number of winning agents. Since all competitive NOM mechanisms must sacrifice the revenue from each agent at some point (by Lemma 3 then this revenue bound is optimal.

Theorem 14. *Mechanism 2 is not obviously manipulable and is $\alpha(1 - 1/\tau)$ -competitive to the optimal revenue, where $\tau = \max_{A \in \mathcal{A}} \{ |\mu(A)| \}$ denotes the maximum cardinality winning set under allocation function a .*

Proof. The mechanism is NOM by Lemmas 3 and 4. It charges first-price payments to all winning bidders apart from the lowest when the number of winners is maximal, meaning the revenue is a $(1 - 1/\tau)$ factor of the social welfare of the returned allocation. Since the allocation function is α -approximate then the social welfare of the allocation is at least an α fraction of the optimal, giving the bound. $\square$

In the next section we generalise the arguments in Lemmas 3 and 4 to handle general outcome spaces, with an additional condition on the allocation function.

7.3.2 GENERAL OUTCOME SPACES

In this section we assume $\mathcal{A}$ represents more general classes of allocations and that agents are multiparameter. Recall that each agent i 's type is now a vector $(v_i^A)_{A \in \mathcal{A}}$ where $v_i^A \in [0, h]$ represents the utility i receives from allocation A . In order to derive similar golden ticket and wooden spoon properties to the previous section we impose an additional requirement on the allocation function of the mechanism that requires the allocation to be able to maximise over a fixed range of allocations. An allocation function a is said to be *maximal-in-range* if there exists some subset $\mathcal{R} \subseteq \mathcal{A}$ such that for all $\mathbf{b} \in D$ the function outputs $a(\mathbf{b}) \in \arg \max_{A \in \mathcal{R}} \text{SW}(A, \mathbf{b})$.

Lemma 5 (Golden ticket for general outcome spaces). *Let $M = (a, p)$ be an maximal-in-range auction with range $\mathcal{R} \subseteq \mathcal{A}$. Then M is BNOM if and only if for every agent $i \in N$ with valuation $v_i = (v_i^A)_{A \in \mathcal{A}}$ there exists a bid profile $\mathbf{b}$ such that $a(\mathbf{b}) \in \arg \max_{A \in \mathcal{R}} v_i^A$ and $p_i(\mathbf{b}) = 0$.*

Proof. Let $A \in \arg \max_{A' \in \mathcal{R}} v_i^{A'}$ be i 's highest-valued allocation amongst those in $\mathcal{R}$ and first note that it must always be possible for i to be allocated A . Take the bid profile $(v_i, \mathbf{0}, \dots, \mathbf{0})$, where $\mathbf{0} = (0, \dots, 0)$ denotes the all-zero type. Since M is maximal-in-range and $A \in \mathcal{R}$ then M must return the allocation A .

($\Rightarrow$) Now suppose that whenever the allocation A is returned i must pay some positive price p . Her best truthful utility is at most $v_i^A - p$. Let b_i be any misreport of agent i with $0 < b_i^A < p$ and $b_i^A = \max_{A' \in \mathcal{R}} b_i^{A'}$. Note that b_i^A is an underbid for allocation A since $p \leq v_i^A$ by individual rationality. Now take the profile $(b_i, \mathbf{0}, \dots, \mathbf{0})$. By the same argument as above i must be allocated A and pay some amount p' where $0 < p' < b_i^A$. Her best utility for this misreport is at least $v_i^A - b_i^A > v_i^A - p$ and hence strictly larger than her best utility for reporting v_i truthfully, breaking BNOM. Hence i must win A for free under some bid profile.

($\Leftarrow$) Since i wins A for free then bidding v_i truthfully can result in a utility of v_i^A , which is clearly unbeatable since M makes no positive transfers. Therefore M is BNOM. $\square$

We must strengthen the requirement of the allocation function being α -optimal to being maximal-in-range. This ensures that bidder i is allocated her most valuable allocation in the range of the auction when the other bidders are excluded from winning. Without this requirement then the auction could return any allocation A that i values at least an α fraction as much as her favourite allocation, which may not imply incentive-compatibility. In this case the performance of the allocation function can be expressed as

$$\alpha = \min_{\mathbf{b}} \frac{\arg \max_{A \in \mathcal{R}} \text{SW}(A, \mathbf{b})}{\arg \max_{A \in \mathcal{A}} \text{SW}(A, \mathbf{b})}. \quad (7.2)$$

The wooden spoon property remains exactly the same and using the exact very simple argument from the proof of Lemma 4 we can show that if each agent can lose the auction then it must be WNOM.

Lemma 6 (Wooden spoon for general outcome spaces). *Let $M = (a, p)$ be an auction for multiparameter agents that is β -competitive against the social welfare. If for every agent $i \in N$ with valuation $v_i = (v_i^A)_{A \in \mathcal{A}}$ there exists a bid profile $\mathbf{b}$ such that $a_i(\mathbf{b}) = \emptyset$ then M is WNOM.*

We note that unlike Lemmas 3 to 5 this is not a complete characterisation since we do not have the necessity of the wooden spoon for auctions in these settings to be WNOM.² However, when designing our NOM auctions in the

²One may weaken the wooden spoon property by, for example, defining it as any profile in which i wins an allocation *other than* her most preferred allocation $\arg \max_{A \in \mathcal{A}} v_i^A$. This gives a necessary condition for an auction satisfying the revenue-competitiveness guarantees, however it is now unclear and unlikely that this condition is sufficient for WNOM. Since our goal is to use these golden ticket and wooden spoon profiles to define simple NOM mechanisms, we prefer to have the sufficiency of the original wooden spoon condition. The question of characterising revenue-competitive WNOM mechanisms in multiparameter settings is an interesting question for further work.

following we will only need the sufficiency of the golden tickets and wooden spoons to show the resulting mechanisms are incentive-compatible.

Definition 25 (Willy Wonka mechanism for general outcome spaces). A *Willy Wonka mechanism for general outcome spaces* is a mechanism $M = (a, p)$ whose allocation and payment functions satisfy the following. For each agent $i \in N$ and each bid $b_i \in D_i$, there exists:

1. A bid profile $\mathbf{b} = (b_i, \mathbf{b}_{-i})$ where i is allocated $\arg \max_{A \in \mathcal{A}} b_i^A$ for free.
2. A bid profile $\mathbf{b} = (b_i, \mathbf{b}'_{-i})$ where i loses the auction.

We only need to modify Mechanism 2 slightly in order to apply to general outcome spaces, which we provide in Mechanism 3. We now assume the allocation function a is maximal-in-range as per Lemma 5, but as before in Mechanism 3 we compute an approximately optimal allocation and then charge first-price payments in most cases. When the allocation has maximal cardinality then we sacrifice the revenue from the lowest winning bidder in order to satisfy the golden ticket property.

Mechanism 3 Willy Wonka auction for general outcome spaces.

Input: Bid profile $\mathbf{b} \in D$.

Output: Allocation $a(\mathbf{b}) \in \mathcal{A}$, payments $p(\mathbf{b}) \in \mathbb{R}_{\geq 0}^n$.

Let $A \in \arg \max \{ \text{SW}(A', \mathbf{b}) : A' \in \mathcal{R} \}$ and let $\mu = \mu(A)$.

for each bidder $i \in N$ **do**

$a_i(\mathbf{b}) \leftarrow A_i$

$p_i(\mathbf{b}) \leftarrow b_i^A \cdot \mathbb{1}[i \in \mu]$

end for

if $A \in \arg \max \{ |A'| : A' \in \mathcal{R} \}$ **then**

Let $i \in \arg \min \{ b_j^A : j \in \mu \}$

$p_i(\mathbf{b}) \leftarrow 0$

end if

return $a(\mathbf{b}), p(\mathbf{b})$

Again it is assumed that the wooden spoon profiles arise from the feasible set itself, that is, for each agent there is some bid profile such that it is not feasible to allocate the agent. If this is not the case we would have to artificially exclude the agent from the winning set. Naturally Mechanism 3 obtains a similar revenue guarantee to Mechanism 2 of $\alpha(1 - 1/\tau)$ where τ is the size of the maximum cardinality winning set for any allocation returned by the mechanism, and α again corresponds to the approximation guarantee of the allocation function with respect to the socially optimal allocation and takes the form given in (7.2).

Theorem 15. *Mechanism 3 is not obviously manipulable and is $\alpha(1 - 1/\tau)$ -competitive to the optimal revenue, where $\tau = \max\{|\mu(A)| : A \in \mathcal{R}\}$ denotes the cardinality of the largest winning set under the maximal-in-range allocation function a .*

In the next section we turn our attention to procurement auctions, where we must now compensate bidders for selecting them in the final allocation. We show similar conditions for the golden tickets and wooden spoons, then study the frugality ratio of the resulting mechanisms, roughly meaning the amount by which these profiles mean we overcompensate the bidders.

7.4 FRUGAL PROCUREMENT

In this section we consider procurement auctions, where for each agent i and each allocation $A \in \mathcal{A}$ we have $t_i(A) \leq 0$. For clarity of exposition we now introduce for each agent i a *cost function* and denote it $c_i : \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$, setting $c_i = -t_i$. Now we have a procurement auction that allocates chores to the agents, hence the agents must be compensated for taking on their allocation. For a procurement auction $M = (a, p)$ given the input $\mathbf{b}$ the utility that agent i with cost c_i will receive is denoted $u_i(M(\mathbf{b}); c_i) = p_i(\mathbf{b}) - c_i(a_i(\mathbf{b}))$.

We are now interested in allocation functions that (approximately) minimise the social cost of resulting allocation. Fix a cost profile $\mathbf{c}$. The social cost of an allocation A with respect to $\mathbf{c}$ is $\text{SC}(A) = \sum_{i \in N} c_i(A_i)$. An allocation function a is said to α -approximate the social cost for $\alpha \geq 1$ if $\text{SC}(a(\mathbf{c}), \mathbf{c}) \leq \alpha C^*$ for every $\mathbf{c}$, where $C^* = \min_{A \in \mathcal{A}} \text{SC}(A, \mathbf{c})$ denotes the social cost of an optimal allocation with respect to $\mathbf{c}$.

We now need to compensate agents for their allocations. All of our mechanisms must be individually rational, hence each bidder must always be paid at least the cost she reports for the resulting allocation. We want to study the extent to which our mechanism overpays with respect to some benchmark. Let $\mathcal{X}$ be some payment benchmark for a procurement auction and $\mathcal{X}(\mathbf{b})$ denote the sum of payments made to the bidders on bid profile $\mathbf{b}$. A payment function p is said to be β -frugal against $\mathcal{X}$ for $\beta \geq 1$ if $\sum_{i \in N} p_i(\mathbf{b}) \leq \beta \mathcal{X}(\mathbf{b})$ for all $\mathbf{b}$. We refer to β as the *frugality ratio* of the payment function (and equally, the mechanism) with respect to $\mathcal{X}$.

We think of the auctions as allocating bundles of chores, and to maintain individual rationality we must now instead reimburse or compensate the bidders allocated a chore with a monetary transfer. For consistency we still refer to the set of such bidders as the set of winners, and every other bidder is therefore still considered a loser of the auction. We also restrict our attention to single-parameter agents.

Lemma 7 (Golden ticket for binary allocation). *Let $M = (a, p)$ be a procurement auction that α -approximates the social cost for any $\alpha > 0$. Then M is BNOM if and only if for every agent $i \in N$ with cost c_i there exists a bid profile where i wins the auction and is paid h .*

Proof. ($\implies$) Suppose for contradiction that M only ever pays bidder i at most $p < h$ when she bids c_i . By individual rationality we must have $p \geq c_i$. Now take an overbid b_i such that $p < b_i < h/\alpha$. Let $\mathbf{b} = (b_i, \mathbf{b}_{-i})$ be a bid profile such that $b_j > \alpha b_i$ for each $j \neq i$. On $\mathbf{b}$ the chore must get allocated to i in order to preserve the α guarantee to the optimal social cost. Since i is allocated the chore she must be compensated at least b_i , resulting in a best-case dishonest utility of at least $b_i - c_i$ which is strictly greater than her best-case truthful utility of $p - c_i$, thus violating BNOM.

($\impliedby$) If there exists a bid profile where i wins and is paid h then when i has cost c_i and bids truthfully her best utility is $h - c_i \geq 0$. Since i can never submit an overbid greater than h then no misreport can get her a greater utility, thus M is BNOM. $\square$

Lemma 8 (Wooden spoon for binary allocation). *Let $M = (a, p)$ be a procurement auction that is β -frugal against the social cost for any $\beta > 0$. Then M is WNOM if and only if for every agent $i \in N$ with cost c_i there exists a bid profile such that i loses the auction.*

Proof. ($\implies$) Assume i always wins the auction. Then on an overbid $b_i > c_i$ she would get utility at least $b_i - c_i$. Now take the bid profile $(c_i, h, \dots, h)$ such that $(n-1)h/\beta > c_i$ where only i may win the auction. The β -frugality of the auction means that i cannot be compensated more than βc_i meaning her best truthful utility is at most $(\beta - 1)c_i$. Taking $b_i > \beta c_i$ results in a worst utility of at least $b_i - c_i > (\beta - 1)c_i$ thus breaking the WNOM constraints. Therefore i must be able to lose the auction.

($\impliedby$) By a similar argument to the proof of Lemma 7 if the monetary payment is independent of the bid then i 's worst truthful and dishonest utilities are both 0, satisfying WNOM. $\square$

We use the above conditions to design frugal Willy Wonka mechanisms for procurement auctions. Similarly to (7.1) fix a player i with bid b_i . On the golden ticket profile for i the optimal Willy Wonka mechanism for frugal procurement should select an allocation A that compensates the agents by

$$\min_{A: i \in \mu(A)} \{ \min_{\mathbf{b}_{-i}} \text{SC}(A, \mathbf{b}_{-i}) + h \},$$

since i will be paid h while the remaining winners are paid their bid. To obtain a good approximation to the above, again consider a mechanism with black box access to an α -approximate allocation function such that on input $\mathbf{b}$ the mechanism selects an α -optimal allocation A that pays the highest winner h and the remaining winners their bid. Now the highest winning bidder (instead of the lowest) is the recipient of the golden ticket, since this minimises the total payment made on the given instance. Then the total cost on allocation A is equal to

$$\alpha \cdot \text{SC}(A, \mathbf{b}) + h - \max_{i \in \mu(A)} b_i.$$

Again, we need not designate a golden ticket profile (giving the highest winning bidder a payment of h) for every bid profile $\mathbf{b}$ as this will degrade the frugality ratio with respect to the sum of (winning) bids. Consider an analogous approach to Mechanisms 2 and 3 for designing frugal Willy Wonka procurement auctions, where the golden ticket profile occurs for each bidder when they have the highest reported cost in an (approximately) optimal allocation of maximum winning set cardinality. Suppose $\mathcal{A}$ contains all allocations with singleton winning sets. Now fix bidder i and take $\mathbf{b} = (b_i, h, \dots, h)$ for some b_i . Observe that the optimal cost of allocating only i is b_i , and at least h for any other allocation. For a given α then a sufficiently small b_i means i must be the sole winner. If the mechanism paid h to bidder i then the resulting frugality ratio would be infinite when we take b_i to zero.

We provide an alternative way of specifying the golden ticket profiles of each bidder in Mechanism 4 and show that although its frugality ratio with respect to the optimal can still be infinite, it will never overpay the bidders by more than twice the cost of the second-best solution, provided there exists an allocation for each agent where they are not the sole winner. Let $k = \max_{A: i \in \mu(A)} |\mu(A)|$ denote the maximum cardinality of winning set containing i for any allocation in $\mathcal{A}$, and let S be the corresponding winning set of any such allocation. Given bid b_i of player i define the golden ticket profile such that for all $j \neq i$ we have $b_j = h/(k-1)$ whenever $j \in S$ and $b_j = h$ otherwise. Denote this golden ticket for b_i as $\gamma_i(b_i)$.

We note that the frugality ratio for Mechanism 4 can vary depending on the size k of the set S .

Claim 4. The frugality ratio FR for Mechanism 4 is equal to 2 whenever $k > 1$, and ∞ otherwise.

Proof. If $k > 1$ then the total payments made on the bid profile $(b_i, \gamma_i(b_i))$ is $h + (k-1)h/(k-1) = 2h$. When $b_i = 0$ then the optimum cost is h ,

Mechanism 4 Willy Wonka procurement auction.

Input: Bid profile $\mathbf{b} \in D$.

Output: Allocation $a(\mathbf{b}) \in \mathcal{A} \subseteq \{0, 1\}^n$, payments $p(\mathbf{b}) \in \mathbb{R}_{\geq 0}^n$.

 Let A be an α -optimal allocation w.r.t. $\mathbf{b}$ and let $\mu = \mu(A)$.

for each bidder $i \in N$ **do**
 $k \leftarrow \max_{A: i \in \mu(A)} |\mu(A)|$ and $h' \leftarrow h/(k-1)$ (when $k > 1$)

 $\gamma_i(b_i) \leftarrow \underbrace{(h', \dots, h')}_{(k-1)}, \underbrace{(h, \dots, h)}_{(n-k)}$
 $a_i(\mathbf{b}) \leftarrow A_i$
 $p_i(\mathbf{b}) \leftarrow b_i \cdot A_i$
if $b_{-i} = \gamma_i(b_i)$ **then**
 $p_i(\mathbf{b}) \leftarrow 0$
end if
end for
return $a(\mathbf{b}), p(\mathbf{b})$

giving a frugality ratio of 2. If $k = 1$ then the sum of payments is h while the optimum is zero, meaning the frugality ratio is infinite. $\square$

Instead let us benchmark the mechanism against the cost of the second-best solution. We denote by $\text{FR}^{(2)}(\mathbf{b})$ the frugality ratio of a given mechanism on input $\mathbf{b}$ with respect to the cost of the second-best solution:

$$\text{FR}^{(2)}(\mathbf{b}) = \frac{\sum_{i \in N} p_i(\mathbf{b})}{\min_A \{ \text{SC}(A, \mathbf{b}) : \text{SC}(A, \mathbf{b}) > C^* \}},$$

where $C^* = \min_A \text{SC}(A, \mathbf{b})$ denotes the cost of an optimum allocation on $\mathbf{b}$. We then take $\text{FR}^{(2)} = \max_{\mathbf{b}} \text{FR}^{(2)}(\mathbf{b})$. With the following we show that Mechanism 4 has good performance with respect to $\text{FR}^{(2)}$ as long as there exists for each agent an allocation where they are not the only winning bidder.

Claim 5. The frugality ratio $\text{FR}^{(2)}$ for Mechanism 4 is equal to 2 whenever $k > 1$, and 1 otherwise.

Proof. Let $A = a(\mathbf{b})$ be the allocation returned by a on input $\mathbf{b} = (b_i, \gamma_i(b_i))$. For all cases where $k \geq 1$ then the cost of the second-best solution is at least h . When $k > 1$ then if $i \in \mu(A)$ the total sum of payments is at most $h + (k-1)h/(k-1) = 2h$, otherwise if $i \notin \mu(A)$ then the total sum of payments is simply h . When $k = 1$ then $\gamma_i(b_i) = (h, \dots, h)$ by definition, and whoever wins is paid h , either because i wins the golden ticket or because some bidder $j \neq i$ wins and must be compensated at least her bid. $\square$

CLOSING REMARKS

In this thesis we have taken the first systematic approach to designing mechanism without obvious manipulations using monetary transfers. After formalising the added flexibility provided by NOM mechanisms in the form of profile labellings, we then characterise the class of social choice functions that admit NOM payment schemes resulting in incentive compatible mechanisms under both direct-revelation and extensive-form models of agent interaction. In both cases, by considering the cycles created in a carefully defined graph we provide a characterisation which gives us a useful tool to study when a social choice function is implementable, as well as a reliable method to design the allocation and payment functions ourselves. By focusing on single-parameter domains we show that all implementable functions must be “overlapping”, a property roughly describing their shape which ensures, that as the mechanism design in charge of deciding the profile labelling, it is always possible to find a monotone restriction of the allocation function that will result in a NOM mechanism. Under these domains we import explicit formulations of the payments to be used, noting that this only applies now to the labelled profiles, for a simple class of mechanisms that we introduce known as “single-line” mechanisms. This allows us to draw parallels between designing strategyproof mechanisms and designing mechanisms without obvious manipulations for the widely studied class of single-parameter domains. With the incentive compatibility constraints automatically handled by these payments, we are free to define the payments for the remaining profiles as desired, with only the labelling constraints to take into account. We extend our techniques to the more general class of indirect mechanisms and find that for the settings of single-parameter domains and mechanisms between two agents in which one agent has a two-value domain, the set of social choice functions implementable as direct-revelation NOM mechanisms is the same as the set of social choice functions implementable as extensive-form NOM mechanisms. Direct mechanisms are simpler to state

and we show that in these two settings we can restrict attention to this class without sacrificing the obvious incentive properties as they relate to NOM.

We apply the techniques developed in the first chapters to the settings of bilateral trade and general auction environments. We study an impossibility result of Troyan and Morrill’s stating that every efficient, individually rational, weakly budget balanced mechanism for bilateral is obviously manipulable. Relaxing weak budget balance to *approximate* weak budget balance, we find that the single-line mechanisms introduced in the previous chapters are unable to achieve all the desired properties, with a potentially infinite factor of the money paid by the buyer required to subsidise trade to the seller. Looking beyond single-line mechanisms we find the negative results persist for BNOM and vanish for WNOM: no labelling can lead to a mechanism with bounded subsidies for BNOM, but we show a labelling achieving exact weak budget balance for WNOM. Then looking at general auction environments that allocate goods as well as bads to the agents, we are able to characterise the class of NOM mechanisms as those satisfying the Willy Wonka properties, which require the existence of so-called “golden ticket” and “wooden spoon” profiles. In single-parameter environments these properties are both necessary and sufficient under fairly weak assumptions on the optimality of the allocation and payment functions. We only (potentially) lose the necessity of the wooden spoon when moving to general domains. In any case, these results allow us to specify simple mechanisms for both goods auctions and procurement auctions in a range of settings. By designating these golden ticket and wooden spoon profiles carefully, we can produce mechanisms that perform very well under revenue-maximisation and frugal procurement.

There are many interesting avenues for further research left open by our research. On the question of implementability, in a similar vein to [Aziz and Lam, 2021; Golowich and S. Li, 2021] it would be interesting to study questions surrounding the computational aspects of NOM. For example, given a social choice function f (for example as a table of values), what is the computational complexity of deciding whether f admits a NOM payment scheme. In other words, due to our characterisation theorems in Theorems 3 and 11, what is the complexity of deciding whether there exists a labelling λ that produces no negative weight cycles in $\mathcal{G}_{i,f}^\lambda$ for some player $i \in N$. One could even alter how f is represented in order to tackle a similar question for Theorem 4 in the context of single-parameter domains. As [Camara, 2022] shows, it is an important practical consideration to study the computational aspects of mechanism design in order to better understand how real-life decision makers may act when interacting with the mechanisms.

Looking towards indirect mechanisms, an obvious question left open by our work is whether there are other domains and settings under which the revelation principle holds. We note that our partial revelation principles are only on the social choice functions, and not the payments, that is, we only guarantee that if a social choice function is implementable by a NOM indirect mechanism then there also exists some direct-revelation mechanism that also implements it. Given these two settings from Sections 6.5.1 and 6.5.2 of Chapter 6 where these partial revelation principles holds, given that the class of implementable social choice functions are the same for both direct-revelation and extensive-form mechanisms, it would be interesting to explore exactly what one can gain by focusing on one class of mechanisms over the other. While direct mechanisms may be simpler to define from the mechanism designer’s perspective, this may not correspond to the mechanism being easy to understand from the point of view of the agents (consider, for example, [S. Li, 2017; Gonczarowski, Heffetz, and Thomas, 2023]), and therefore we may want to focus on indirect mechanisms since they may be easier to follow by the cognitively limited agent. Conversely, perhaps direct-revelation mechanisms have advantages over extensive-form mechanisms in other regards such as the possible class of NOM payment functions they admit? These would all be interesting avenues for further research, and particularly salient to the study of bounded rationality and simplicity in mechanism design.

In the setting of bilateral trade, it would be useful to better understand the implications of the trading windows characterisation of Theorem 10, when we drop the efficiency requirement on the allocation function (keeping individual rationality, weak budget balance, and of course NOM). Moreover, with Theorem 9 we show that no BNOM mechanism is able to guarantee even approximate weak budget balance for any bounded subsidy when we require efficiency and individual rationality, hence nor can any NOM mechanism. As an interpolation between Theorem 10 and full efficiency, one could study what is possible under an approximate efficiency property.

For the auction settings covered in Chapter 7, we firstly note that the performance of the payment rules we design is dependent on the number of winners in an optimal allocation. Can we design payment rules whose performance does not depend on the structure of the feasible set, to better deal with instances where this results in poor performance for the mechanism? We also note that, given a bid b_i of player i , the best and worst outcomes of the mechanism always occur on the same profile, regardless of her true type. These are the golden ticket and wooden spoon profiles, respectively, and the resulting mechanisms are therefore single-line, as already noted, and

therefore the payment schemes are simple to state (Theorem 5). Are golden ticket and wooden spoon payment rules the only types of single-line payment rules leading to good performance from the mechanism? NOM also permits much more general payment schemes – can we design non single-line payment rules that yield better performance that can be achieved by a single-line mechanism? Finally, can we design (efficient) allocation functions ourselves for general settings so that our mechanisms do not rely on them as black boxes, and can we use in conjunction with an appropriate payment rule to yield a NOM mechanism?

Our characterisations in Chapter 7 are only a starting point for the design of NOM mechanisms in a range of settings, and we expect there to be many more highly relevant contexts where similar approaches will produce positive results on the performance of NOM mechanisms over other solution concepts. One could relax strategyproofness to NOM to study potential gains in the setting of budget-feasible procurement auctions, which introduces an interesting interaction between the publicly-known values of the agents’ services and their privately-held costs, especially given that this provides a nice middle ground between strictly single-parameter domains and more general multiparameter ones. An allocation function in this setup aims to maximise the procured value while minimising the cost, subject to compensatory payments never exceeding some given budget, and it would be interesting to explore to what extent the various approximation bounds are due to strategyproofness (Klumper and Schäfer, 2022; Amanatidis, Kleer, and Schäfer, 2022; Amanatidis, Klumper, et al., 2023). This introduces the question of extending NOM to the case of divisible allocations, and we note that in this thesis we have largely (though not always) studied binary outcomes, hence it would be interesting to study how both the class of implementable functions as well as their payments change under this more general setting. In a related manner (reading divisible allocations as probabilities), one could further extend the definition of NOM to handle randomised mechanisms. Given the very restrictive nature of strategyproofness in the scheduling problem on unrelated machines, with an associated approximation bound of n on the optimal makespan for n machines (Christodoulou, Koutsoupias, and Kovács, 2023), applying NOM to this setting provides not only a well-studied multiparameter setting with much room to provide advantages over strategyproofness, but also a chance to consider NOM under the objective of optimising the egalitarian (as opposed to utilitarian) social welfare. We note that [Psomas and Verma, 2022] considers optimising all of the utilitarian, egalitarian, and Nash social welfare in the context of fair division without obvious manipulations, however there are of course many

more settings to which these objective functions find use in the literature which prove ripe ground for further research.

8.1 IS NOM THE RIGHT MODEL FOR BOUNDED RATIONALITY?

Although NOM is a well-justified notion of incentive compatibility for agents with imperfect rationality, given that obvious manipulations are exactly those that are able to be recognised by agents who can only understand the range of the mechanism, conditional upon their report, and not the full specification of the mechanism (Trojan and Morrill, 2020), it is of course not the only manner in which one can define bounded rationality for the imperfectly rational agent. Indeed, Li’s notion of obvious strategyproof is another solution concept that interprets the same idea of agents being unable to perform fine-grained contingent reasoning on the reports of each of the other agents in a more pessimistic light. We have seen that NOM is, on the whole, a flexible solution concept. There is no reason to believe that NOM as defined by Trojan and Morrill is exactly the right notion of incentive compatibility for imperfect rationality, even if we take an optimistic perspective on when agents will attempt to manipulate. For example, we can assume agents have some additional understanding of the mechanism and suppose that, instead of comparing the outcomes of the mechanism at the extremes of their utility function, they compare the outcomes at certain *quantiles* of their utility function (still taken over all possible bid profiles of the other players). This would result in a slightly less brittle solution concept by taking into account, in a sense, how the agent’s utility function is distributed over the range of possible outcomes. Therefore, if proportionally more bid profiles lead to a lower utility for i with type t_i when bidding b_i then she is able to take this into account and act accordingly. We provide the following informal definition of a potential “quantile NOM”.

Definition 26 (Quantile NOM). Let $Q_{\mathbf{b}_{-i}}^p(b_i)$ denote the p -quantile of i ’s utility function ranging over all $\mathbf{b}_{-i}$ when she bids b_i for $p \in [0, 1]$. Mechanism M is p -quantile NOM for $p \in [0, 1]$ if for every agent $i \in N$ and every pair of types $t_i, b_i \in D_i$ it holds that:

$$Q_{\mathbf{b}_{-i}}^p(t_i) \geq Q_{\mathbf{b}_{-i}}^p(b_i).$$

Note that 0-quantile NOM is equivalent to WNOM and 1-quantile NOM is equivalent to BNOM.

Another possible strengthening of NOM is to assume that each agent has a fixed set of reports of the other agents over which they can perform

contingent reasoning. For a set of agents $Y \subseteq N$ we denote by $\mathbf{b}_Y = (b_j)_{j \in Y}$ the (partial) bid profile where each agent $j \in Y$ submits bid $b_j \in D_j$. Accordingly given a player $i \in N$ and set $Y \subseteq N$ we decompose the bid profile $(b_i, \mathbf{b}_{-i})$ as $(b_i, \mathbf{b}_Y, \mathbf{b}_Z)$ where $Z = N \setminus Y \setminus \{i\}$.

Consider an interpolation between the perfect rationality modelled by strategyproofness and the absence of all contingent reasoning modelled by NOM whereby an agent is able to perform contingent reasoning on a fixed subset of the other agents.

Definition 27 (Z -NOM). Let $i \in N$, $Y = Y_i \subseteq N$, and $Z = Z_i = N \setminus Y_i \setminus \{i\}$. Mechanism (f, p) is Z_i -NOM for player i if for each $\mathbf{b}_Y \in D_Y$ and each $t_i, b_i \in D_i$ it holds that:

$$\begin{aligned} \sup_{\mathbf{b}_Z \in D_Z} \{u_i(M(t_i, \mathbf{b}_Y, \mathbf{b}_Z); t_i)\} &\geq \sup_{\mathbf{b}_Z \in D_Z} \{u_i(M(b_i, \mathbf{b}_Y, \mathbf{b}_Z); t_i)\}, \\ \inf_{\mathbf{b}_Z \in D_Z} \{u_i(M(t_i, \mathbf{b}_Y, \mathbf{b}_Z); t_i)\} &\geq \inf_{\mathbf{b}_Z \in D_Z} \{u_i(M(b_i, \mathbf{b}_Y, \mathbf{b}_Z); t_i)\}. \end{aligned}$$

If (27) holds then (f, p) is said to be Z_i -BNOM and if (27) holds then it is said to be Z_i -WNOM. If for each i there is a set Z_i of size at least k such that the mechanism is Z_i -NOM then it is said to be k -NOM.

Here the set Y_i represents the fixed set of agents over whose actions player i can perform contingent reasoning. Note that when $Y_i = [n] \setminus \{i\}$ we recover the definition of strategyproofness, and for $Y_i = \emptyset$ we recover the original definition of NOM.

Both the proposed extensions of NOM above provide a way to parameterise the extent to which agents are perfectly (or imperfectly) rational. There will be no one solution concept that is applicable to all settings, nor would it be possible to define a notion of bounded rationality that is “perfect”. The characterisation theorems of S. Li (2017) and Troyan and Morrill (2020) of course provide strong justification that their models of imperfect rationality are indeed useful, however they are by no means the only useful ones possible. One can define a wide range of solution concepts that assume agents have imperfect rationality, and rather than focusing on defining the “right one” (an impossible task), it is instead more important to understand that it is possible and indeed vital to incorporate the cognitive limitations of the agents when designing mechanisms for real world agents.

8.2 FINAL THOUGHTS

Overall this thesis finds that NOM is a solution concept with much potential in the study of bounded rationality in mechanism design. It has been intro-

duced as a salient notion of incentive compatibility for agents with zero contingent reasoning skills, and is far from a trivial solution concept to impose on the requirements of the mechanism. We have formalised the notion that NOM mechanisms provide much more flexibility over other more restrictive solution concepts in the literature, and especially the highly stringent twin solution concept for boundedly rational agents of obvious strategyproofness. NOM mechanisms therefore deserve to be better understood, and there is increasing interest in their study appearing in the literature. With this thesis we have applied NOM to only a fraction of the settings touched by mechanism design, and we hope this work spawns many more interesting ideas in the study of mechanism design for agents with bounded rationality.

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